

RH850/U2 – SERIES U2Bx

MICROCONTROLLER INTRODUCTION

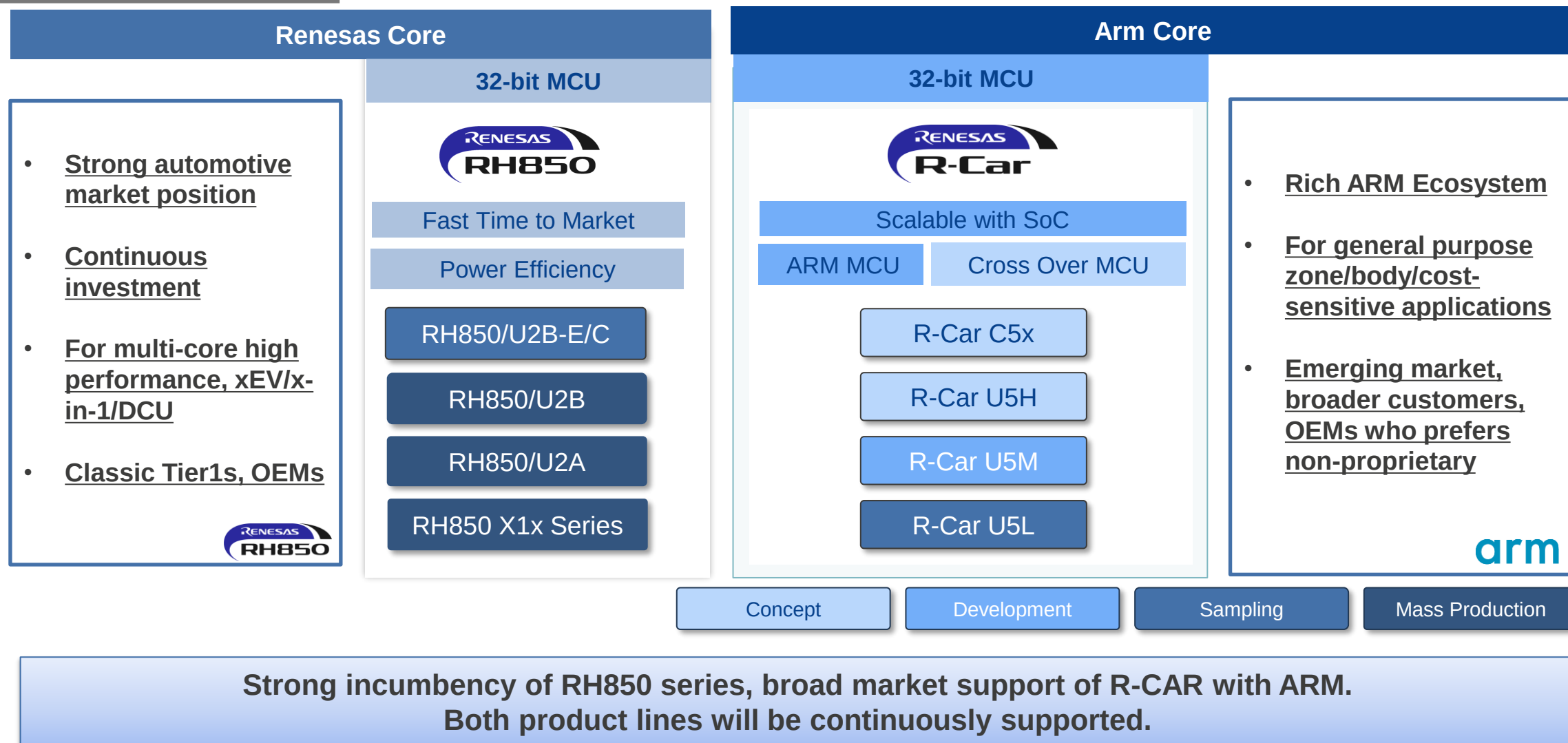
MAR-2025 (REV 1.19)

HMBM3

HPCSG

RENESAS ELECTRONICS CORPORATION

DUAL CPU LINE-UP STRATEGY FOR AUTO MCU MARKET



WORLD-LEADING FLASH TECHNOLOGY

Changing Industry Needs Driving Automotive MCUs

- Multi-Core (High Performance & Robustness)
- Low Power Consumption (Overheating Prevention)
- Increasing Program Capacity
- High Integration (Cutting-Edge Features)



Green-Car



Autonomous Driving



Zone ECU



Development & Production Partnership of World Standard 28nm MCU

Increasing Supply Security

- 28nm dual fab (Taiwan & Japan) since 2024
- Long process availability, since 2 years producing
- More dice per wafer due to smaller process
- Lower power consumption at same performance
- High SW reuse from prior generation Renesas MCUs

→ Easy migration from 40nm to 28nm MCU

RH850 CAPACITY EXPANSION THROUGH MULTI-FAB

WHY RENESAS?



- ✓ Secure and Safe supply
- ✓ Strategic partnership with TSMC

RH850 40nm MCU

RH850 28nm MCU



Naka factory



TSMC Taiwan



JASM

- JASM: Japan Advanced Semiconductor Manufacturing
- TSMC will build a US\$8.6 billion plant in Kumamoto
 - Production started
 - Increase monthly production capacity to 55,000 12-inch wafers



Kumamoto



MICROCONTROLLER QUALITY – RH850 IN 40NM AND 28NM

RH850 (40nm + 28nm) family continue to maintain high quality in shipment quantities of 700M pcs or more.



Note: Total shipping qty and failure rate as of 2023.

EXECUTIVE SUMMARY – AUTOMOTIVE MARKET LANDSCAPE

Industry trends are increasing demand for our MCU and SoC portfolio



Mass deployment of advanced ADAS



Data and software-oriented E/E architecture



Rapid HEV/EV powertrain evolution

Personalized cockpit
Cabin awareness monitoring
Greater driving autonomy
Vehicle network optimization
Affordable electrification

OEM centric model: differentiate and stay ahead throughout vehicle lifecycle



New mobility revenue through new digital experiences



Taking control of critical semiconductor selection



Higher expectation for e2e SDK experience

Define strategy for domain integration (SoC) & real-time processing partitioning (MCU)
Innovation via software: own application framework & AI/ADAS

HPC solves customer problems with broadest MCU and SoC portfolio



Purpose built for automotive applications



Technology leadership with 3nm and chiplet



Scale real-time processing capability w/ ASIL-D

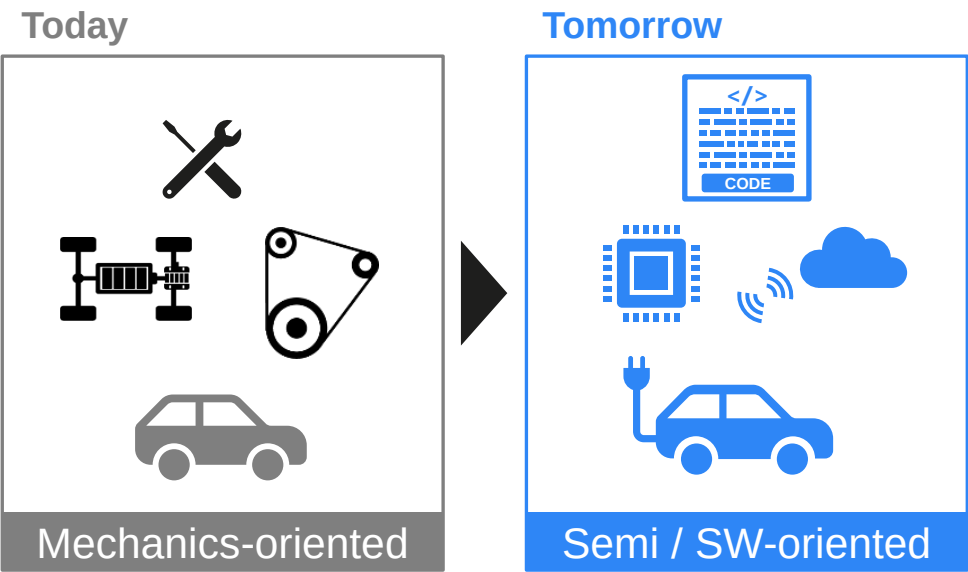
Flexible one-stop-shop & portfolio scaling for entire vehicle fleet
R&D efficiency via software reuse
Centralized & domain compute
Zonal control & edge MCUs

AUTO-SEMI MARKET EXPANSION

The value of future automotive will increasingly depend on semiconductors, leading to significant expansion in the semiconductor market.

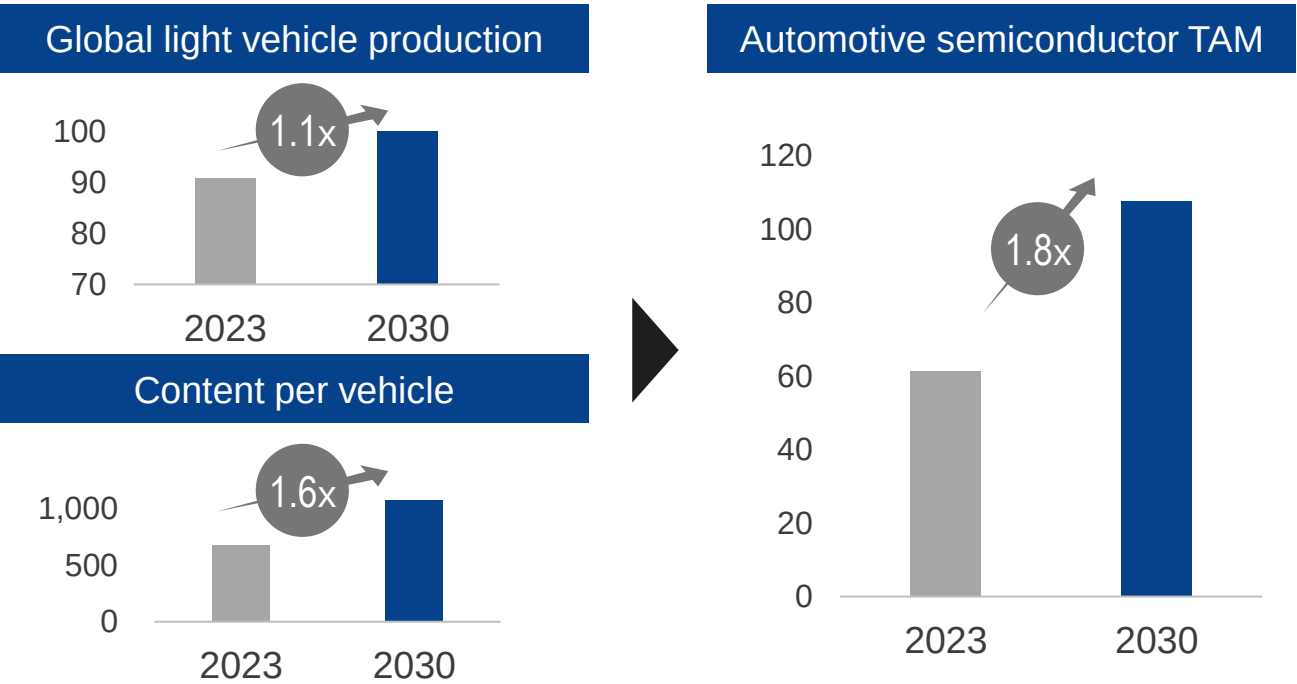
Shift of value

- The acceleration of the CASE trend is shifting the core value of automotives from mechanical components to semiconductors / software.



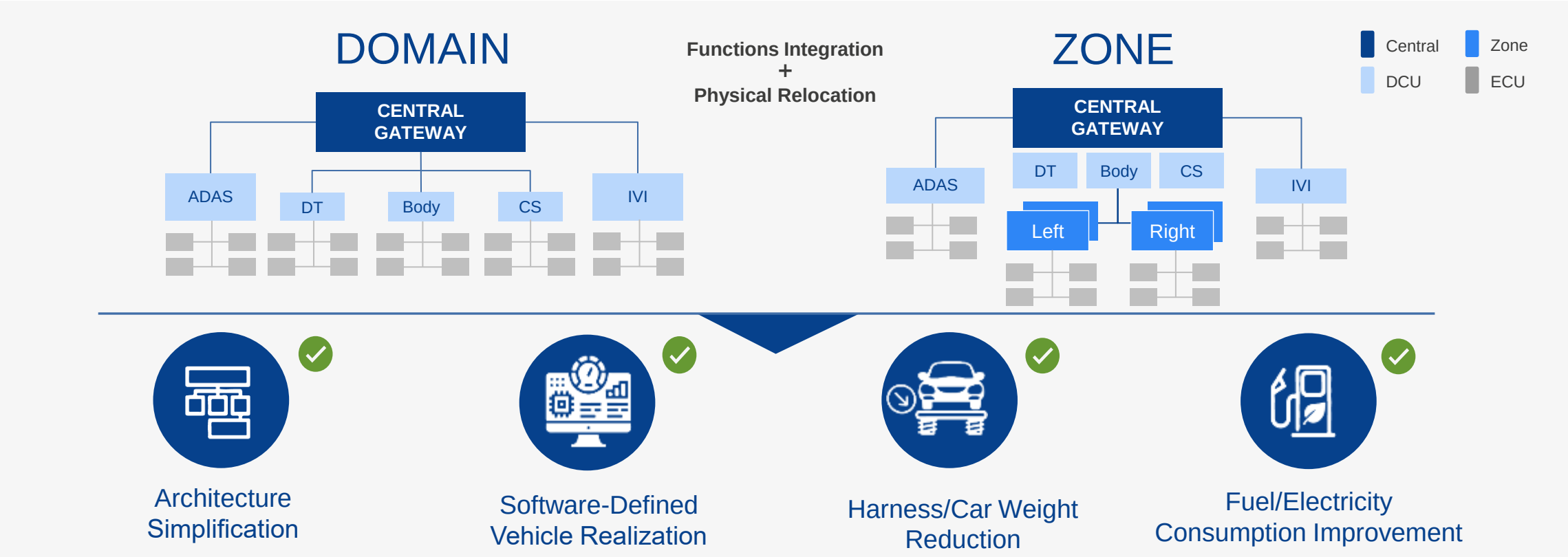
Semiconductor market

- Due to the increase in semiconductor content per vehicle, the semiconductor market is expected to expand even with a slight vehicle production growth.



EVOLUTION OF E/E ARCHITECTURE

As automotive become more semiconductor-oriented, the evolution of the E/E architecture within vehicles is progressing

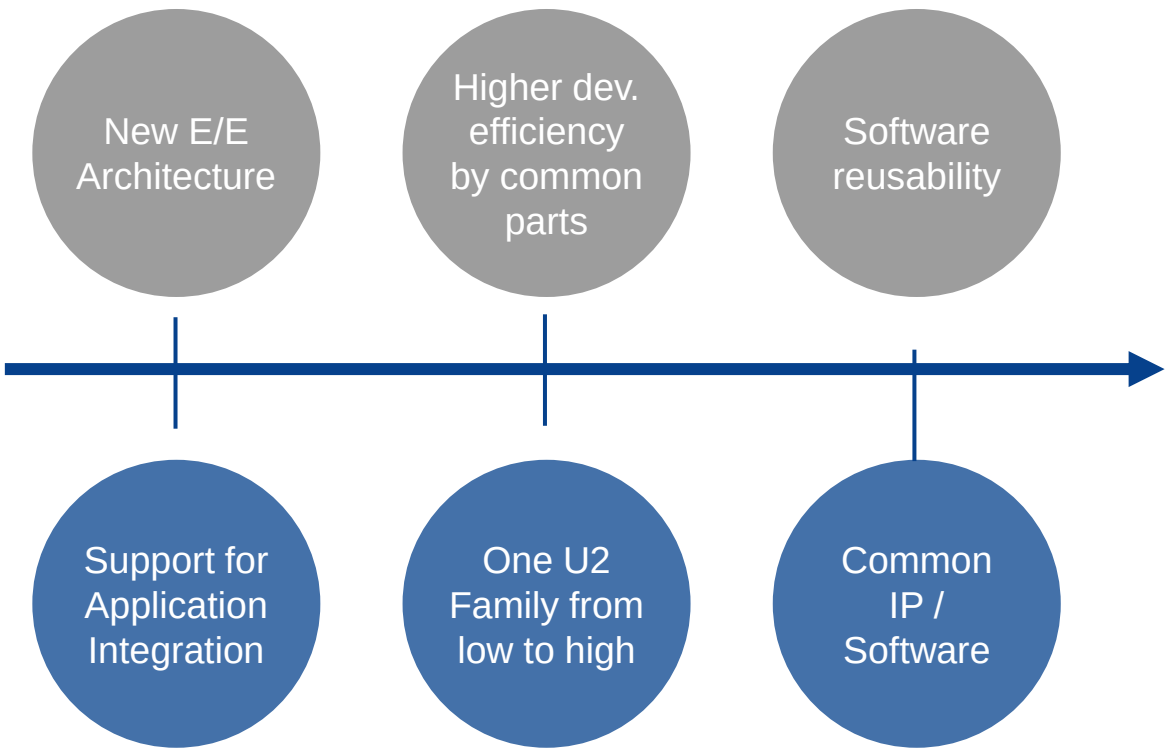


DCU: Domain Control Unit, ECU: Electric Control Unit, DT: Drivetrain (Powertrain), CS: Chassis, IVI: In-Vehicle Infotainment

RH850/U2X OVERVIEW

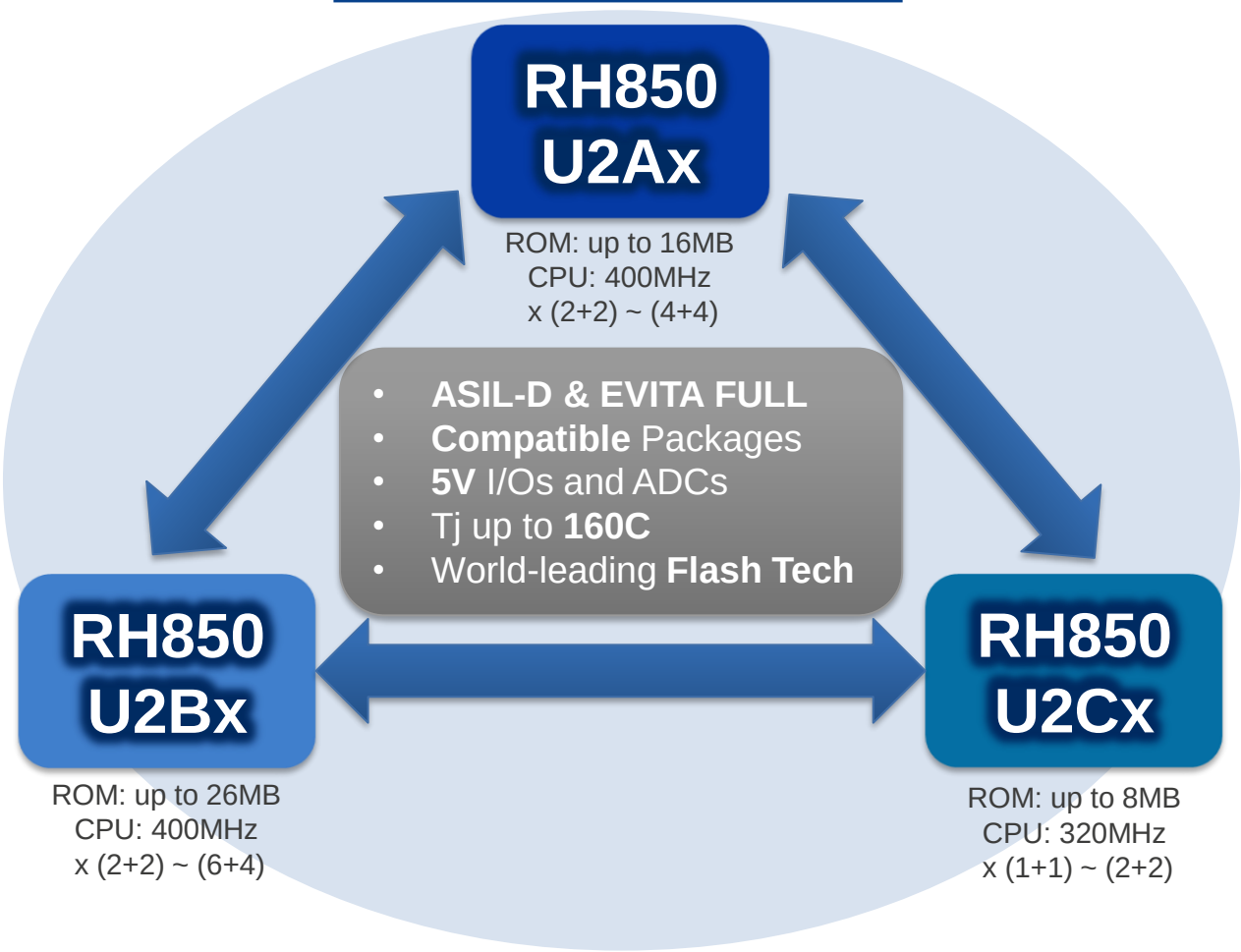
RH850/U2X FAMILY - CONCEPT

Market requirement



Renesas' platform strategy

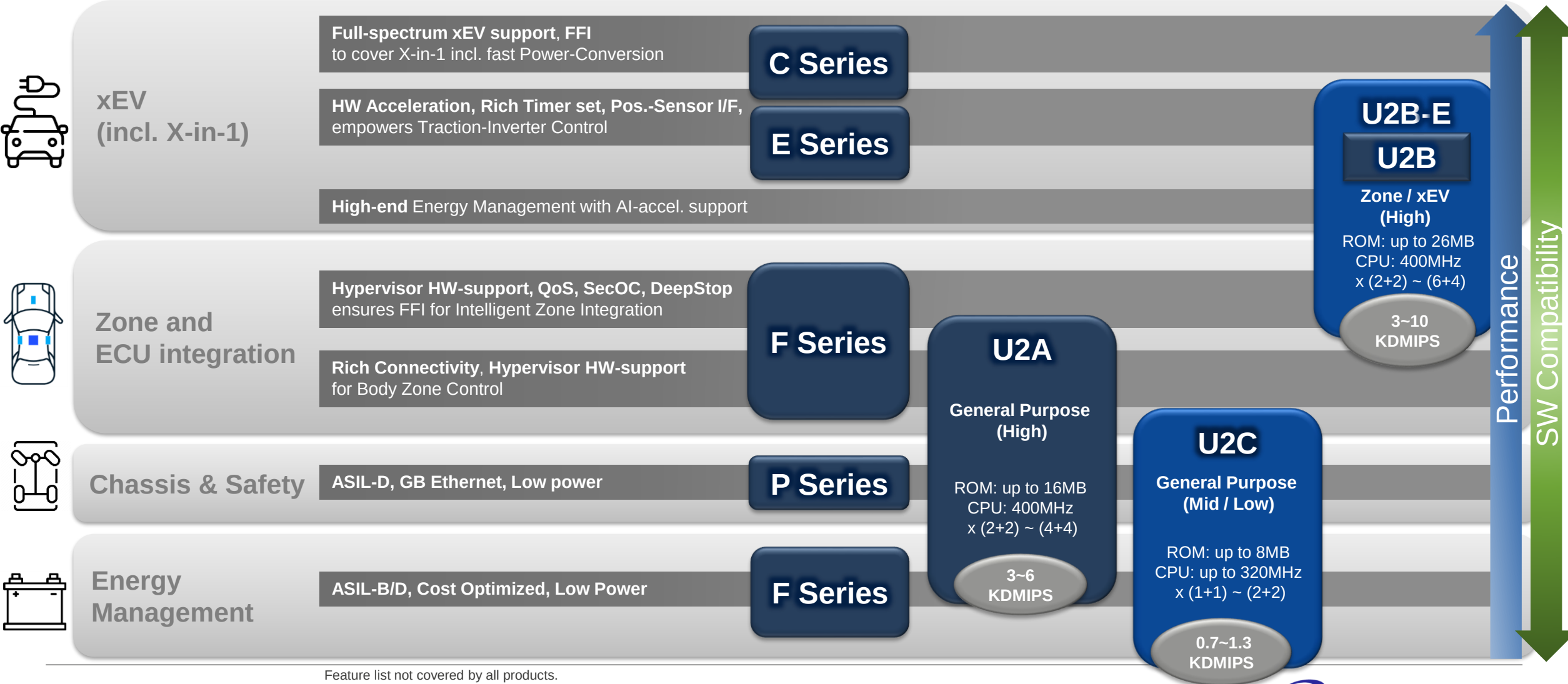
The Most Scalable ASIL-D Platform From 2MB to 26MB



RH850 FAMILY – PRODUCT POSITIONING

Mass Production / Dev chip

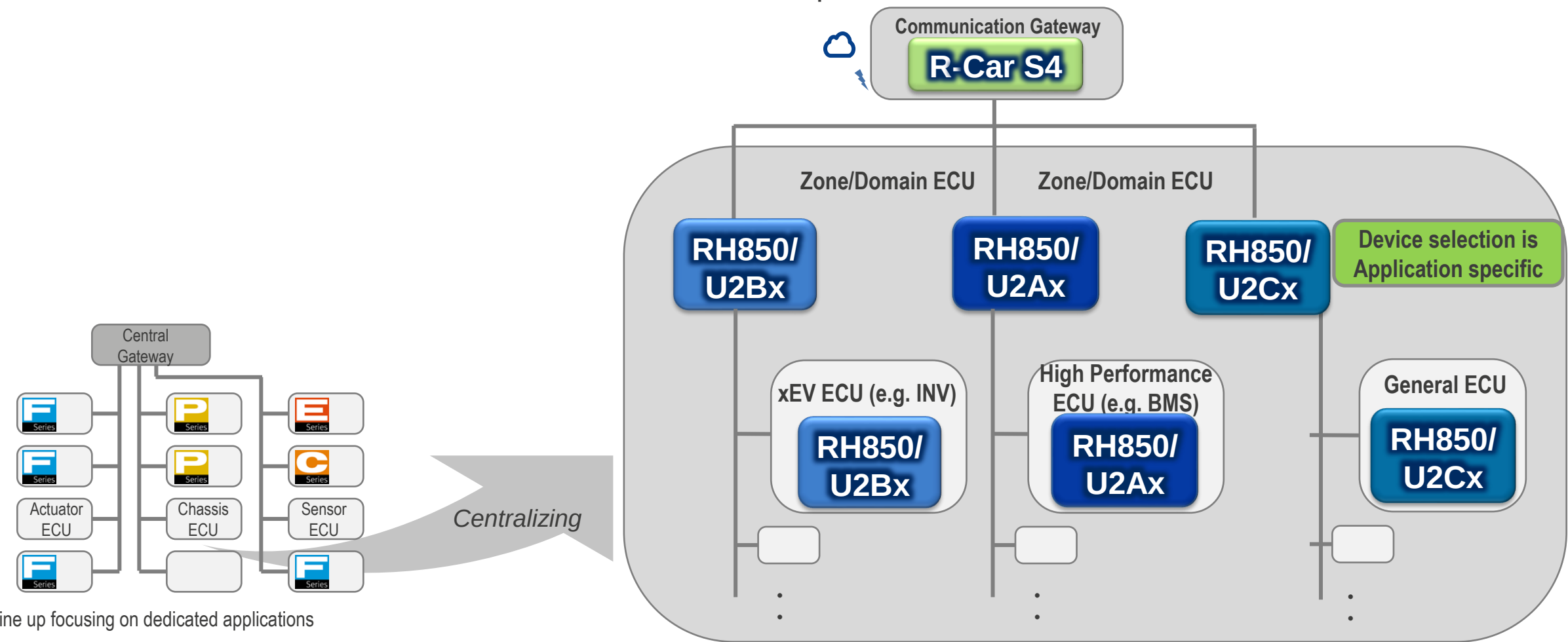
Sampling



Feature list not covered by all products.

RH850/U2X SERIES COVERS E/E ARCHITECTURE CHANGES

The RH850/U2x series covers Zone / DCU where MCU requirements are diversified



Line up focusing on dedicated applications

RH850/U2x KEY FEATURES

Inherits the strengths of legacy products

- ✓ Easy-to-use peripherals (Timer, ADC)
- ✓ Legacy communication interfaces (CAN / CAN-FD, I2C, LIN, etc.)

Provides multifaceted scalability

- ✓ ASIL-D
- ✓ Various pin packages*
 - BGA 156 / 292 / 373 / 468 / 516
 - QFP 100 / 144 / 176
- ✓ (1+1) up to (6+4) CPU cores @400MHz
- ✓ Multi Core with rich set of peripherals

Succession

Scalability

RH850/U2x

Evolution

Robustness

Support for latest technologies

- ✓ Gigabit Ethernet TSN*
- ✓ 10Mbps Interfaces*:
CAN-XL, Ethernet 10Base T1S
- ✓ HW Accelerators*

Provides Safety and Security

- ✓ Safety mechanisms for ISO26262
- ✓ ISO21434 support*
- ✓ Evita-Full + China crypto by HSM*

RENESAS MATCHING ALL REQUIREMENTS FOR MCU

28NM STATE-OF-THE-ART MCU SOLUTIONS



- State-of-the-art architecture for upcoming Zone Architecture to fulfill Safety, Security, A.I. & xEV requirements in Automotive ✓
- Huge market success at various OEMs with RH850/U2x ✓
- Full auto qualified 28nm flash, for 2 years in mass production ✓
- Support platform strategy to speed up for system integration ✓

- Pin group compatibility to replace other vendors U2B-E + other vendors able to share one PCB design ✓
- Differentiation on system level (e.g. Traction inverter IP) ✓
- Highest quality requirements based on 40nm with 0.18ppm* ✓
- Most reliable semiconductor partner for supply chain (proven by OEMs & Tier1s) ✓

#1 in Global
MCU
Solutions

RH850 40nm MCU

RH850 28nm MCU



Naka factory



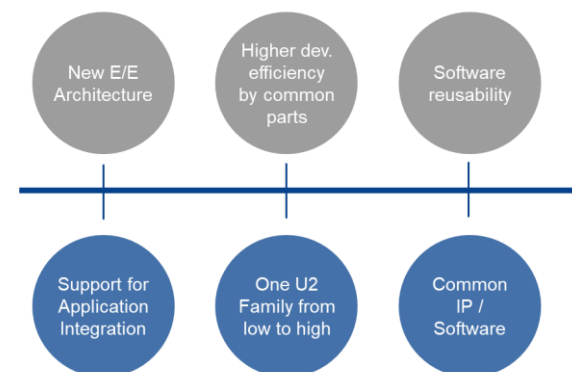
TSMC Taiwan



JASM (E/2024)

- ✓ Secure and Safe dual supply
- ✓ Strategic partnership with TSMC

Market requirement



Renesas' platform strategy

* Data as of 31-MAR-2023

PERFORMANCE

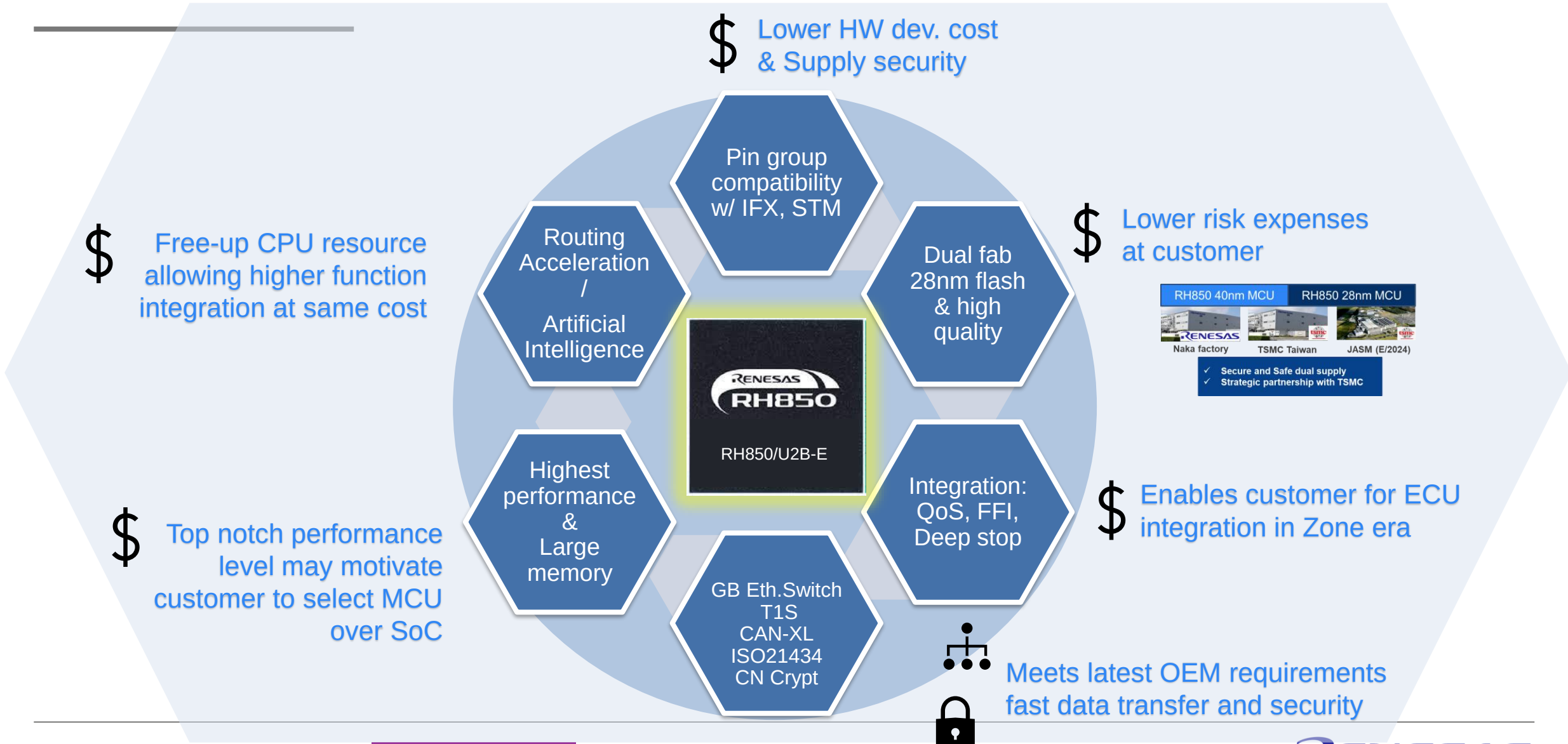
PERFORMANCE BENCHMARKS RH850/U2x

- Standard Benchmarks

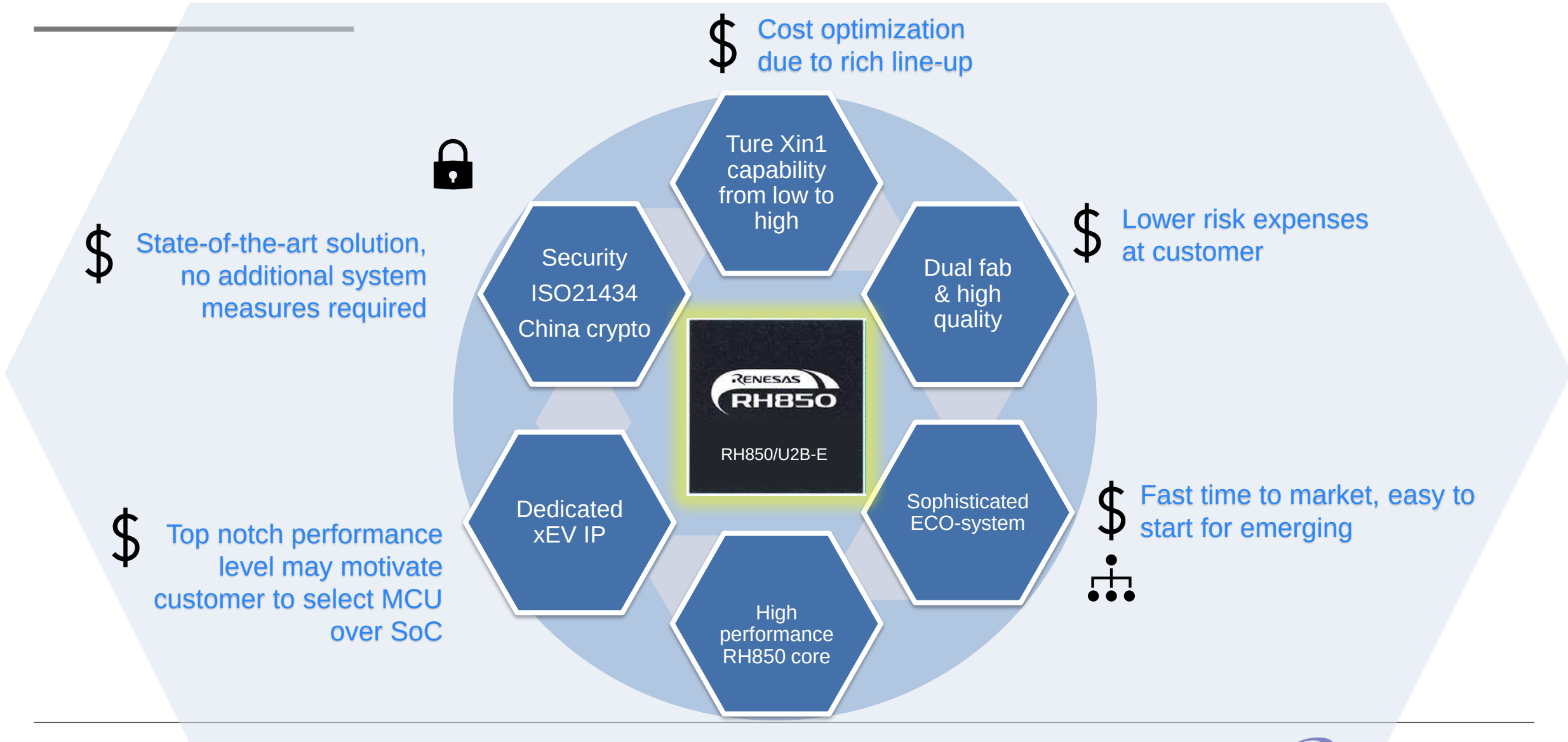
Device	Core	FPU	Operating Speed	Testbench	Compiler/ Revision	Result (absolute)	Result (relative)
RH850/E2x-FCC2 (measured, EVA-board)	G4MH	Yes	400 MHz	CoreMark 1.0	GHS ccv850 v7.1.4 / 2017.1.3	1.668k CM / core	4.17 CM / MHz
				Dhrystone 2.1		1.696k DMIPS / core	4.24 DMIPS / MHz
				AutoMark		564 AMK / core	1.41 AMK / MHz

Note: U2A/U2B/U2B-E implement G4MH core.

VALUE PROPOSITION FOR HIGH-END ZONE



VALUE PROPOSITION FOR XEV



RH850/U2B KEY FEATURES

Inherits the strengths of legacy products

- ✓ Easy-to-use peripherals (Timer, ADC)
- ✓ Legacy communication interfaces (CAN / CAN-FD, I2C, LIN, etc.)

Provides multifaceted scalability

- ✓ ASIL-D
- ✓ 292 / 373 / 468 / 516 pin packages
- ✓ (2+2) up to (6+4) CPU cores @400MHz
- ✓ Multi Core with rich set of peripherals

Succession

Scalability

RH850/U2B

Evolution

Robustness

Support for latest technologies

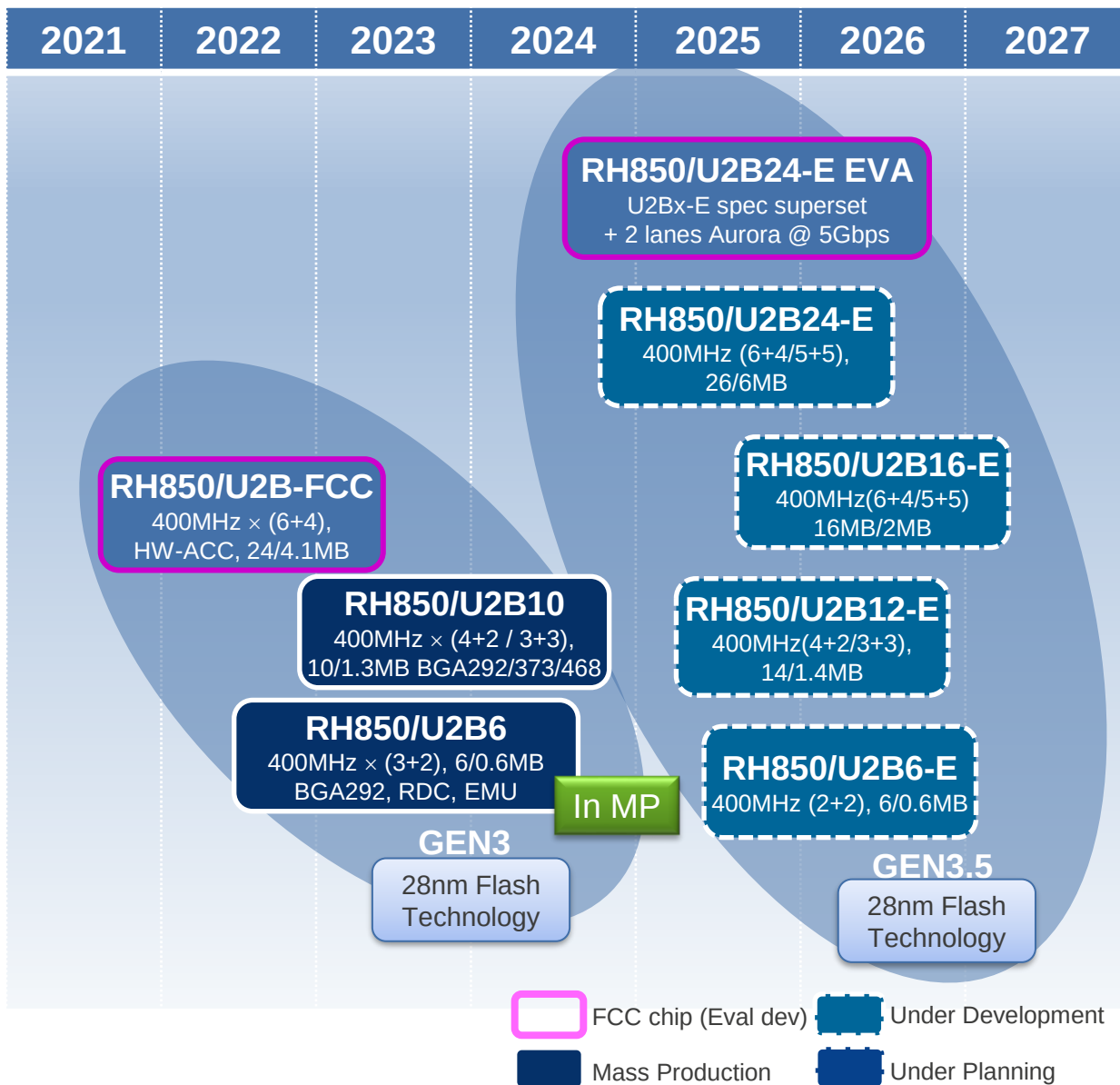
- ✓ Gigabit Ethernet TSN
- ✓ 10Mbps Interfaces:
CAN-XL, Ethernet 10Base T1S
- ✓ HW Accelerators

Provides Safety and Security

- ✓ Safety mechanisms for ISO26262
- ✓ ISO21434 support
- ✓ Evita-Full + China crypto by HSM

RH850/U2B/U2B-E MCU ROADMAP

Preliminary



Key Values **U2Bx-E(Extended)**

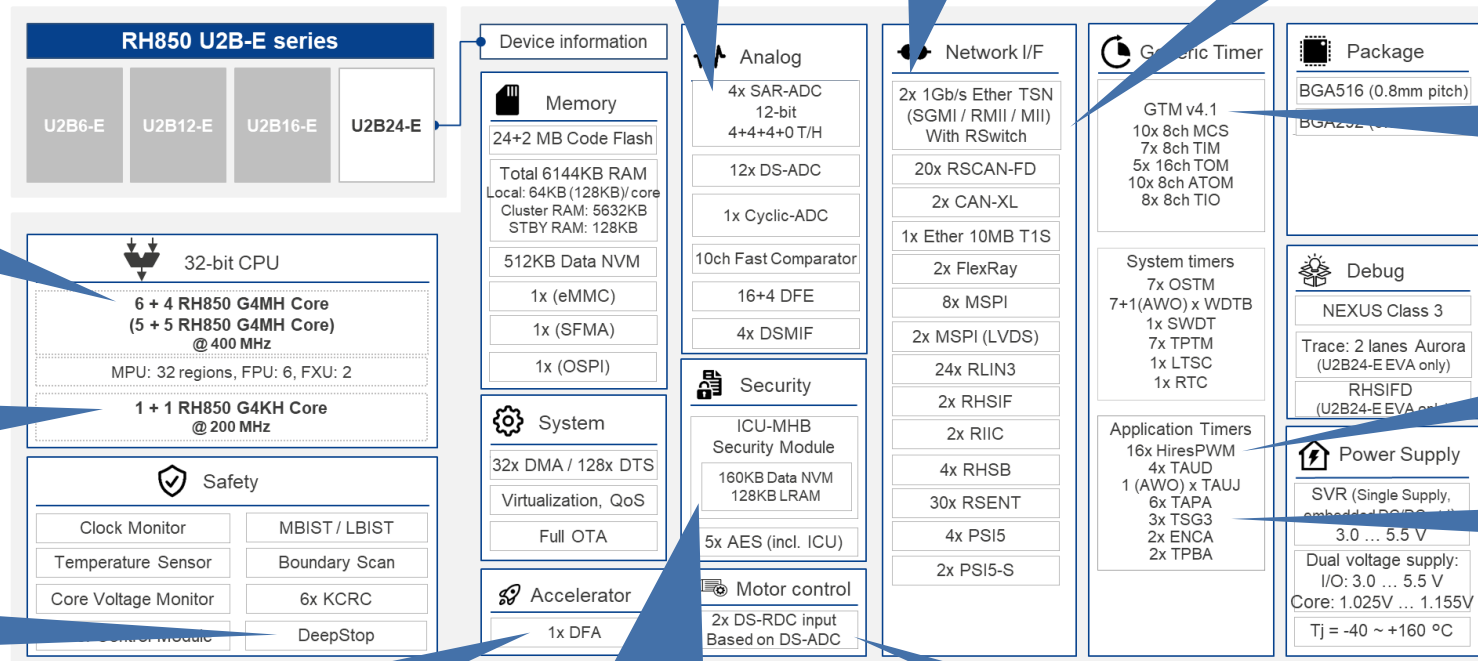
- **More Efficient and Precise xEV Control**
 - Enhanced Timers and Analog
 - Dedicated SIMD Accelerator for AI/ML
- **Enabling Future E/E Architecture**
 - Dedicated Core for Routing Acceleration (CAN-2-X)
 - Ethernet Switch (L2/L3) to reduce BOM and improve latency
 - Future Proof Comms (10B-T1S & CAN-XL)
- **Lower consumption to extend driving range**
 - Deep Stop Support
- **Enhanced Security & SecOC**
 - Dedicated AES modules to reduce CAN/ETH authenticating latency
 - Chinese algos and ISO21434 support
 - Post-quantum cryptography (PQC) covered
- **Early development**
 - Additional U2B24-E EVA support Trace & Calibration

Fully pin and SW compatible

RH850/U2B24-E

KEY FEATURES

RH850 U2B24-E / U2B24-EVA OVERVIEW



Abundant ADCs with T/H for multiple parallel measurements

L2/L3 Switch with 2x ETH 1Gb MAC

Future Proof COMMS (CAN-XL & 10B-T1S)

Flexible Cores Configs to address different SW/Safety req.

Companion CPU for COMMS offloading

DeepStop for ultra low current consumption

GTM Timers with 10x Programmable Cores & advanced Motor Control Features

High Resolution PWM

Renesas Advanced Motor Control Timers

AI/ML/NN Accelerator (SIMD Acceleration)

Fast CAN/10BT1s authentication (SecOC Support) Fast secure boot

3rd Generation RDC Resolver & Inductive Position Sensors I/F

U2Bx-E SERIES – KEY FEATURES FOR CUSTOMER VALUE



Performance → Highest level at Low Power Consumption

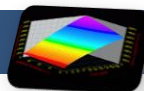


- Multi-core: Up to 6 CPUs @ 400MHz
- HW accelerators: Routing Accelerator, System Manager, AI / ML

Customer value

- Unified CPU layout reduces development effort for system optimization
- Freed-up main CPU resources allow higher function integration at same cost

Scalability & Flexibility



- Line-up with 6~26MB code flash, multi core (2+2) ~ (6+4)
- IP consolidation among all 28nm RH850 Series
- BGA292/516 pin group compatibility with other market solutions

Customer value

- Scalability allows selection of the best device → Lower cost overhead
- High level of SW reusability → Reduced development effort for standard SW
- Shared PCB for several MCU vendors → Greatly reduced PCB cost

Technologies



- Over-The-Air update → *No-wait* / address independent OTA in field
- HW-based virtualization & Quality of Service for Integration Platform
- Additional CPU for Routing Engine + Gbit Ethernet + CAN-XL + Eth T1S
- Flexible eMotor position sensing (RDC) with DS-ADC
- *FMONOS* → 28nm, 100MHz fast flash technology, quality

Customer value

- No need for external storage devices → Reduced cost and security risk
- Freedom-From-Interference guaranteed by HW reduces validation efforts
- Flexibility by SW solutions & high performance due to dedicated CPU / IP
- BOM cost reduction by reduced chipset for Traction Inverter applications
- High speed flash needs less cache → Lower power consumption

Safety & Security → *ASIL D* and *Evita-Full*



- ICU for *EVITA-Full* including Elliptic Curve Cryptography, AES
- Up to 4x Cryptographic Engine Units (ACEU, lockstep)
- Support ISO21434 + China Crypto SM3/4, PQC covered

Customer value

- Flexible Security concept → Lower risk by enabling future adaptations
- Secure communication & Secure boot covered by high performance HW
- Comply to latest security standard and global market requirement

System Development Environment → Model-based design

- Virtual Platform for early SW development and performance analysis
- Collaborating with all relevant 3rd parties in Zone area



Customer value

- Development schedule improvement by pre-silicon SW development start
- Huge variety of ecosystem solutions to match individual cost targets

RH850/U2B SERIES LINE-UP

PACKAGE VS MEMORY

- U2Bx-E provides pin group compatibility with selected competitive devices in the same application area.

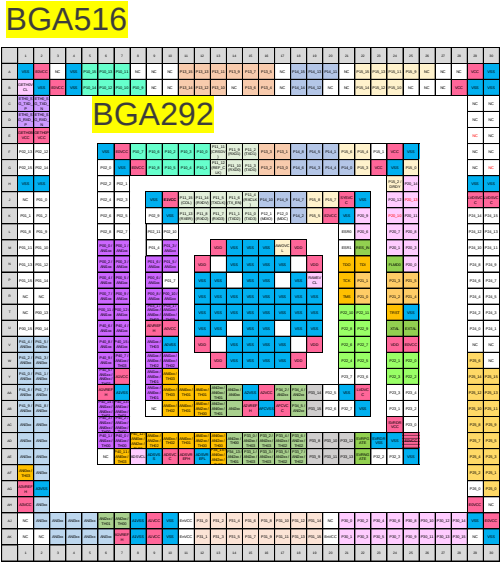


BGA516 25x25mm 0.8mm pitch							
BGA468 25x25mm 0.8mm pitch	 FC-BGA						
BGA373 21x21mm 0.8mm pitch	 FC-BGA						
BGA292 17x17mm 0.8mm pitch	 FC-BGA						
PKG Product	U2B-FCC	U2B6 400MHz x 3+2 ROM: 6 MB	U2B10 400MHz x 4+2 (3+3) ROM: 10 MB	U2B6-E 400MHz x 2+2 ROM: 6 MB	U2B12-E 400MHz x 4+2 (3+3) ROM: 14 MB	U2B16-E 400MHz x 6+4 (5+5) ROM: 16 MB	U2B24-E/EVA 400MHz x 6+4 (5+5) ROM: 26 MB
	U2B Series			U2B-E Series			

RH850/U2B-E SERIES LINE-UP

PACKAGE VS MEMORY

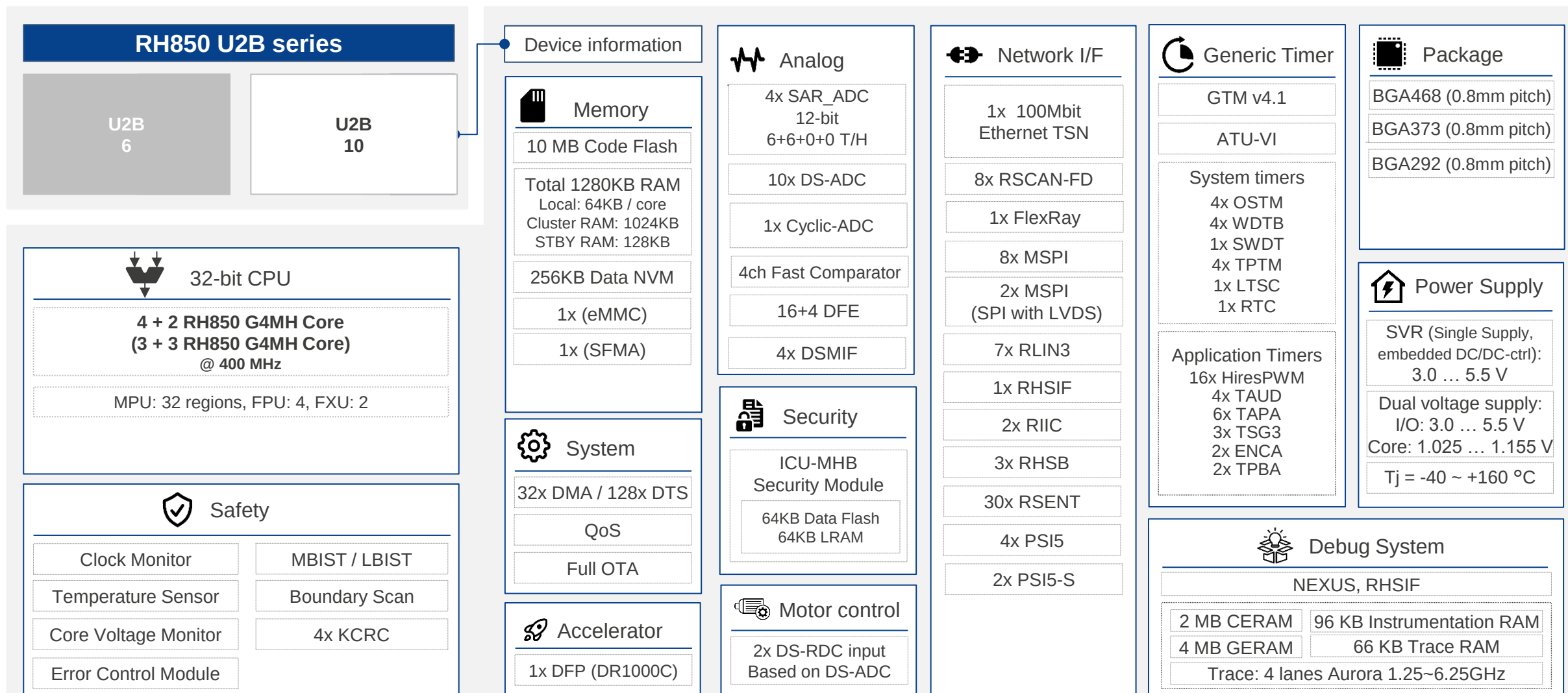
BGA516 25x25mm 0.8mm pitch					
BGA292 17x17mm 0.8mm pitch					
PKG	U2B6-E 400MHz x 2+2 ROM: 6 MB	U2B12-E 400MHz x 4+2 (3+3) ROM: 14 MB	U2B16-E 400MHz x 6+4 (5+5) ROM: 16 MB	U2B24-E 400MHz x 6+4 (5+5) ROM: 26 MB	U2B24-E EVA 400MHz x 6+4 (5+5) ROM: 26 MB
Product	U2B-E Series				



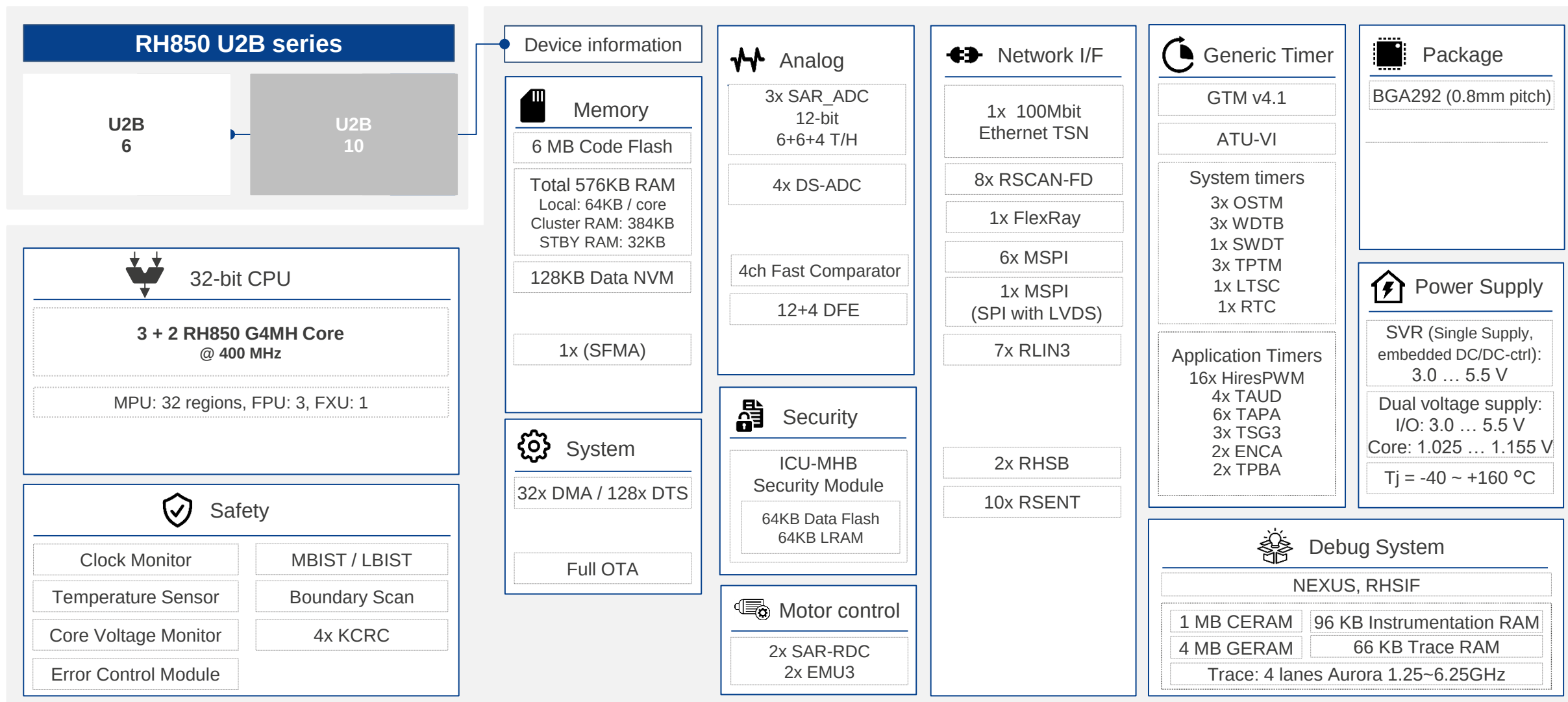
U2B16-E are fully pin, pinmux and SW compatible with all U2B-E devices → one line-up

GEN3: U2BX BLOCK DIAGRAMS

RH850 U2B10 / U2B10-FCC OVERVIEW

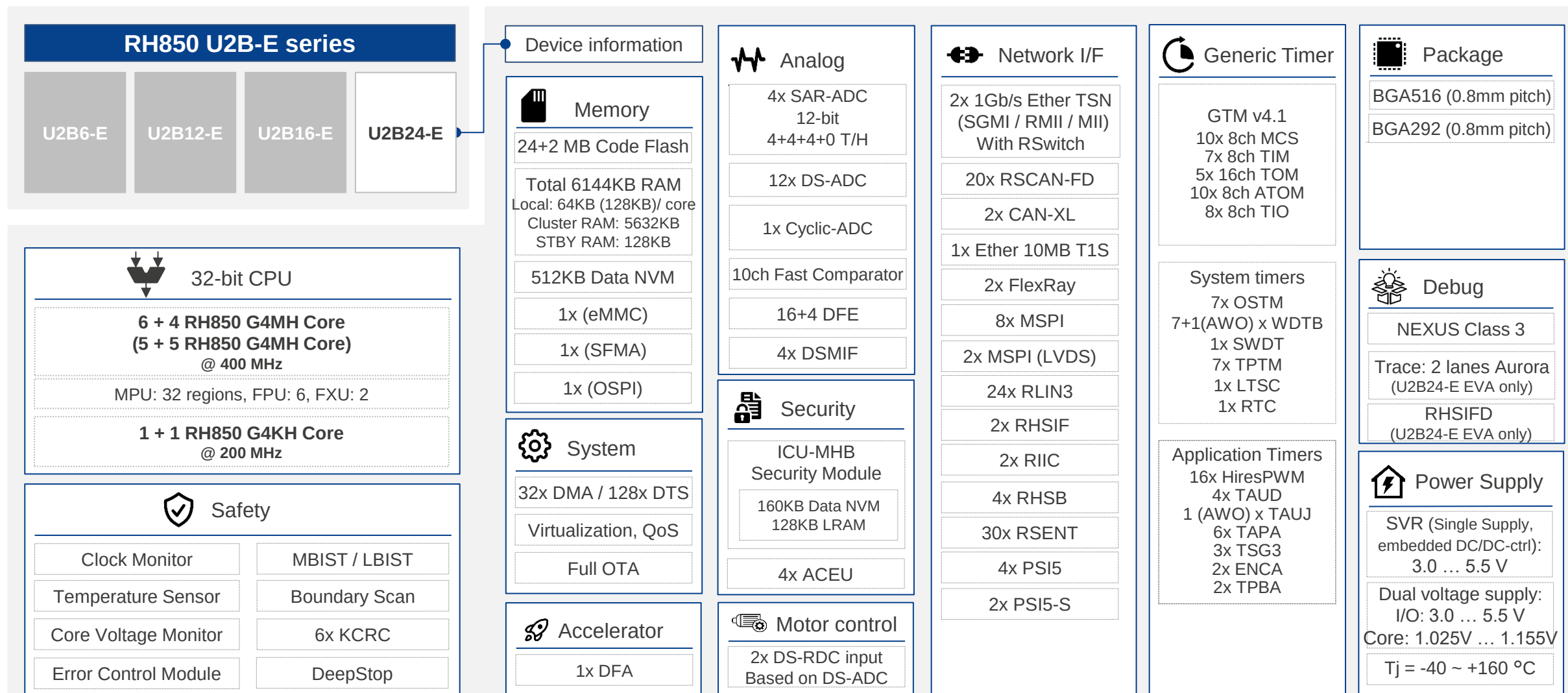


RH850 U2B6 / U2B6-FCC OVERVIEW

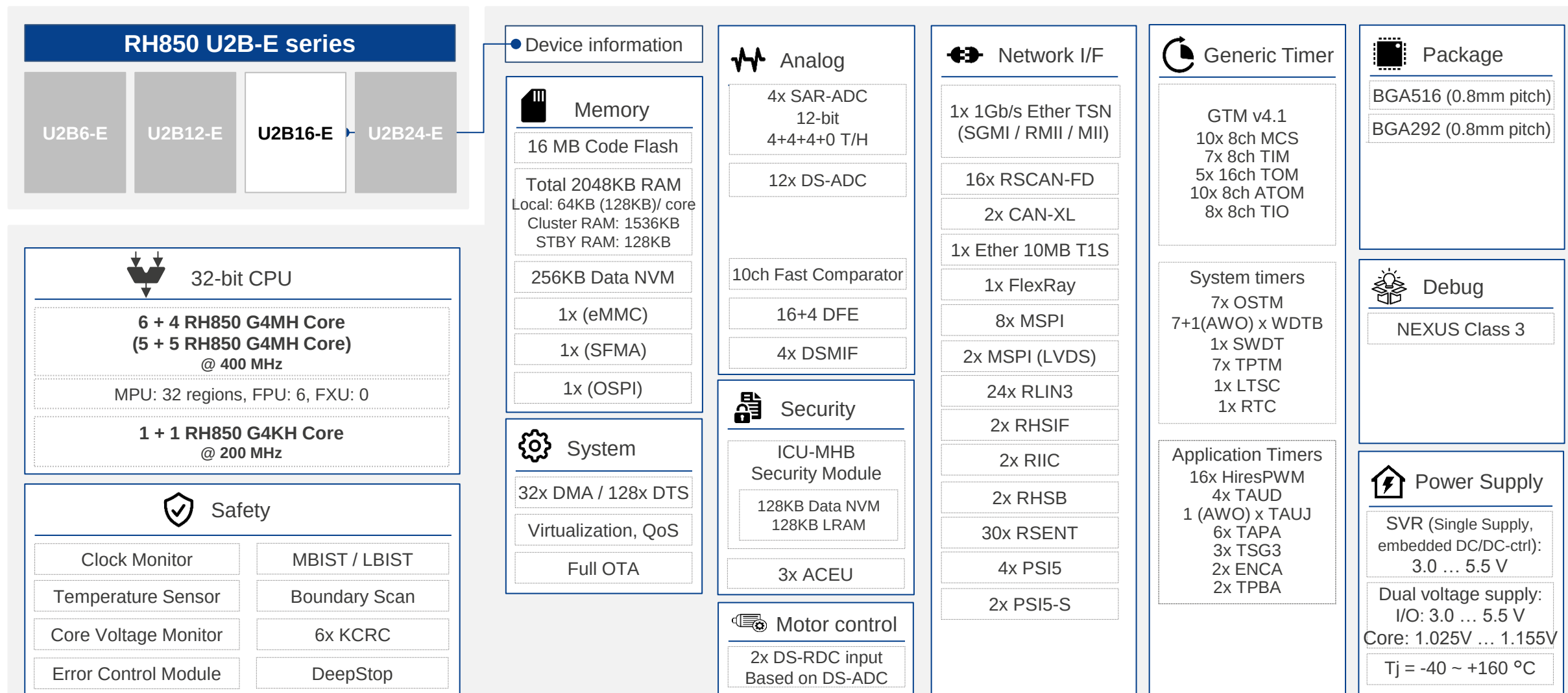


GEN3.5: U2BX-E BLOCK DIAGRAMS

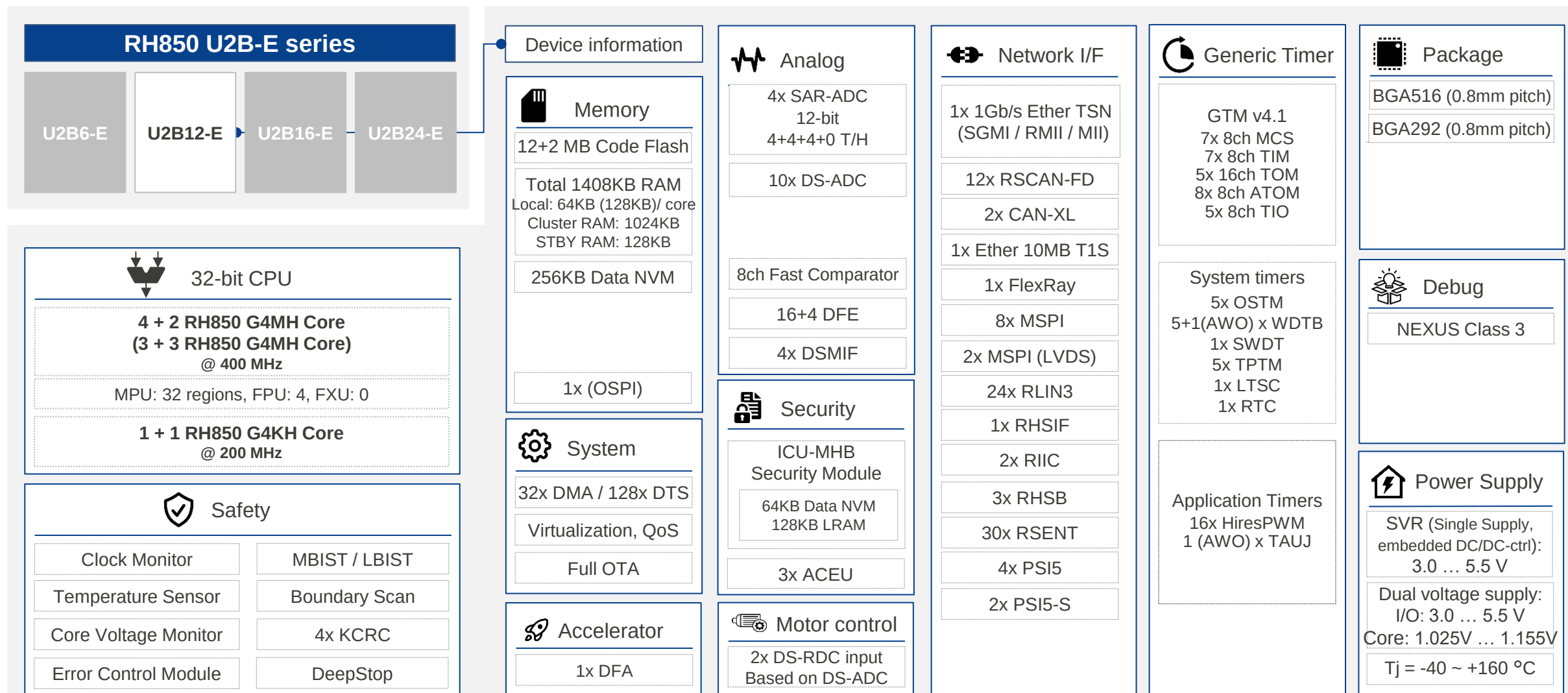
RH850 U2B24-E / U2B24-EVA OVERVIEW



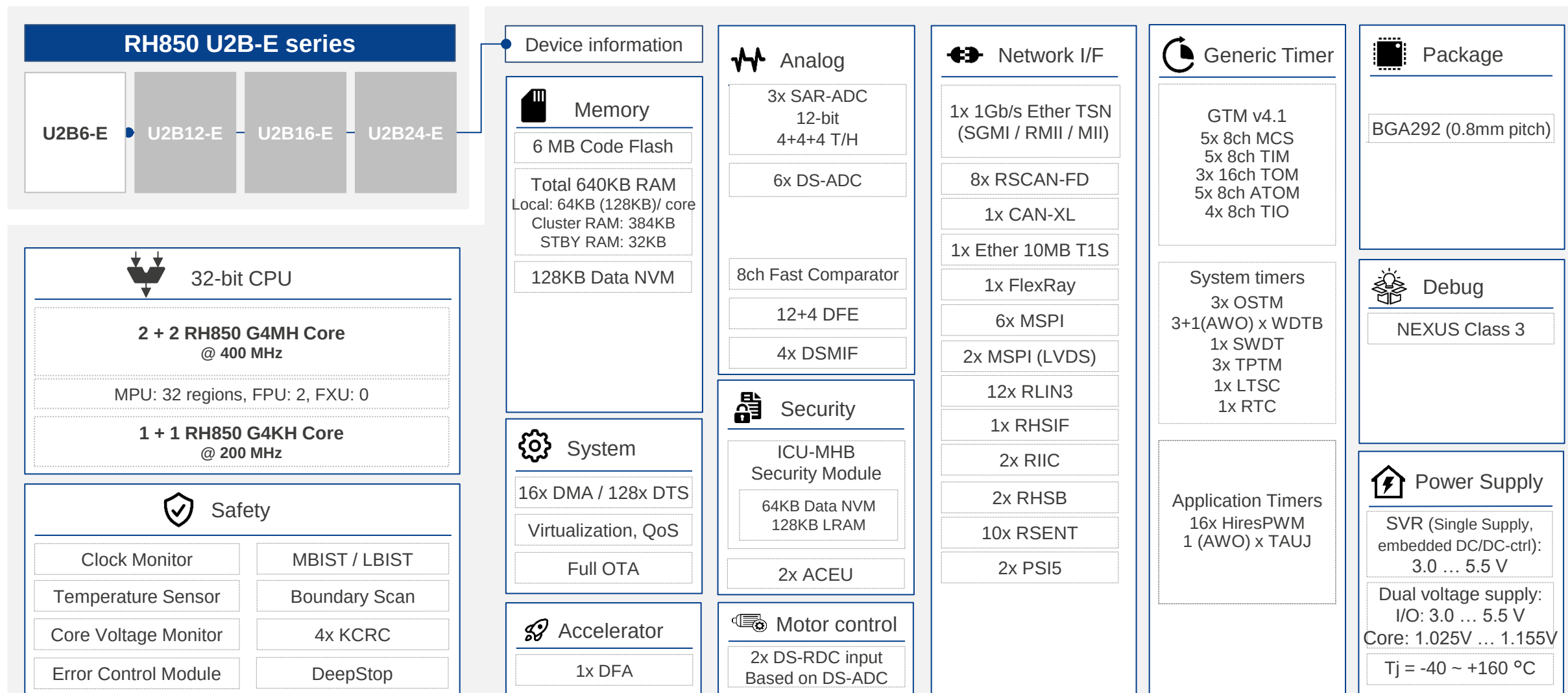
RH850 U2B16-E OVERVIEW



RH850 U2B12-E OVERVIEW



RH850 U2B6-E OVERVIEW



Device specification overview (U2Bx/U2Bx-E) [1/6]

			U2B/U2B-E Series (mass-produced products)									
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)		U2B12-E (Gen3.5)		U2B16-E (Gen3.5)		U2B24-E (Gen3.5)	
Processor	CPU	Clock freq.	400 MHz	400 MHz	400 MHz		400 MHz		400 MHz		400 MHz	
		Main core	2	3	4	3	4	3	6	5	6	5
		LSDC	2	2	2	3	2	3	4	5	4	5
		FPU	2	3	4		4		6		6	
		FXU (FP-SIMD)	0	1	2		0		0		2	
		MPU Region	32	32	32		32		32		32	
	Cache	Instruction	16 KB (4 Way)	16 KB (4 Way)	16 KB (4 Way)		16 KB (4 Way)		16 KB (4 Way)		16 KB (4 Way)	
		data	4 line (256 bit/line)	4 line (256 bit/line)	4 line (256 bit/line)		4 line (256 bit/line)		4 line (256 bit/line)		4 line (256 bit/line)	
	Data Flow Processor (*1)		-	-	DR1000C		-		-		-	
	Data Flow Architecture (*2)		yes	-	-		yes		-		yes	
System	Virtualization support function		yes	yes	yes		yes		yes		yes	
	QoS (Quality of service)		yes	-	yes		yes		yes		yes	
Memory	Flash	Code Flash	6 MB	6 MB	10 MB		12 MB + 2MB		16 MB		24 MB + 2MB	
		Data Flash (Except for Security)	128 KB	128 KB	256 KB		256KB		256KB		512KB	
	RAM	Total	512 KB + 128KB (G4KH LRAM)	576 KB	1280 KB		1280 KB + 128KB (G4KH LRAM)		1920 KB + 128KB (G4KH LRAM)		6016 KB + 128KB (G4KH LRAM)	
		Cluster RAM (incl. Standby RAM)	384 KB (32 KB)	384 KB (32 KB)	1024 KB (128 KB)		1024 KB (128 KB)		1536KB (128 KB)		5632KB (128 KB)	
		Local RAM	64 KB x 2	64 KB x 3	64 KB x 4		64 KB x 4		64 KB x 6		64 KB x 6	
		Emulation RAM (only FCC device)	-	5 MB (U2B6-FCC)	6 MB (U2B10-FCC)		-		-		4 MB (U2B24-E EVA CRAM)	
DMAC	sDMAC	Channel	16	16	16 x 2		16 x 2		16 x 2		16 x 2	
	DTS (*3)	Channel	128	128	128		128		128		128	

(*1): Produced by NSITEXE, Inc., (*2): Produced by Robert Bosch GmbH, (*3): Hardware Redundancy

Device specification overview (U2Bx/U2Bx-E) [2/6]

			U2B/U2B-E Series (mass-produced products)					
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
Timer	GTM	Version	v4.1	v4.1	v4.1	v4.1	v4.1	v4.1
	ATU		-	ATU-VI	ATU-VI	-	-	-
	ENCA (Encoder Timer)		-	2	2	-	2	2
	HRPWM (High Resolution PWM)	Channel	16	16	16	16	16	16
	TAPA	Channel	-	6	6	-	6	6
	TAUD	Channel	-	4 (*4)	4 (*4)	-	4 (*4)	4 (*4)
	TAUJ	Channel	1 (AWO)	-	-	1 (AWO)	1 (AWO)	1 (AWO)
	TPBA	Channel	-	2	2	-	2	2
	TSG3	Channel	-	3	3	-	3	3
	PWM-Diag	Channel	-	-	-	-	-	-
	CMT (OSTM)	Channel	3	3	4	5	7	7
	LTSC	Module	1	1	1	1	1	1
	RTC	Module	1 (AWO)	-	-	1 (AWO)	1 (AWO)	1 (AWO)
	TPT	Module	3	3	4	5	7	7
	WDT	Channel	3 + 1(AWO)	3	4	5+1(AWO)	7+1 (AWO)	7+1 (AWO)
	SWDT (Secure WDT)	Channel	1	1	1	1	1	1
Motor Control IP	EMU	Channel	-	2	-	-	-	-
	SAR-RDC	Module	-	2	-	-	-	-
	DS-RDC	Module	2 (uses DS-ADC)	-	2 (uses DS-ADC)	2 (uses DS-ADC)	2 (uses DS-ADC)	2 (uses DS-ADC)

(*4): Supports SENT communication diagnosis for Powertrain

Device specification overview (U2Bx/U2Bx-E) [3/6]

			U2B/U2B-E Series (mass-produced products)					
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
ADC	ADC (SAR ADC)	Module	3	3	4	4	4	4
		Analog inputs			BGA468: 96 BGA373: 80 BGA292: 64	BGA516: 91	BGA516: 91	BGA516: 91
		T&H	4+4+4	6+6+4	6+6+0+0	4+4+4	4+4+4+0	4+4+4+0
			BGA292: 62	BGA292: 64		BGA292: 62	BGA292: 62	BGA292: 62
	DSADC ($\Delta\Sigma$ ADC)	Module	6	4	10	10	12	12
		Analog inputs	24	18	BGA468: 42 BGA373: 42 BGA292: 34	BGA516: 42 BGA292: 24	BGA516: 46 BGA292: 24	BGA516: 46 BGA292: 24
		DSM-IF	4	-	4	4	4	4
	CADC (Cyclic ADC)	Module	-	-	1	-	-	1
		Analog inputs	-	-	8	-	-	8
	Fast Comparator	Channel	8	4	4	8	10	10
Digital Filter	DFE	Filter type	6th order IIR or 64-tap FIR (*5)	6th order IIR or 64-tap FIR (*5)	6th order IIR or 64-tap FIR (*5)	6th order IIR or 64-tap FIR (*5)	6th order IIR or 64-tap FIR (*5)	6th order IIR or 64-tap FIR (*5)
		Channel	12+4	12+4	16 + 4	16 + 4	16 + 4	16 + 4

(*5): When using a 64-tap FIR, resources of 2 ch are used.

Device specification overview (U2Bx/U2Bx-E) [4/6]

			U2B/U2B-E Series (mass-produced products)					
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
Internal Chip Communication	SPI	Channel	6 (MSPI)	6 (MSPI)	8 (MSPI)	8 (MSPI)	8 (MSPI)	8 (MSPI)
	SPI with LVDS	Channel	1 (MSPI)	1 (MSPI)	2 (MSPI)	2 (MSPI)	2 (MSPI)	2 (MSPI)
	RHSB	Channel	2	2	3	3	2	4
	RHSIF	Channel	1	-	1	1	2	2
	I2C	Channel	2	-	2	2	2	2
Onboard Communication	SCI3	Channel	-	-	-	-	-	-
	RLIN3	Channel	11+1	6+1	6+1	23+1	23+1	23+1
	RS-CANFD	Channel	8 (ver.4.1)	8 (ver.4.1)	8 (ver.4.1)	12 (ver.4.1)	16 (ver.4.1)	20 (ver.4.1)
		Total msg buffer	768	768	768	1152	1152	1920
	CAN-XL	Channel	1	-	-	2	2	2
	Ethernet	module	1 x Gb/s Ether TSN (SGMII/RMII/MII)	1 x 100Mb/s Ether TSN	1 x 100Mb/s Ether TSN	1 x Gb/s Ether TSN (SGMII/RMII/MII)	1 x Gb/s Ether TSN (SGMII/RMII/MII)	2 x Gb/s Ether TSN (SGMII/RMII/MII)
	Ethernet T1S	module	1 x 10MB T1S	-	-	1 x 10MB T1S	1 x 10MB T1S	1 x 10MB T1S
Sensor I/F	FLXA (FlexRay)	module	1	1	1	1	1	2
	RSENT	Channel	10	10	30	30	30	30
	PSI5	Channel	2	-	4	4	4	4
External Memory I/F	PSI5-S	Channel	-	-	2	2	2	2
	EXBC (External BUS)		-	-	-	-	-	-
	eMMC	Channel	-	-	1	-	1	1
	SFMA	Channel	-	1	1	-	1	1
	O-SPI	Channel	-	-	-	1	1	1

Device specification overview (U2Bx/U2Bx-E) [5/6]

			U2B/U2B-E Series (mass-produced products)					
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
Partition manager	Partition manager core		RH850/G4KH (1+1) @200MHz	-	-	RH850/G4KH (1+1) @200MHz	RH850/G4KH (1+1) @200MHz	RH850/G4KH (1+1) @200MHz
		Local RAM	128KB	-	-	128KB	128KB	128KB
Security	Security module		ICU-MHB (1 x AES LS) + + 2x ACEU	ICU-MHB	ICU-MHB	ICU-MHB (1 x AES LS) + 3 x ACEU	ICU-MHB (1 x AES LS) + + 3x ACEU	ICU-MHB (1 x AES LS) + + 4x ACEU
	Evita Level		Full	Full	Full	Full	Full	Full
	Data Flash for Security		64 KB	64 KB	64 KB	64 KB	128 KB	160 KB
	Local RAM for Security (*6)		128 KB	64 KB	64 KB	128 KB	128 KB	128 KB
Functional Safety	ASIL		D	D	D	D	D	D
	CRC	module	4 (KCRC)	4 (KCRC)	4 (KCRC)	4 (KCRC)	6 (KCRC)	6 (KCRC)
	Error Control Module		Yes	Yes	Yes	Yes	Yes	Yes
	LBIST/MBIST		Yes	Yes	Yes	Yes	Yes	Yes
	Clock Monitor		Yes	Yes	Yes	Yes	Yes	Yes
	Core Voltage Monitor		Yes	Yes	Yes	Yes	Yes	Yes
	Temperature Sensor		Yes	Yes	Yes	Yes	Yes	Yes
Power Mgmt	LPS		Yes	-	-	Yes	Yes	Yes
	STBC		Yes	Yes	Yes	Yes	Yes	Yes
	Stop mode		Yes	Yes	Yes	Yes	Yes	Yes
	Deep stop mode		Yes	-	-	Yes	Yes	Yes
	Cyclic run / cyclic stop mode		Yes	-	-	Yes	Yes	Yes
	Power off standby mode		Yes	Yes	Yes	Yes	Yes	Yes

(*6) : Secure RAM can be extended by using CRAM. This CRAM portion can be configured for the same protection level as the exclusive Secure RAM.

Device specification overview (U2Bx/U2Bx-E) [6/6]

		U2B/U2B-E Series (mass-produced products)					
		U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
PLL	Main OSC	1	1	1	1	1	1
	Internal Ring OSC	3	3	3	3	3	3
	PLL	3	2	3	3	3	3
Power Supply		4.5-5.5V	4.5-5.5V	4.5-5.5V	4.5-5.5V	4.5-5.5V	4.5-5.5V
		3.0-3.6V	3.0-3.6V	3.0-3.6V	3.0-3.6V	3.0-3.6V	3.0-3.6V
		1.025-1.155V	1.025-1.155V	1.025-1.155V	1.025-1.155V	1.025-1.155V	1.025-1.155V
	SVR	Yes	Yes	Yes	Yes	Yes	Yes
Trace: Aurora I/F	1 lanes @ 2.5Gbps ea.	-	U2B-FCC supported	U2B-FCC supported	-	-	-
	2 lanes @ 5Gbps ea.	-	-	-	-	-	U2B24-E EVA supported
	4 lanes @ 6.25Gbps ea.	-	U2B-EVA supported	U2B-EVA supported	-	-	-
Operating Temperature		Tjmax=160°C	Tjmax=160°C	Tjmax=160°C	Tjmax=160°C	Tjmax=160°C	Tjmax=160°C
PKG	FCBGA516-25x25-0.8	-	-	-	-	-	-
	FCBGA468-25x25-0.8	-	-	-	-	-	-
	FCBGA373-21x21-0.8	-	-	-	-	-	-
	BGA516-25x25-0.8	-	-	-	Yes (Combi layout) (*7)	Yes (Combi layout) (*7)	Yes (Combi layout) (*7)
	BGA468-25x25-0.8	-	-	Yes	-	-	-
	BGA373-21x21-0.8	-	-	Yes	-	-	-
	BGA292-17x17-0.8	Yes (Combi layout) (*7)	Yes	Yes	Yes (Combi layout) (*7)	Yes (Combi layout) (*7)	Yes (Combi layout) (*7)

(*7): Pin group compatibility with selected competitor devices in the same application area.

APPENDIX:

ATU-VI AND GTM V4.1 SPECIFICATIONS

Device specification Timer (ATU)

			U2B Series (mass-produced products)	
			U2B6	U2B10
ATU-VI	Timer A	Channel	6	8
	Timer B	Channel	1	1
	Timer C + (internal timer)	Channel	16 + (4)	36 + (8)
	Timer D		16	32
	Min Guard / Multi shot	Channel	8	20
	Not supported above	Channel	8	12
	Timer E	Channel	20	36
	Timer F	Channel	7	16
	Timer G + (internal timer)	Channel	8 + (2)	10 + (4)

Device specification Timer (GTM)

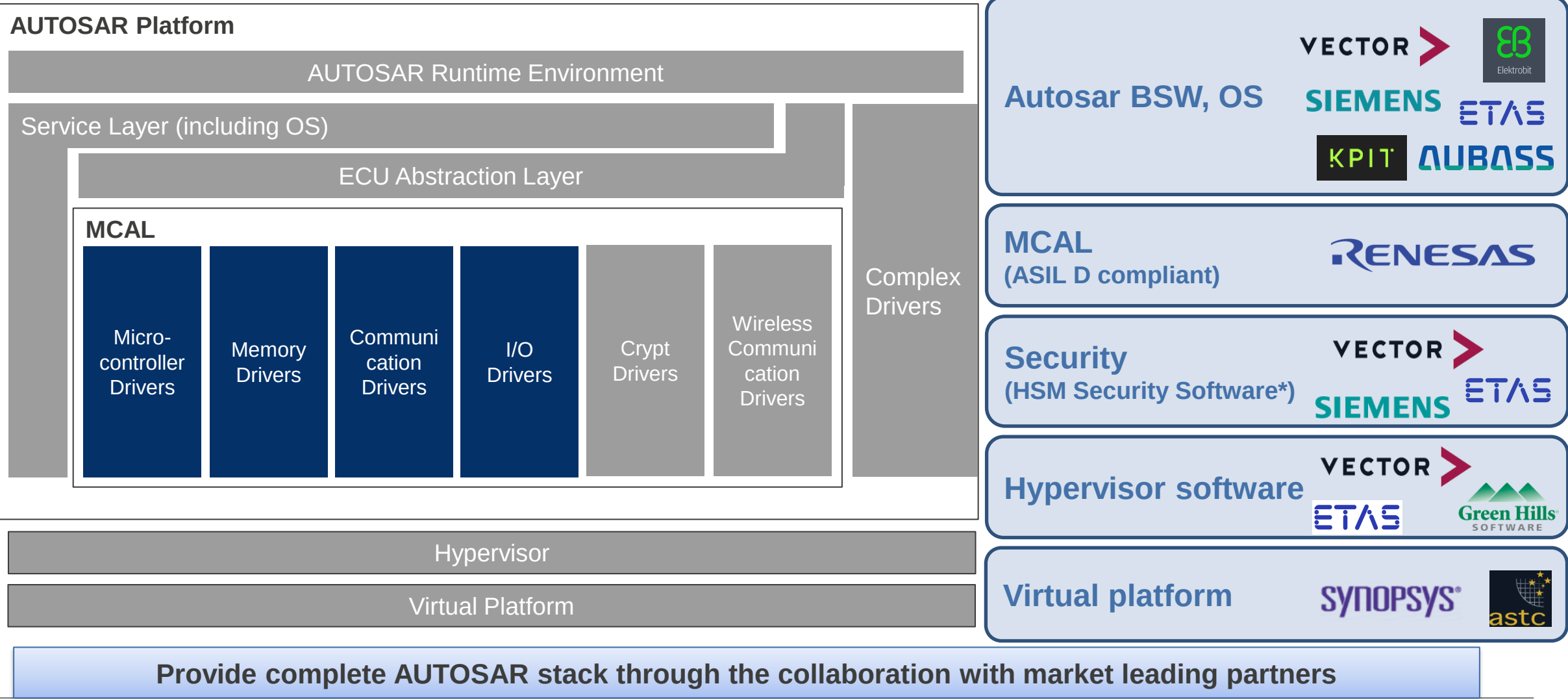
			U2B Series (mass-produced products)					
			U2B6-E (Gen3.5)	U2B6 (Gen3)	U2B10 (Gen3)	U2B12-E (Gen3.5)	U2B16-E (Gen3.5)	U2B24-E (Gen3.5)
GTM v4.1	ARU	Unit	1	1	1	1	1	1
	BRC	Unit	1	1	1	1	1	1
	FIFO/PSM	Unit	2	3	3	2	3	3
	CMU	Unit	1	1	1	1	1	1
	CCM(200MHz / 100MHz)	Unit	5(5/0)	4(3/1)	8(3/5)	8(8/0)	10(10/0)	10(10/0)
	TBU	Unit x Channel	1 x 4ch	1 x 4ch	1 x 4ch	1 x 4ch	1 x 4ch	1 x 4ch
	TIM	Unit x Channel	5 x 8ch	3 x 8ch	7 x 8ch	7 x 8ch	7 x 8ch	7 x 8ch
	TOM	Unit x Channel	3 x 16ch	2 x 16ch	4 x 16ch	5 x 16ch	5 x 16ch	5 x 16ch
	ATOM	Unit x Channel HiRes Channels	5 x 8ch 2 x 4ch	4 x 8ch 2 x 4ch	8 x 8ch 2 x 4ch	8 x 8ch 2 x 4ch	10 x 8ch 2 x 4ch	10 x 8ch 2 x 4ch
	TIOp8	Unit x Channel	4 x 8ch	2 x 8ch	4 x 8ch	5 x 8ch	8 x 8ch	8 x 8ch
	DTM	Unit	18 x 4ch	12 x 4ch	20 x 4ch	20 x 4ch	26 x 4ch	26 x 4ch
	MCS	Unit x Channel	5 x 8ch	4 x 8ch	7 x 8ch	7 x 8ch	10 x 8ch	10 x 8ch
	MCFG	Unit	-	-	-	-	-	-
	MAP	Unit	1	1	1	1	1	1
	DPLL	Unit	1	1	1	1	1	1
	SPE	Unit	2	2	3	3	5	5
	ICM	Unit	1	1	1	1	1	1
	CMP	Unit	1	1	1	1	1	1
	MON	Unit	1	1	1	1	1	1
	MCS RAM		5 x 24KB	2 x 12KB + 2 x 6KB	2 x 12 KB + 5 x 6 KB	5 x 24 KB + 2 x 12 KB	5 x 24 KB + 5 x 12 KB	5 x 24 KB + 5 x 12 KB
	DPLL RAM		1.5KB + 6.0KB	1.5KB + 3.0KB	1.5 KB + 6.0 KB	1.5 KB + 12.0 KB	1.5 KB + 12.0 KB	1.5 KB + 12.0 KB
	FIFO / PSM RAM		2 x 3.625 KB	3 x 3.625 KB	3 x 3.625 KB	2 x 3.625 KB	3 x 3.625 KB	3 x 3.625 KB
	Bus Master (MCS/slots) / Slave		AXIM (5/4) / AEI	AXIM (4/4) / AEI	AXIM (7/4) / AEI	AXIM (7/4) / AEI	AXIM (10/4) / AEI	AXIM (10/4) / AEI

SOFTWARE

```
function use_array(a) {  
  for (var c = 0; c < a.length; c++) {  
    use_array(a[c], a);  
  }  
  return a;  
}  
  
function new_user(a) {  
  for (var b = "", c = 0; c < a.length; c++) {  
    b += " " + a[c] + " ";  
  }  
  return b;  
}  
  
document.getElementById("DOMAttrModified textInput input change key").  
setAttribute("INTOUE: " + a.unique);
```

RH850 SW COMPONENTS AND PARTNERS

* Low Level Security Driver (sample code) provided by Renesas



RH850/U2X IP SW COMPATIBILITY

MAJOR PERIPHERAL IP'S

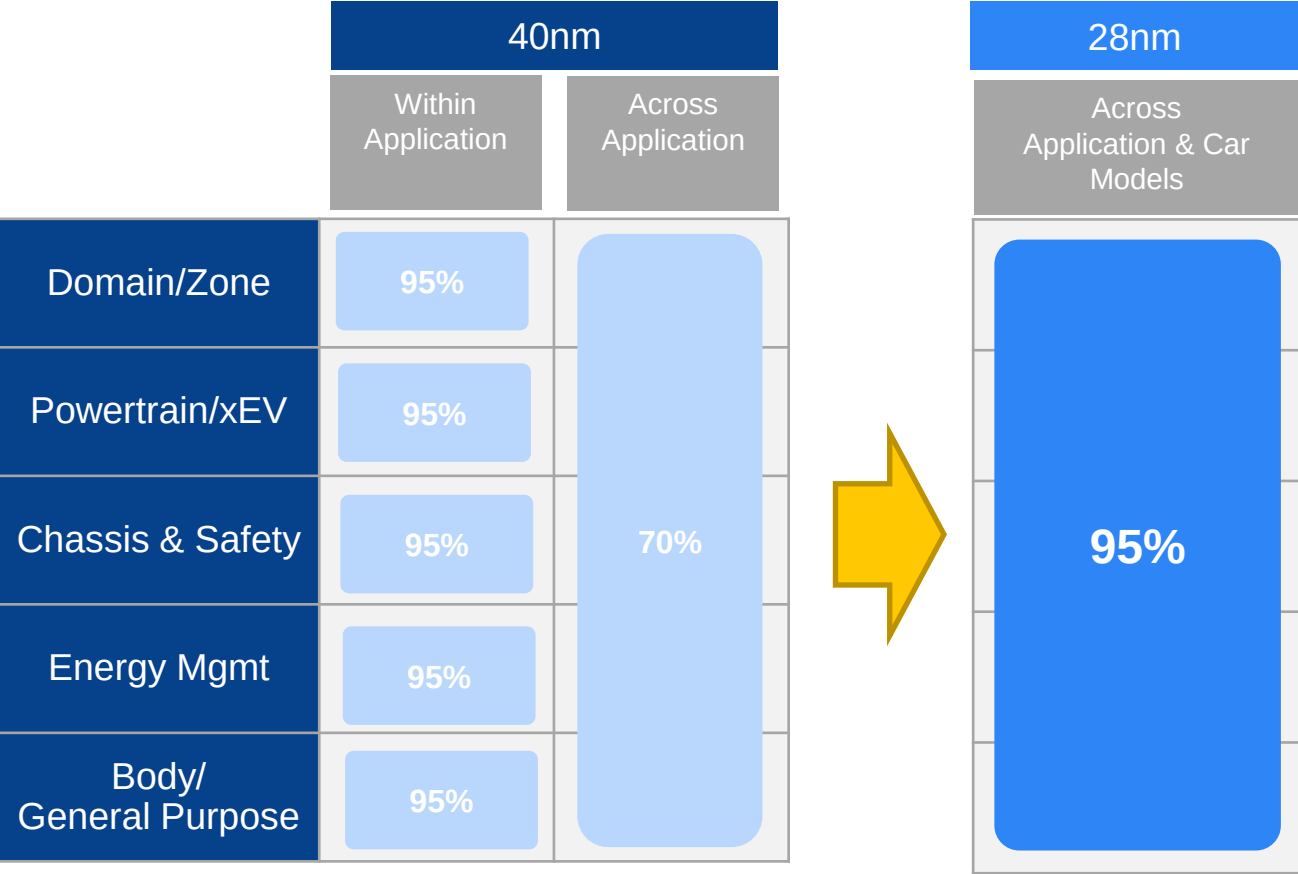
IP SW Compatibility due
to a common Design Platform
+ Common FuSa & Security
Concept

No	IP	UM Naming	U2C8	U2B24-E	U2B-FCC	U2A16	Remark
1	CAN-FD	RSCANFD	Same	Same	Same	Same	
2	CAN-XL	CAN-XL	New	New	-	-	
3	SPI	MSPI	Same	Same	Same	Same	
4	LIN	RLIN3	Same	Same	Same	Same	
5		FlexRay	Same	Same	Same	Same	
6		Ethernet TSN/AVB	TSN	RSW2.3(TSN)	RSW1(TSN)	AVB	IP Upgrade
7		10Base-T1S	New	New	-	-	
8	I2C	RIIC	-	Same	Same	Same	
9		I3C	New	-	-	-	
10		PSI5	Same	Same	Same	Same	
11		PSI5S	Same	Same	Same	Same	
12	SENT	RSENT	Same	Same	Same	Same	
13		CXPI	Same	-	-	Same	
14	Zipwire	RHSIF	-	Same	Same	Same	
15	Generic Timer	GTM	V4.x	V4.1	V4.1	V3.5	IP Upgrade
16	Timer 16 Bit	TAU-D	Same	Same	Same	Same	
17	Timer 32-Bit	TAU-J	Same	Same	Same	Same	
18	Motor Control Timer 18-Bit	TSG3	Same	Same	Same	Same	
19	Encoder Timer 16-Bit	ENCA	Same	Same	Same	Same	
20	LED/Ligh Control	PWM-Diag	Same	Same	Same	Same	
21	OS Timer 32-Bit	OSTM	Same	Same	Same	Same	
22	Real Time Counter	RTCA	Same	Same	Same	Same	
23	Watchdog Timer	WDT	Same	Same	Same	Same	
24	Long Term Counter 64-Bit	LTSC	Same	Same	Same	Same	
25	SAR ADC	ADC	Same	Same	Same	Same	
26	SDHC	eMMC	Vers. TBD	V4.x	V4.x	V4.x	
27	Security HSM	ICUM	ICUMHB*	ICUMHB	ICUMHB	ICUMHA	Function Upgrade U2B/C

HIGH SOFTWARE REUSABILITY

WITH RENESAS CROSS DOMAIN 28NM MCU PLATFORM

Software First – Renesas Platform Approach



High integration density of 28nm enables wide application coverage with one family

- Fewer MCU variants and SW platforms to develop
-> from 5 to 3
 - Fewer MCU variants: simple & robust supply chain
- Double digit MUSD reduction of development cost
- Reduction of development lead-time by 6 months
- Re-use of SW within the 28nm family

Safe & long-lasting process

- 28nm is already in production at Renesas: safe launch!
- Long process availability:
 - Last node before 16nm FINFET
 - Already becoming the industry standard for industrial

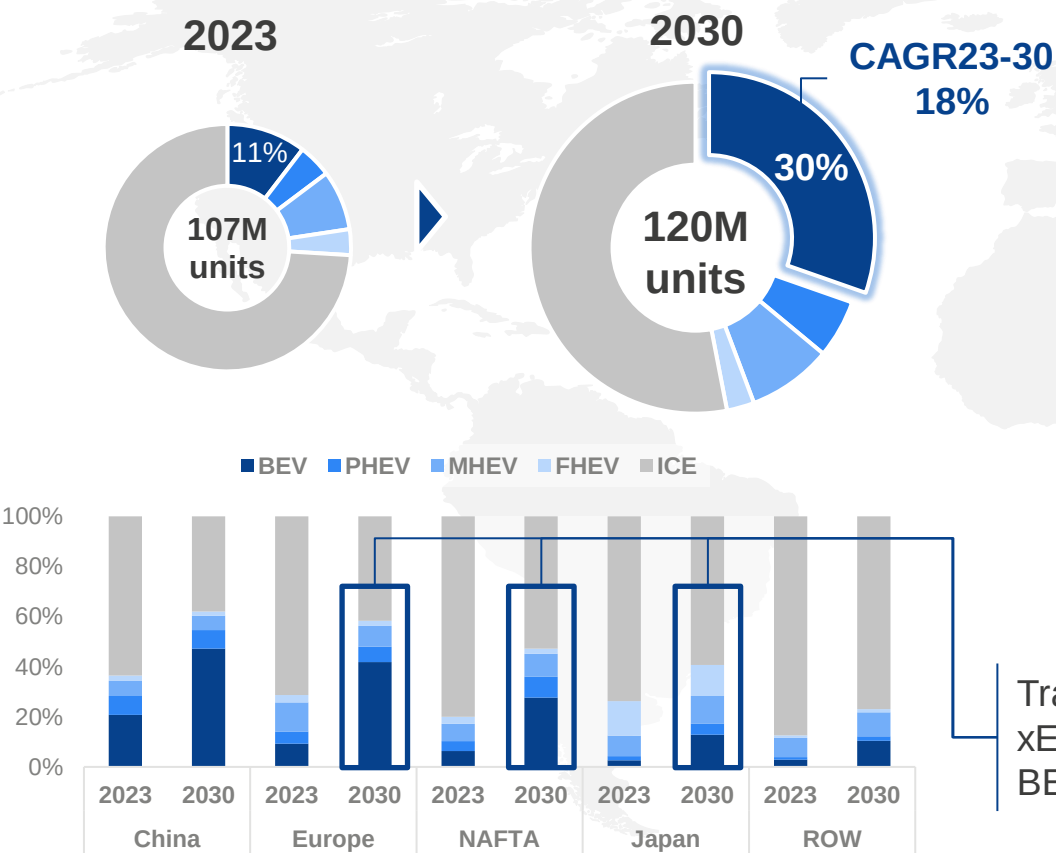
XEV MARKET TREND

MARKET TREND CHANGE

xEV MARKET

- Based on carbon neutral initiatives in governments, transition to Electric Vehicles (esp. BEV) is increasing with high growth

Vehicle production



Carbon Neutral Initiatives in each Country



Increase the proportion of NEVs in new car sales to over 50% by 2035. All ICE in new car sales change to HV



After 2035, the sale of ICE vehicles will be prohibited, except for vehicles that use e-fuels



Aim to increase the ratio of BEVs in light vehicles to 60% by 2030 and to 67% by 2032



By 2035, the goal is to achieve 100% of new car sales, including HEV, as EVs in the passenger car segment



Aim to achieve 70% electrification of commercial vehicles, 30% of private vehicles, and 40% of buses by 2030

Traditional OEMs will accelerate xEV adoption following Chinese BEV OEMs

SYSTEM TRENDS

Vehicle trend

xHEV



Tighter CO₂ reduction targets

BEV



Traction system trend

ICE + Transmission + Inverter + Motor



IGBT

Compact, Modularization

High efficiency

E-Axle

In-wheel motor

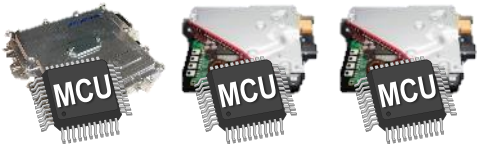


SiC

ECU integration trend

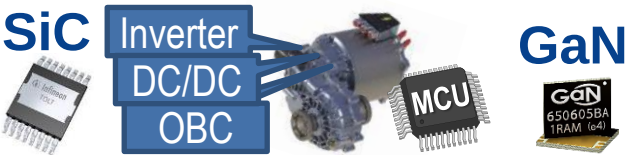
ECU

Inverter + DC/DC + OBC



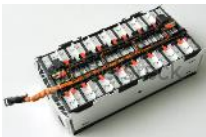
Compact, Cost optimization

X-in-1 E-Axle



BMS trend

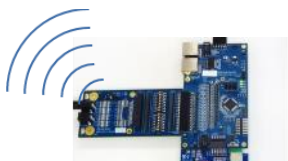
Small, Centralized



Large area battery
Battery pack reuse / change

Distributed

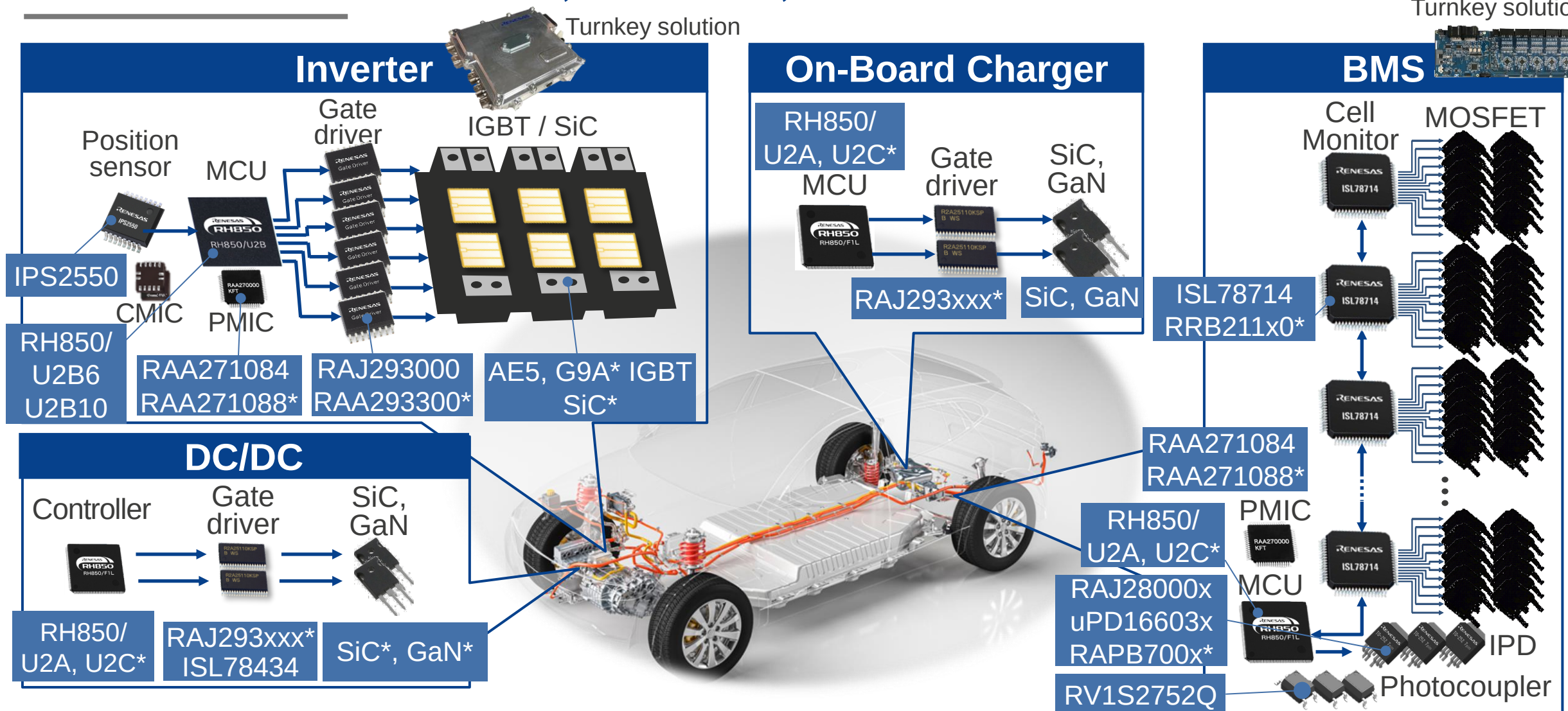
Wireless



REN XEV SOLUTION

RENESAS HAS ALL MCU, ANALOG, POWER AND SYSTEM SOLUTION

Turnkey solution



BMS: Battery Management System / PMIC: Power Management IC / MMIC: Monolithic Microwave IC / IPD: Intelligent Power Device

MCUs FOR xEV SOLUTIONS

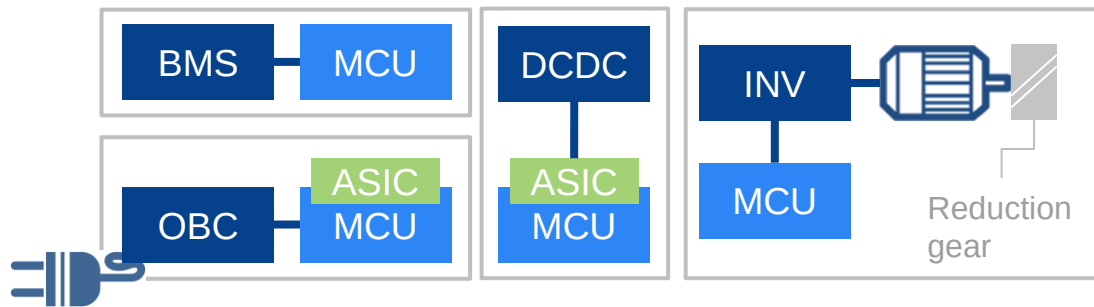
U2B FOR XEV SYSTEM INTEGRATION

xEV MARKET TREND

Xin1 integration

- The first step is to combine the hardware of different xEV systems into a “Single Box”, which significantly **lowers system costs**. Control boards usually remain separate, using multiple microcontroller units (MCUs).
- The subsequent step involves integrating the control boards and utilizing a powerful MCU instead of multiple MCUs/DSPs

Today

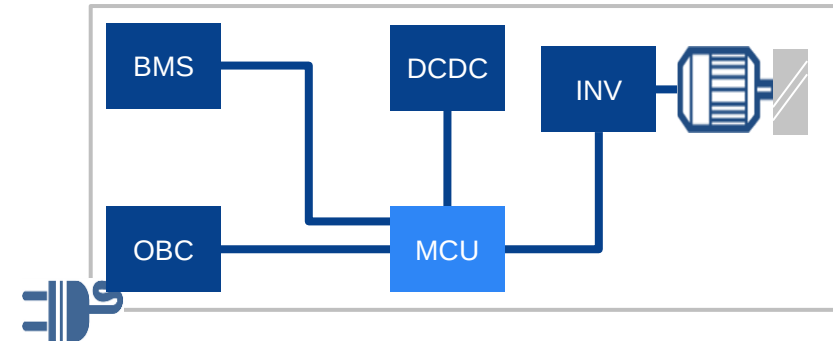


Separated Systems

Maintaining distinct cooling and mechanical systems raises both costs and weight.

It also complicates supplier management and sourcing.

Tomorrow



Integrated Control & Mechanics

Merge control functions into one MCU to reduce bill of materials costs;

RENESAS MCU IN SYSTEM INTEGRATION

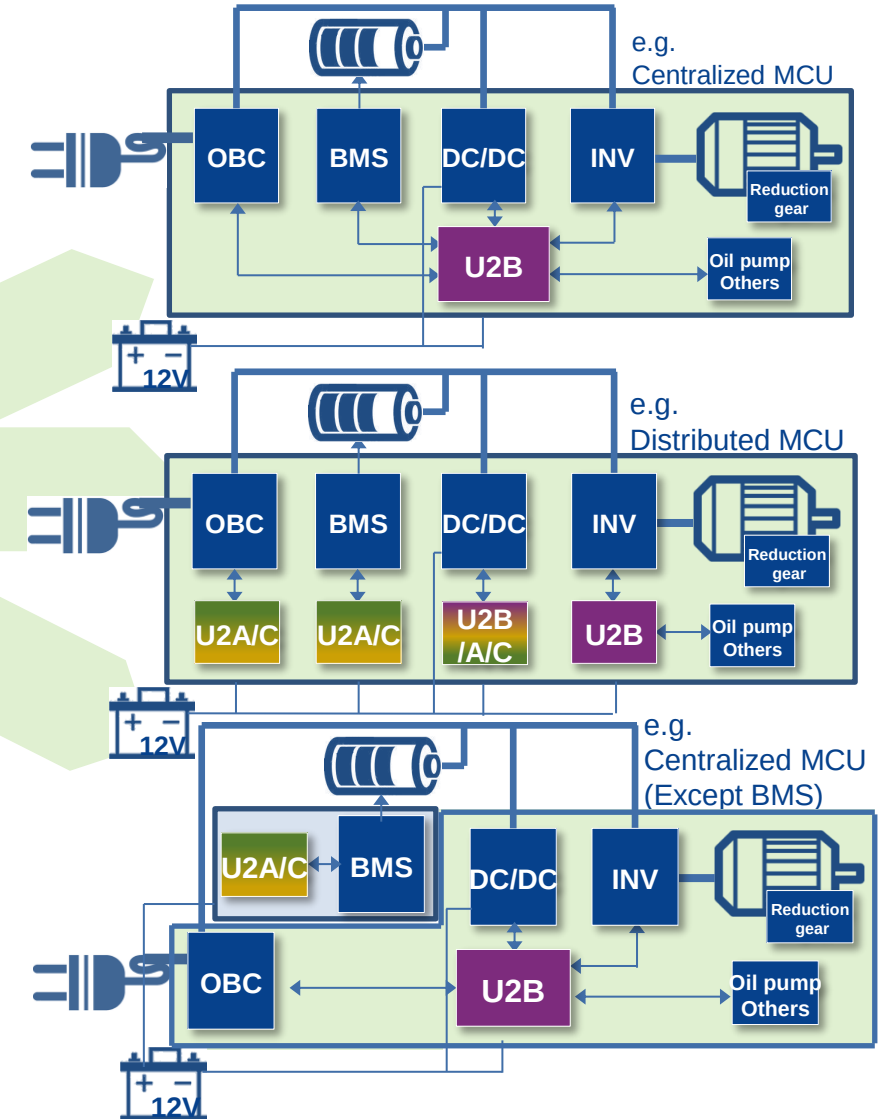
E-Axle

A motor, reduction gear, and inverter are integrated
Use 4 MCUs

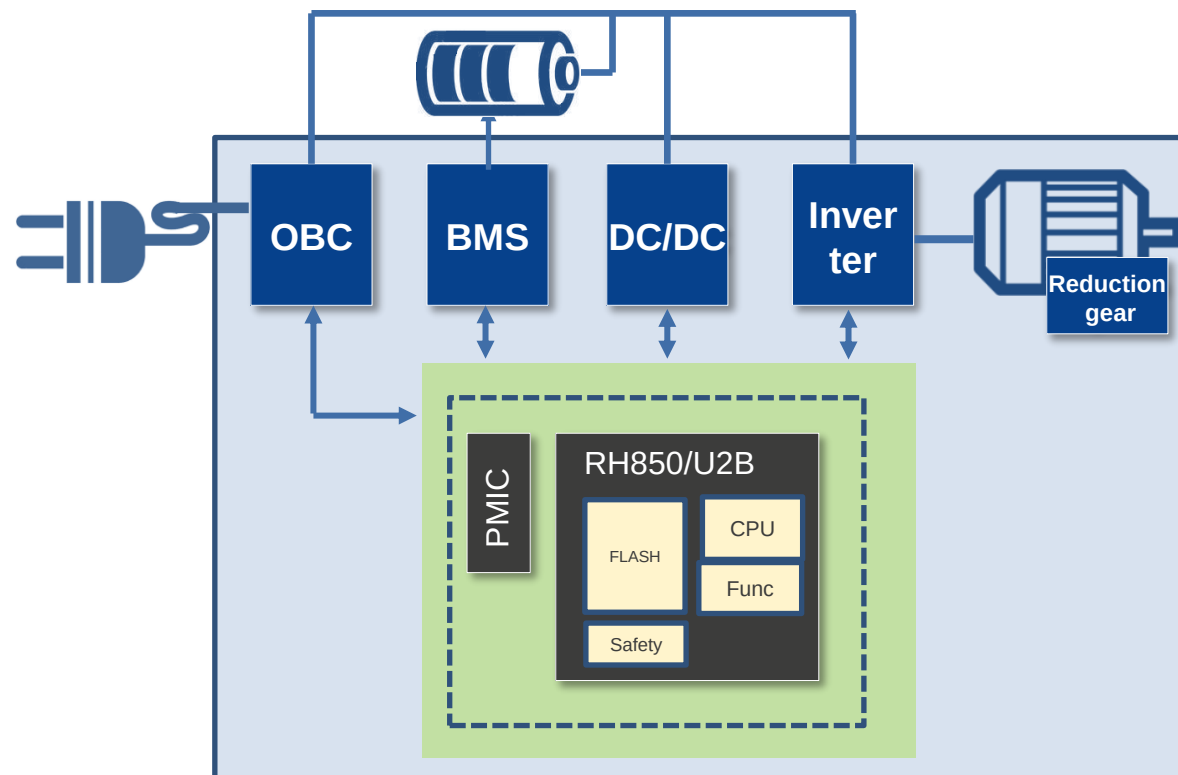
System Integration

X in 1 system

Motor, inverter, BMS, DCDC, etc.
integrated into one package
There are several integrated variations
Use MCU 1 to 4



RH850/U2B ADVANTAGE FOR X-IN-1



CPU

- Processing distribution using high-performance multi-core CPU

Peripheral functions

- BOM reduction with built-in R/D converter
- High-frequency switching with high resolution PWM
- Scalability within the U2B series

Flash

- No performance degradation due to built-in Flash

Safety

- Supports ASIL-D

Ease of development:

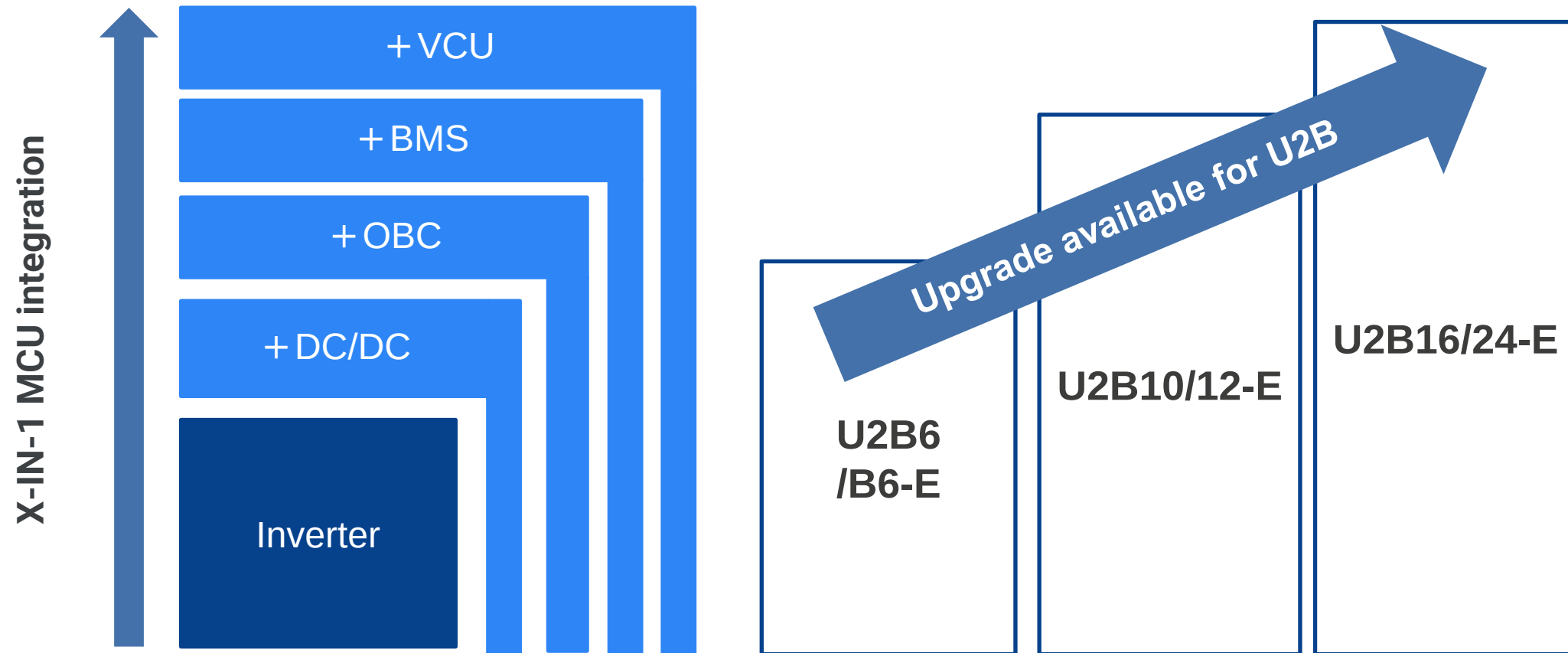
- Reduce man-hours with model-based design environment

Quality

- High-quality automotive track record

RH850/U2B ADVANTAGE FOR X-IN-1 ~ SCALABILITY ~

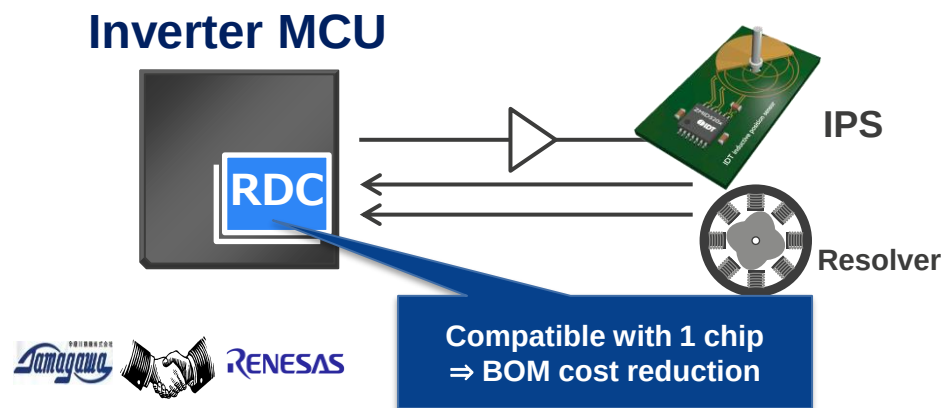
Scalability within the U2B series for X-in-1



RH850/U2B ADVANTAGE FOR X-IN-1

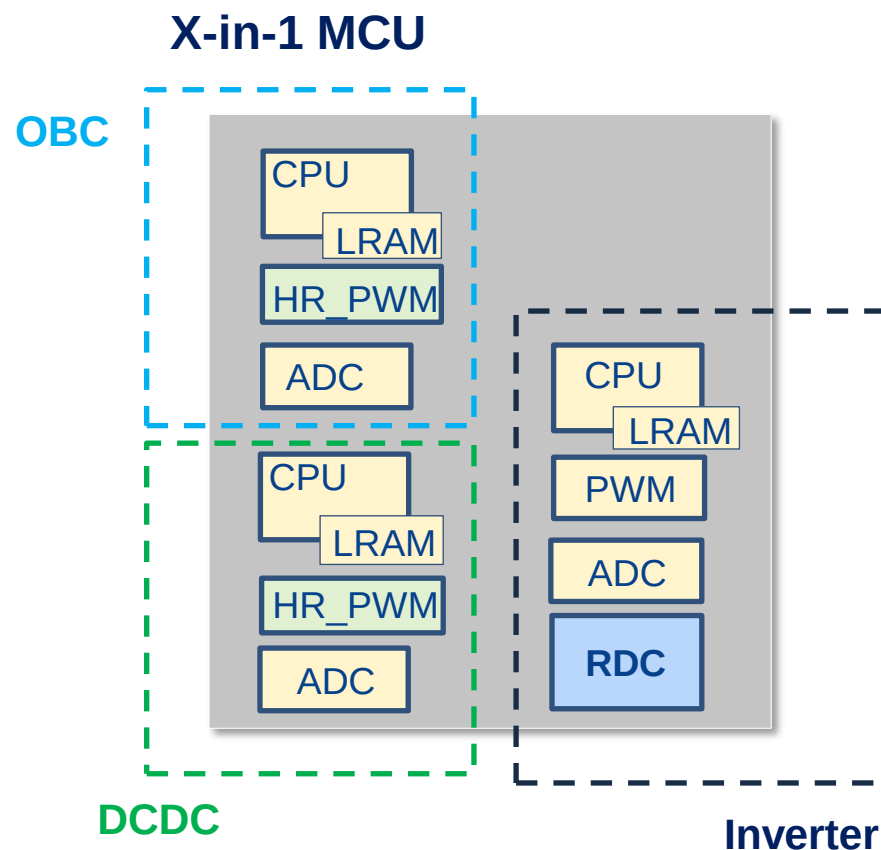
~ BOM COST SAVING ~

- Built-in RDC reduces ;
 - External components.
- Functional Safety concerns



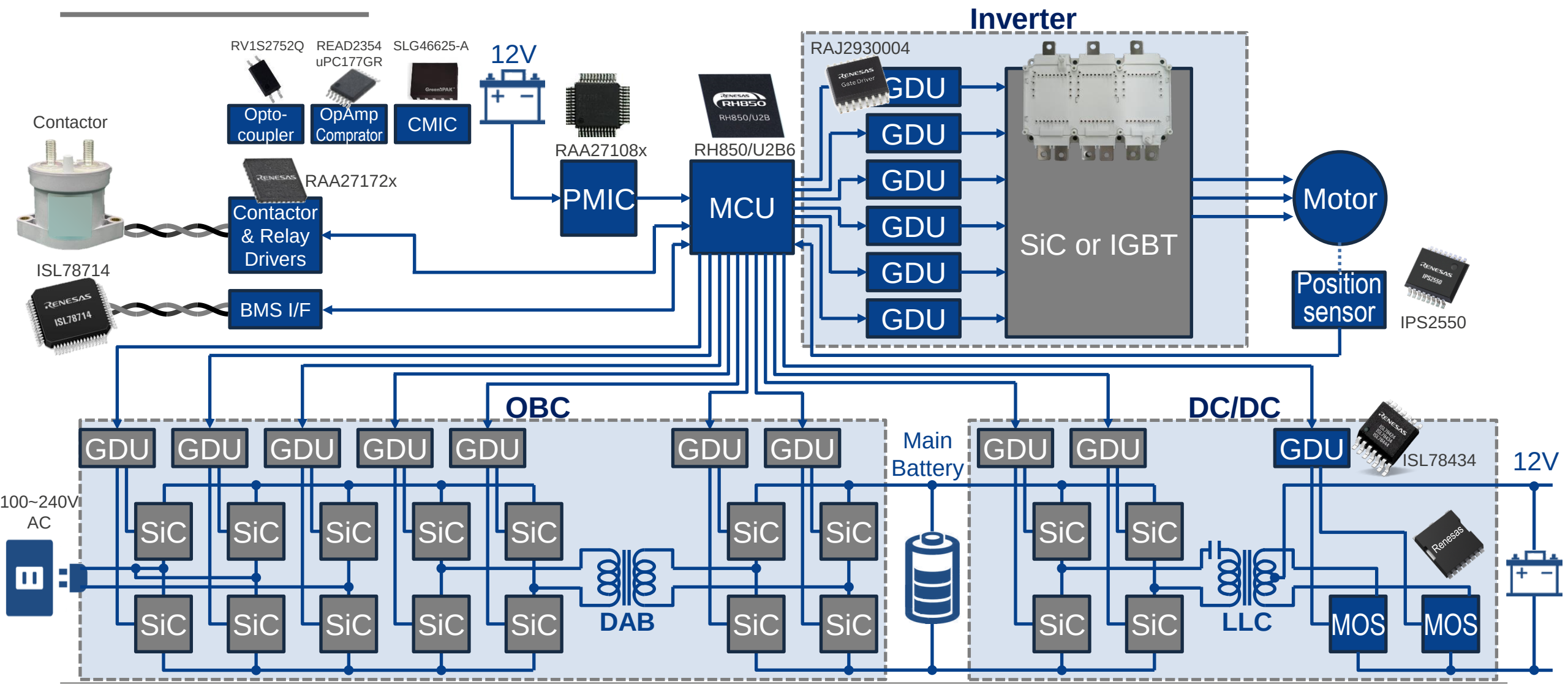
Achieved high performance, high reliability, and noise resistance through collaboration with Tamagawa Seiki.

Distribute processing by assigning functions to each application



X-IN-1 BY A SINGLE RH850 MCU & RENESAS POWER PRODUCTS

RENESAS PROVIDES ALL KEY COMPONENTS AND SYSTEM SOLUTION



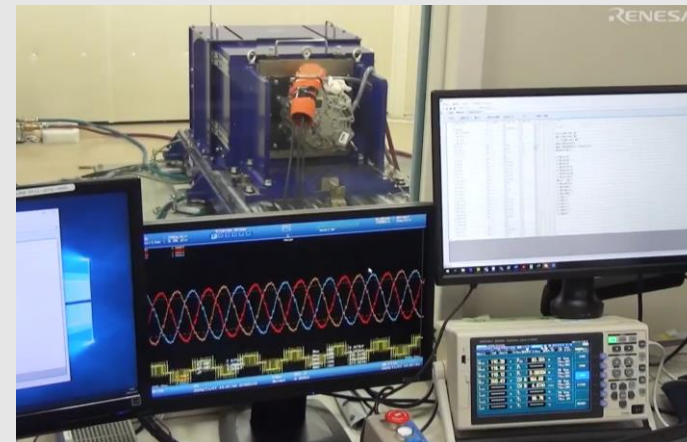
X-IN-1 SOLUTION DEVELOPMENT AND VALIDATION ENVIRONMENT

X-in-1 PoC solution



- System solution verified with dyno and loads.
- Software suite: MCAL, complex drivers, and application software
- MATLAB simulation, HILS, Autocode ready

System-level support



- Support by inverter application engineers
- Verification environment by dyno benches, electronic load, HILS

System-level knowledge and support capability.

We can reproduce customer's troubles in our lab and propose solutions.

X-IN-1 SOLUTION DELIVERABLES

Hardware



✓ Inverter, Design data, Test results

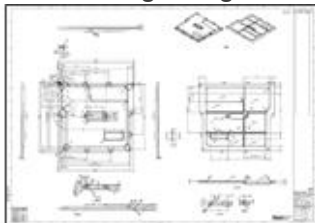
Schematic



PCB



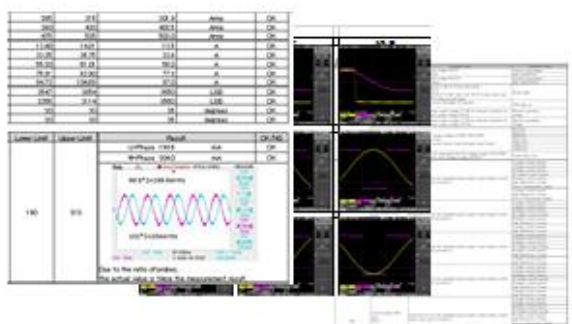
Housing design document



Design Specification



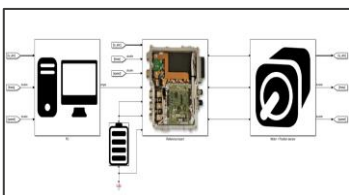
Test specification, Evaluation result report



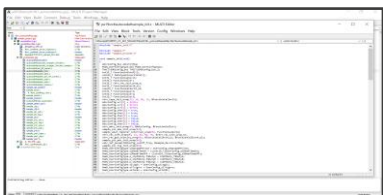
Software

✓ MATLAB/Simulink model for simulation and Autocode, Sample software. Application note, Tools, Evaluation report

Sample model



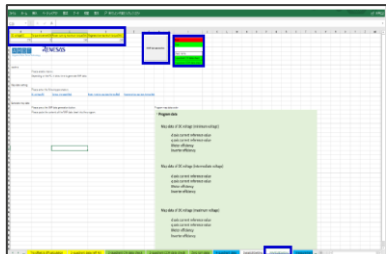
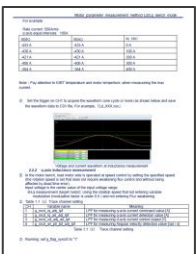
Sample software



Application note



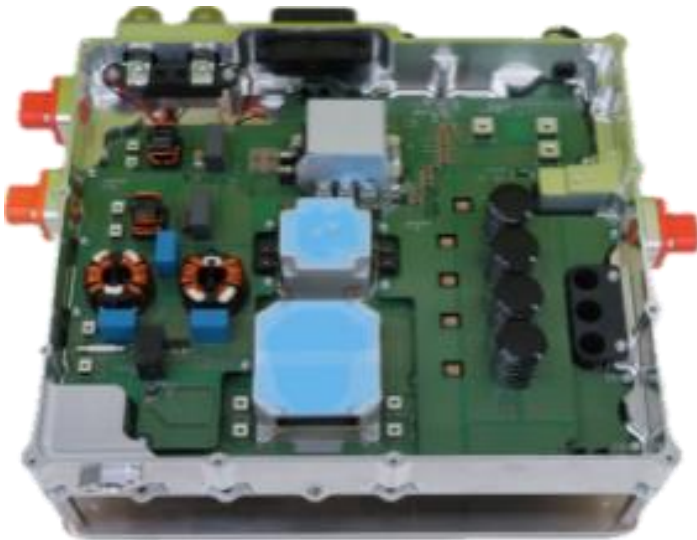
Measurement tool and document for Motor parameter, Torque map, Deadtime table, Inductance map



Evaluation result

X-IN-1 (INVERTER + DC/DC + OBC) SPECIFICATION OVERVIEW

Renesas provides system solution that achieves high efficiency and high integration



Device	
MCU	RH850/U2B6
PMIC	RAA271084
HV-GDU	RAJ2930004
Power device	IGBT AE4
Power module (6in1)	RBZ900H075VT0HVP

Items	X-in-1 (inverter + DC/DC + OBC) specifications
Inverter power	Up to 100kW
Inverter current	Up to 500A _{RMS}
DC link voltage	Up to 400V
Rotation speed	Up to 30,000rpm mechanical (120,000rpm electrical)
Cooling	Liquid cooling (50% LLC mixture)
Sensors	• Current sensor : U/V/W-phase current sensor • Temperature sensor : Power module, Liquid, Motor • Position sensor : Resolver / Inductive position sensor
Command mode	Torque control, Speed control
DC/DC	1.5kW (400V→12V) LLC
OBC	6.6kW (AC → Battery) Bridgeless PFC + DAB 3kW (Battery → AC)

U2B FOR TRACTION & GENERATOR MOTORS

U2B CONCEPT FOR TRACTION MOTOR

- INTEGRATION CONCEPT -

1. High performance & Safety

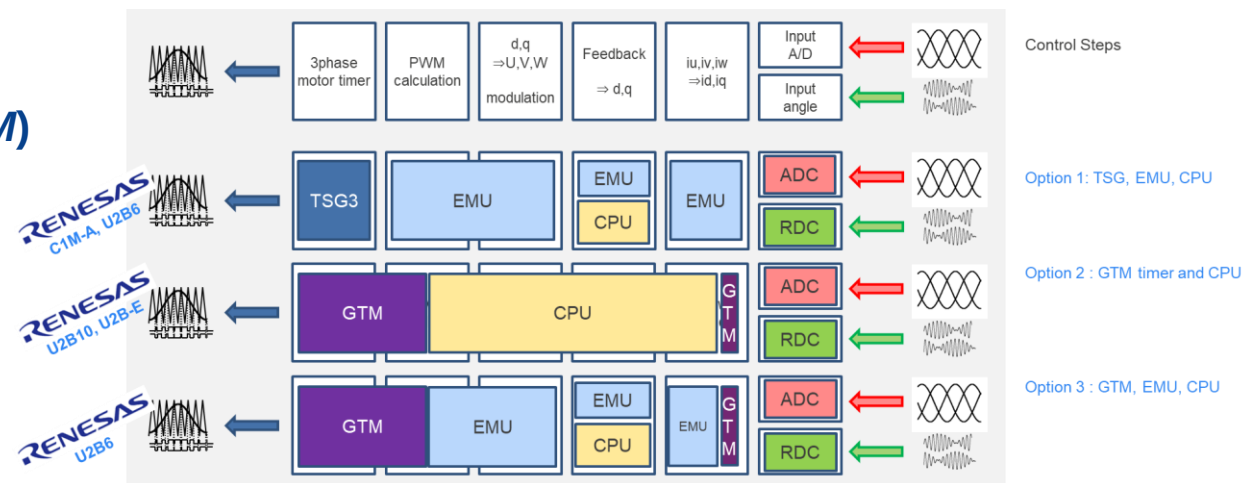
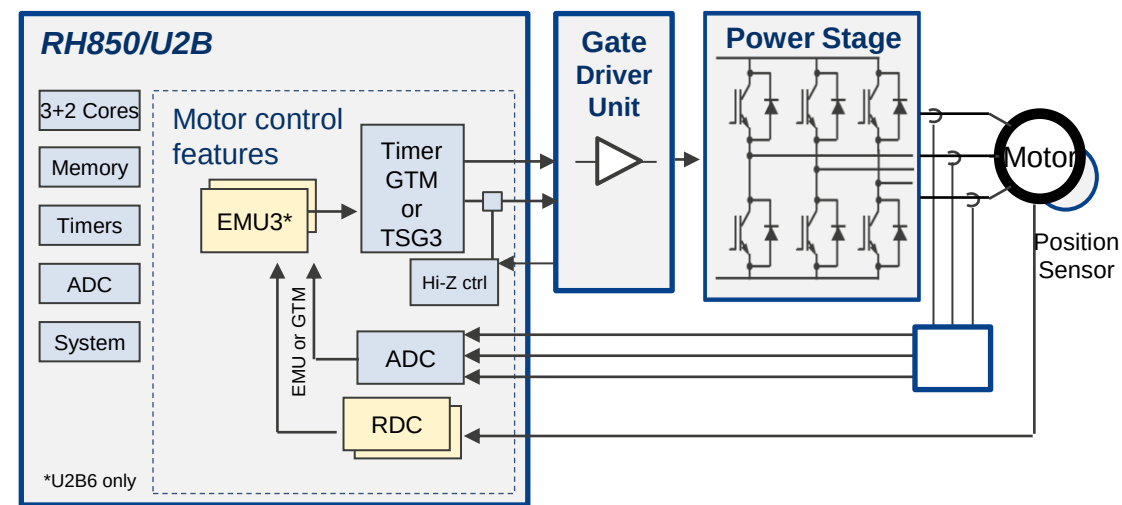
- Multi-core, 400MHz
- Enhanced diagnostic functions
- Functional Safety

2. Built-in redundant Resolver to Digital Converter (RDC)

- Reduces system cost with an integrated RDC
- Improvement of failure diagnosis

3. Built-in redundant Motor Control Support (EMU3 or GTM)

- Built-in flexible, optional hardware logic for motor control
- Able to control up to 2 motors independently in conjunction with the timers



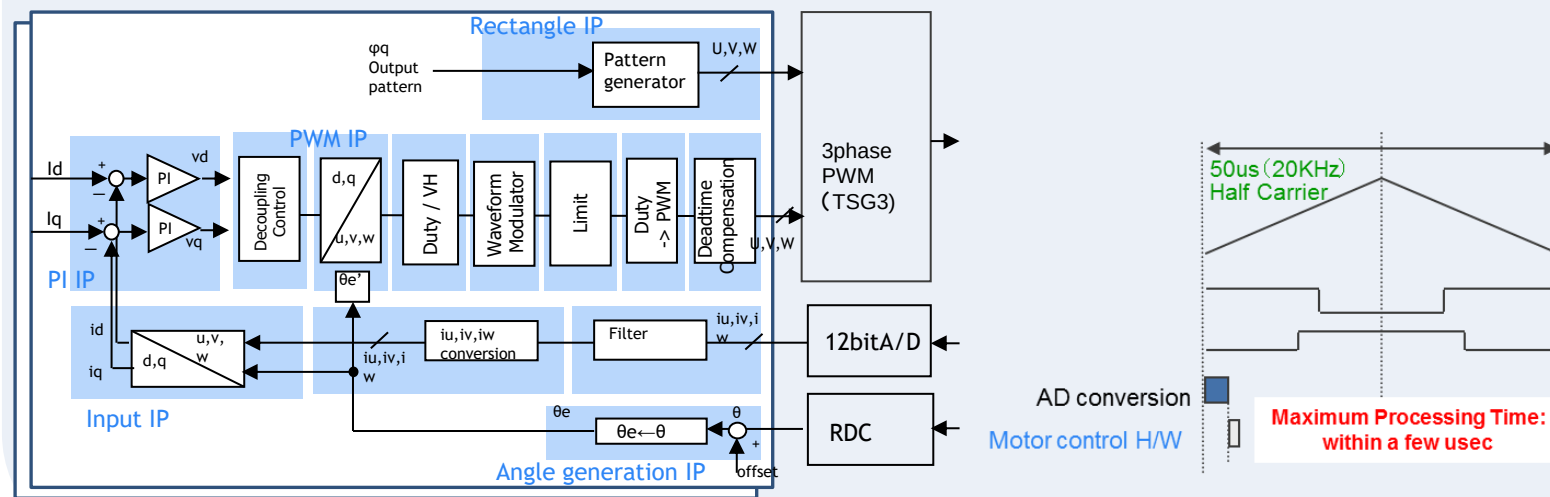
U2B CONCEPT FOR TRACTION & GENERATOR MOTORS

- INTEGRATION CONCEPT -

➤ Ideal solution for E-drive to control traction & generator motors with 1 MCU

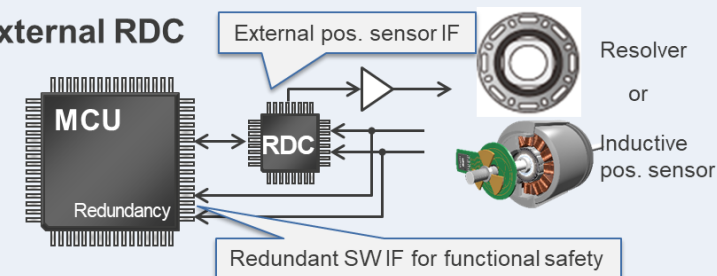
- Control 2 motors with 1 MCU based on high performance multi core CPU
- Reduce CPU load with H/W-accelerator (EMU3) for high-speed motor control
- Reduce ECU BOM cost by an embedded Position sensor interface to connect either resolver- or inductive position sensors
- ISO-26262 ASIL-D compliant

EMU3 solution

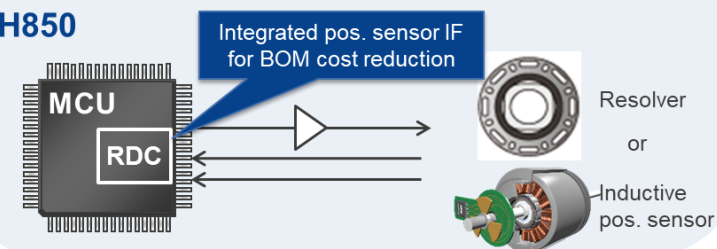


Integrated RDC solution

External RDC



RH850



U2Bx - EV/HEV APPLICATION FOCUS

			RH850/U2Bx/U2Bx-E											
			U2B6		U2B10		U2B6-E	U2B12-E		U2B16-E		U2B24-E		
CPU	Clock freq.		400MHz		400MHz		400MHz		400MHz		400MHz		400MHz	
	Main cores (G4MH)		3		4	3	2		4	3	6	5	6	5
	Lockstep cores		2		2	3	2		2	3	4	5	4	5
Flash	Code Flash		6 MB		10 MB		6 MB		12+2 MB		16 MB		24+2 MB	
GTM			v4.1		v4.1		v4.1		v4.1		v4.1		v4.1	
ADC	SAR A/D		3 modules		4 modules		3 modules		4 modules		4 modules		4 modules	
		T/H	6 / 6 / 4		6 / 6 / 0 / 0		4+4+4		4+4+4		4 / 4 / 4 / 0		4 / 4 / 4 / 0	
	Delta-Sigma A/D		4 modules		10 modules		6 modules		10 modules		12 modules		12 modules	
RDC interface support			2 modules (SAR-RDC) with tracking loop		2 modules (use DS-ADC)									
Motor control accelerator EMU			2 modules		-		-		-		-		-	
Features for motor control			- Reuse of existing user's assets (C1M-A successor) - Reduction of BOM cost (Built-in RDC) - High performance by motor control accelerator (EMU)		- Sharing user assets with other applications (PT,HV) - Reduction of BOM cost (Built-in RDC using DS-ADC) - 4 channels External Delta-Sigma Modulator Interface (DSM-IF)									

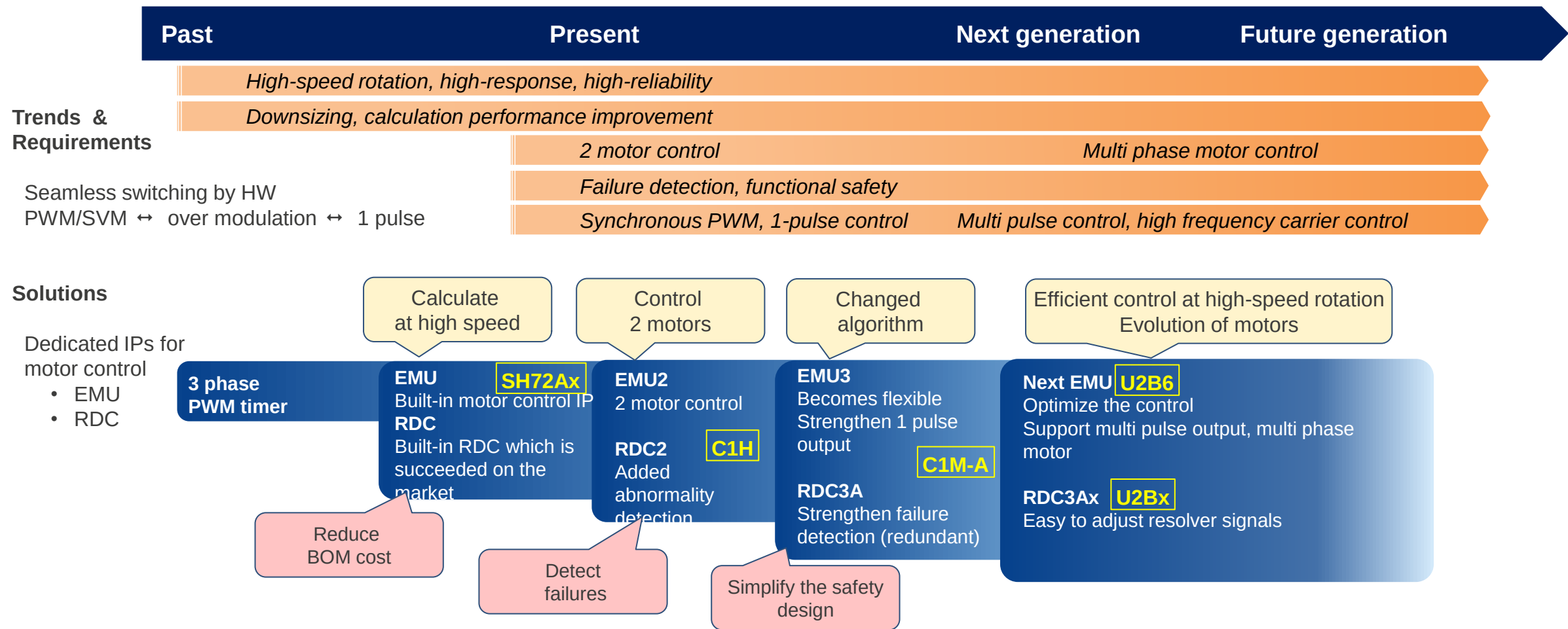
U2B6: Motor control features are compatible with predecessor devices (C1x-family)

U2B10 / U2B6-E / U2B12-E / U2B16-E / U2B24-E: Generic motor control features for Powertrain and xEV applications

- **RDC (RESOLVER TO DIGITAL CONVERTER)**
- **EMU (E-MOTOR CONTROL UNIT)**

EMBEDDED MOTOR CONTROL (EMU) AND R/D CONVERTER (RDC)

→ STARTED 2011 → 3RD GENERATION IN THE MARKET NOW → 4TH GENERATION PLANNED



CORRESPONDING TO DIVERSIFYING ANGLE SENSORS

BUILD-IN RDC CAN SUPPORT RESOLVERS AS WELL AS INDUCTIVE POSITION SENSORS

- Resolvers continue being used in the mainstream of traction motor applications
- Various sensors(inductive, magnetic) are used in addition to resolvers
- Renesas' RDC supports these sensors
 - ✓ Angle data can be input with an analog signal for sin, cos (dealing with signals without modulation)
 - ✓ Immune to noise (differential analog input, noise elimination by tracking loop)
 - ✓ S/W commonization (Not many changes to the S/W even if sensors other than a resolver are used)

HIGH-SPEED INDUCTIVE POSITION SENSOR : IPS2550

Interface: sin/cos single ended or differential

Automotive AEC-Q100 Grade-0 Designed & Qualified

Temperature range: -40° to 160° C ambient

Functional Safety: supports ASIL-C @ single IC

Voltage Supply: 3.3V $\pm$ 10% or 5.0V $\pm$ 10%

Rotational Speed: up to 600,000 (el) rpm

Propagation delay: 5 μ s

Overvoltage, reverse polarity, short-circuit protected

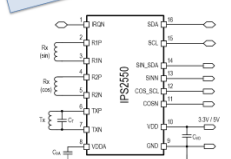
Programming over digital interface or analog outputs

Diagnostics interrupt to external MCU

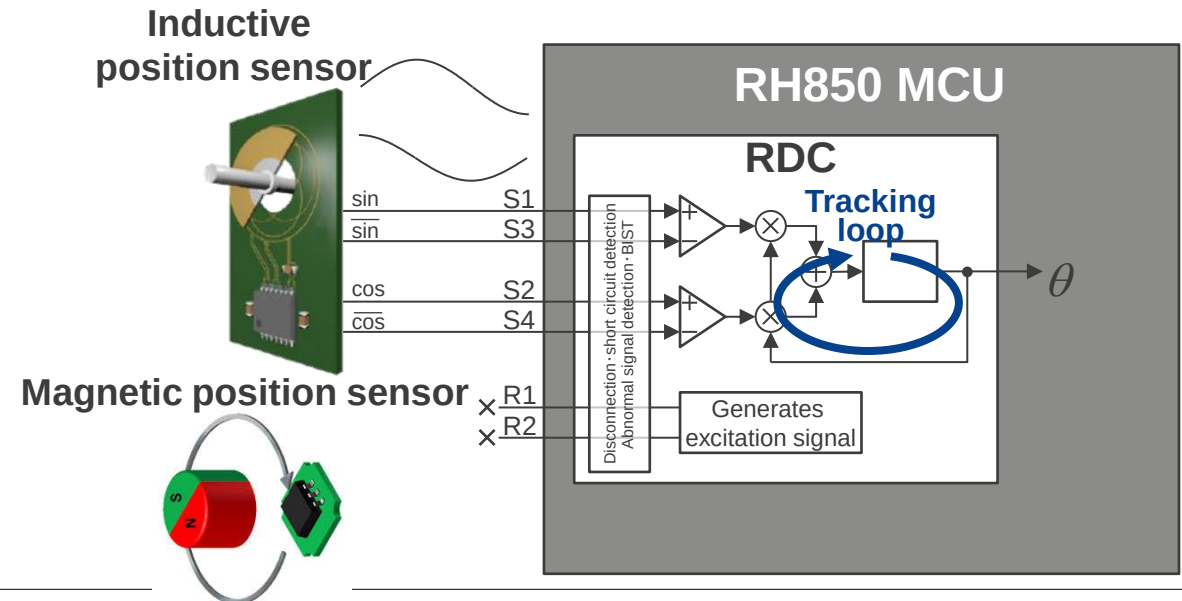
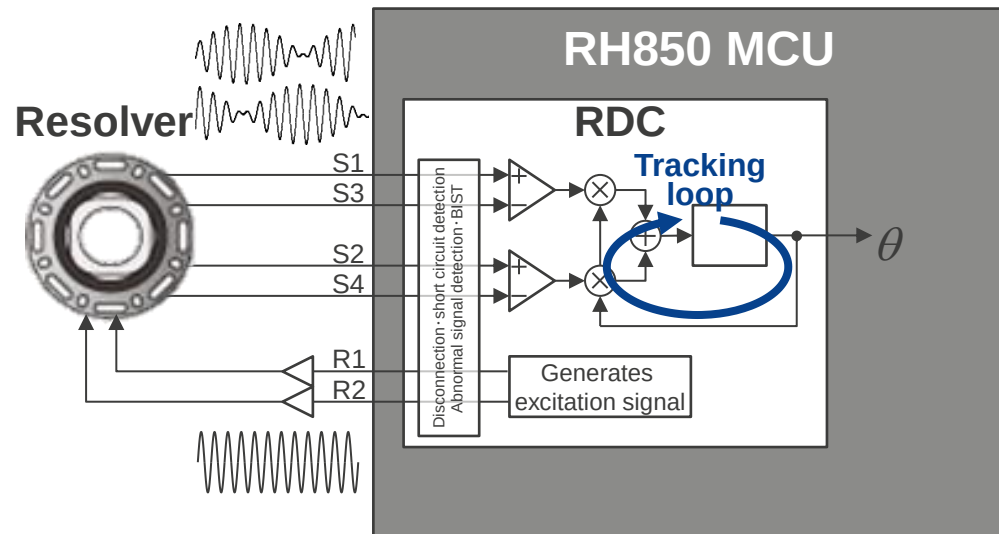
AGC to compensate air-gap variations

TSSOP-16 with exposed pad

Samples available December 2019



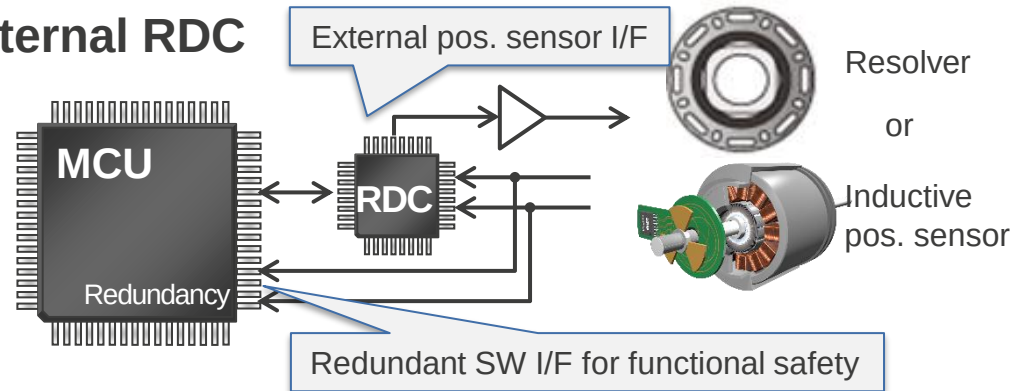
© 2019 Renesas Electronics Corporation. All rights reserved. RENESAS CONFIDENTIAL 12



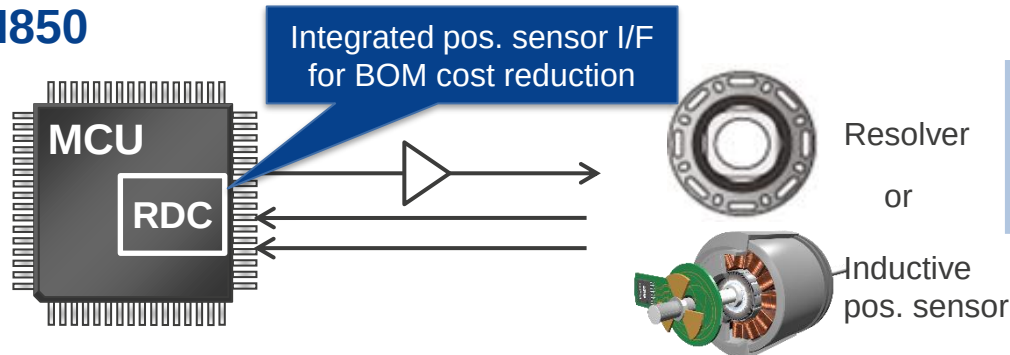
BUILT-IN RESOLVER-TO-DIGITAL CONVERTER (RDC)

→ BOM COST AND SPACE REDUCTION

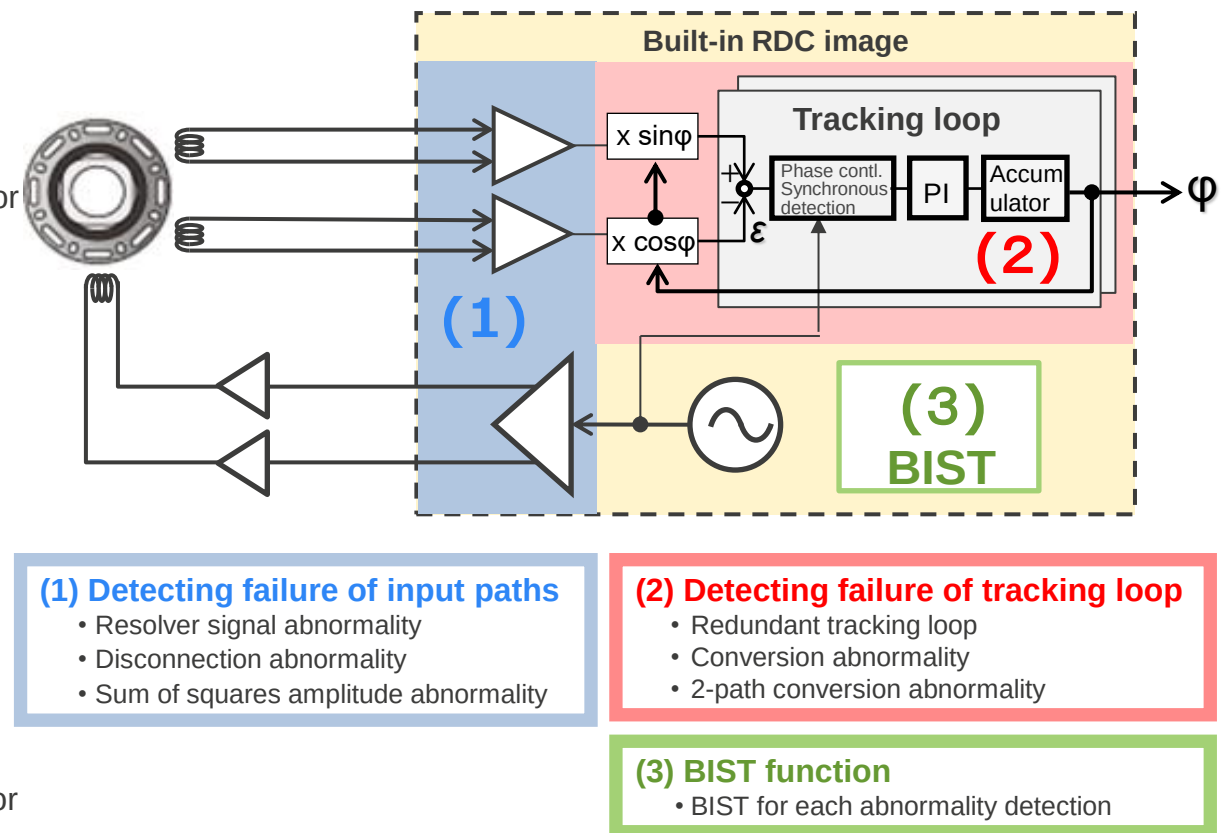
External RDC



RH850



RH850/U2B/U2B-E



- Input abnormality detection and double structure tracking loop → Redundant RDC can be removed

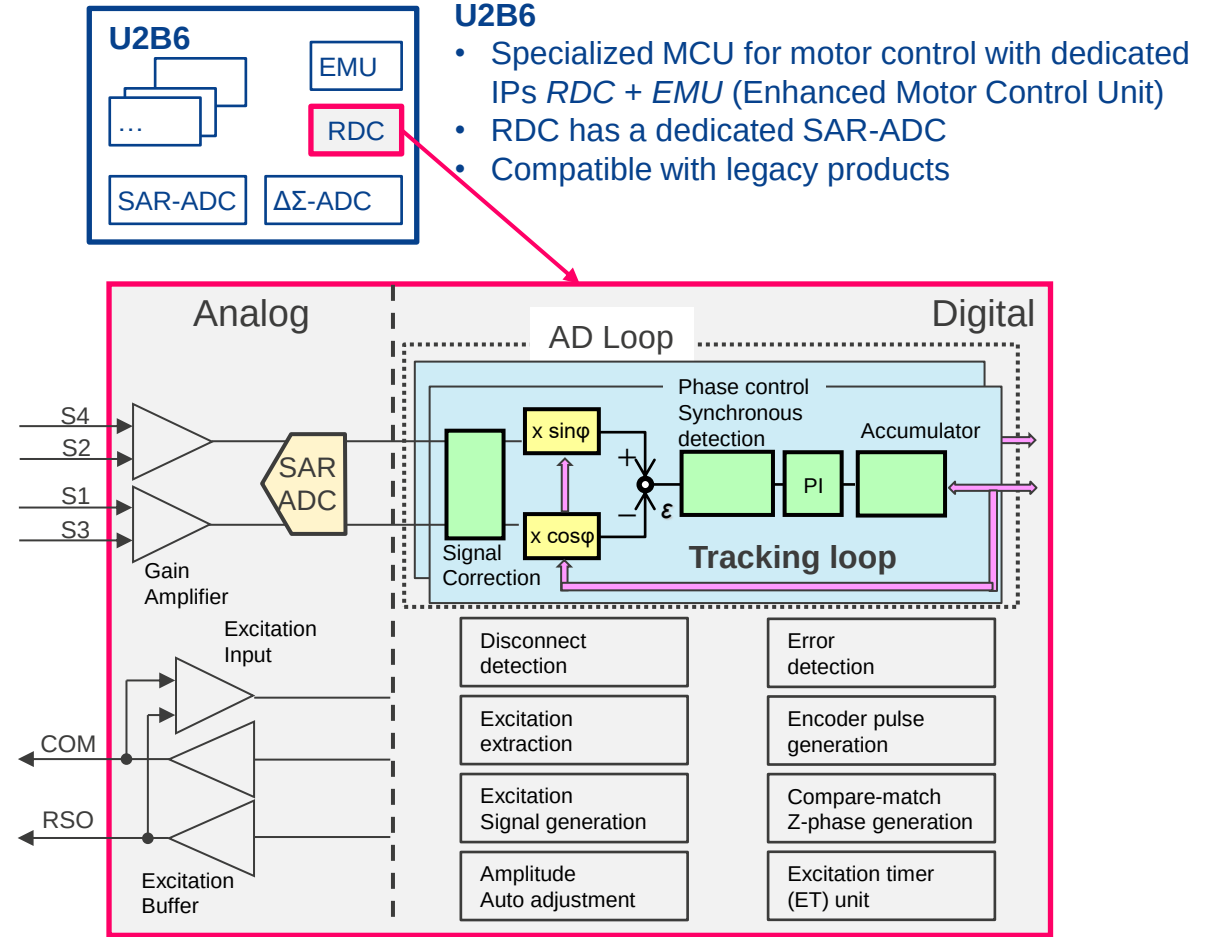
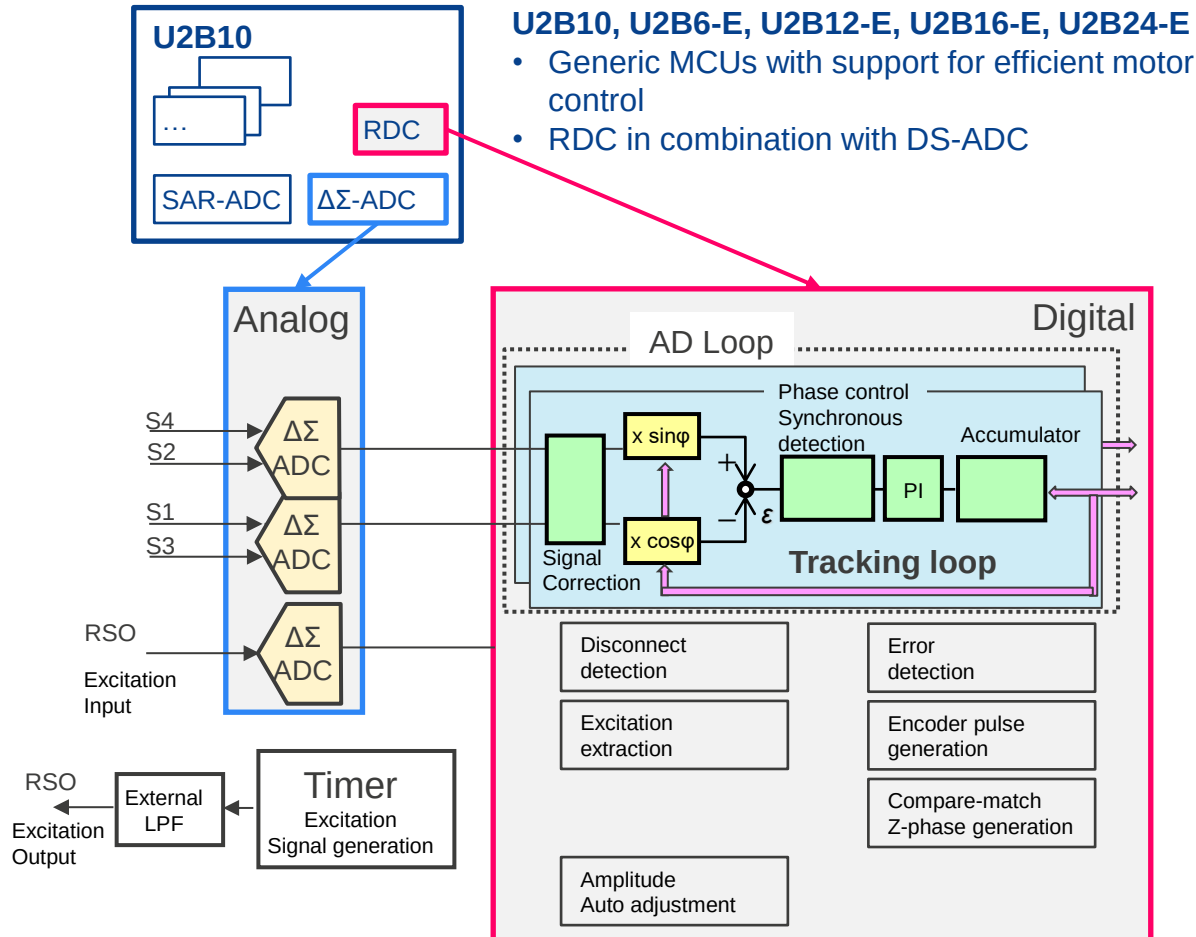
BUILT-IN RDC

Item	External RDC		Software RDC		Built-in RDC	
CPU load	None	😊	Heavy	😞	None	😊
Velocity	High rotation speed	😊	Depends on software	😞	High rotation speed	😊
Delay in response (in rotation)	Guaranteed in specifications		Predict and correct depending on velocity (error factor)	😞	Guaranteed in specifications	
Angle accuracy	Accuracy guaranteed	😊	Depends on software Error between ADC channels	😞	Accuracy guaranteed	😊
Trigger by angle value	Compare match in MCU	😊	Predict timing and correct (error factor)	😞	Compare match	😊
Error detection	Hardware detection	😊	Depends on Software		Hardware detection	😊
Safety	Non-redundant / no BIST	😞	Depends on Software / none		Redundant RDC architecture / BIST	😊
Board Space	More space required	😞	Saves space	😊	Saves space	😊

The Built-in RDC is better than an External RDC or Software RDC

U2BX FOR MOTOR CONTROL

RESOLVER-TO-DIGITAL CONVERSION (RDC)

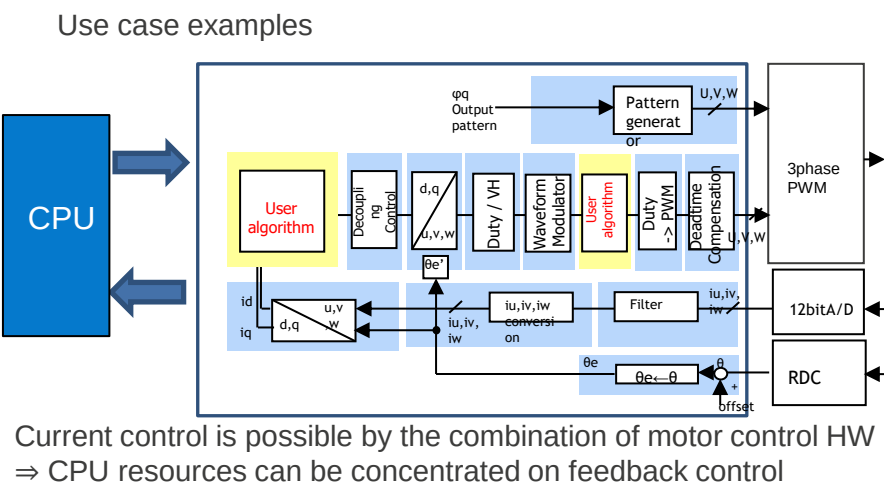
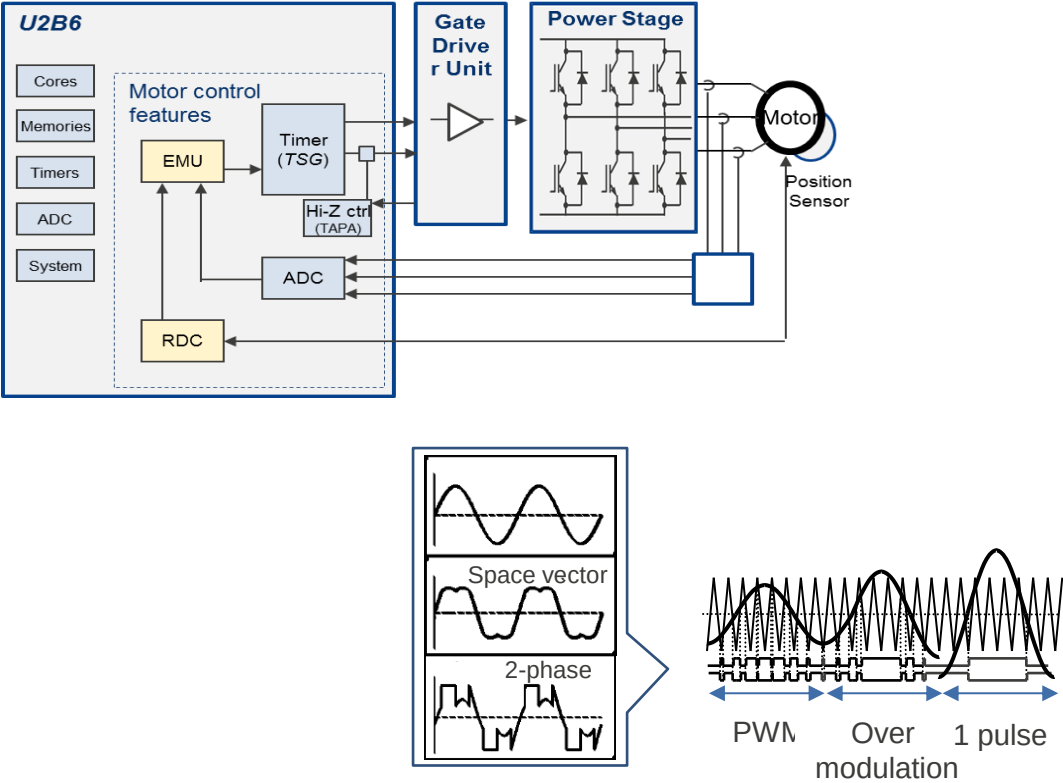
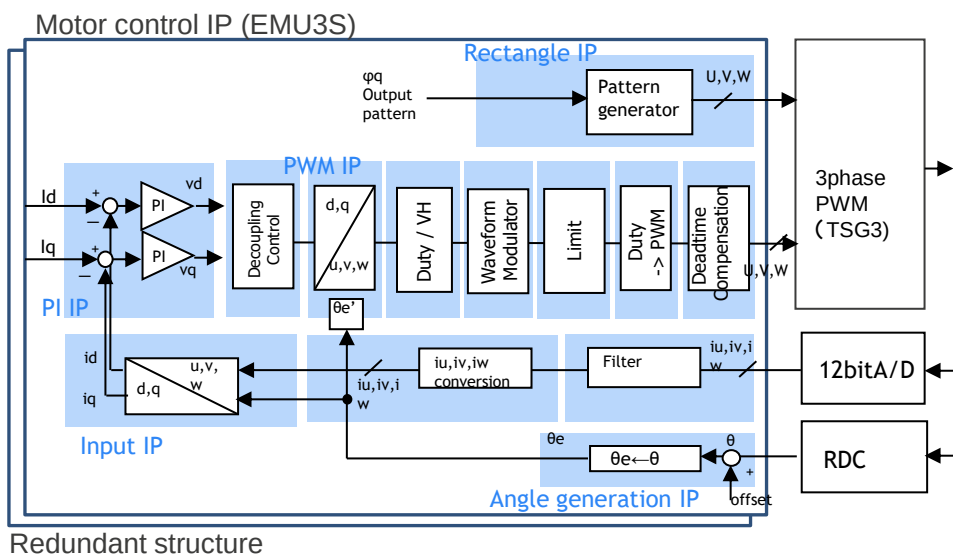


MOTOR CONTROL IP

EMU OVERVIEW

- Motor Control can be managed in parallel to the CPU
- Various modulations can be switched seamlessly (PWM, Space Vector Modulation, 2-phase modulation, overmodulation, 1-pulse, etc.)
- Selectable IPs for calculation

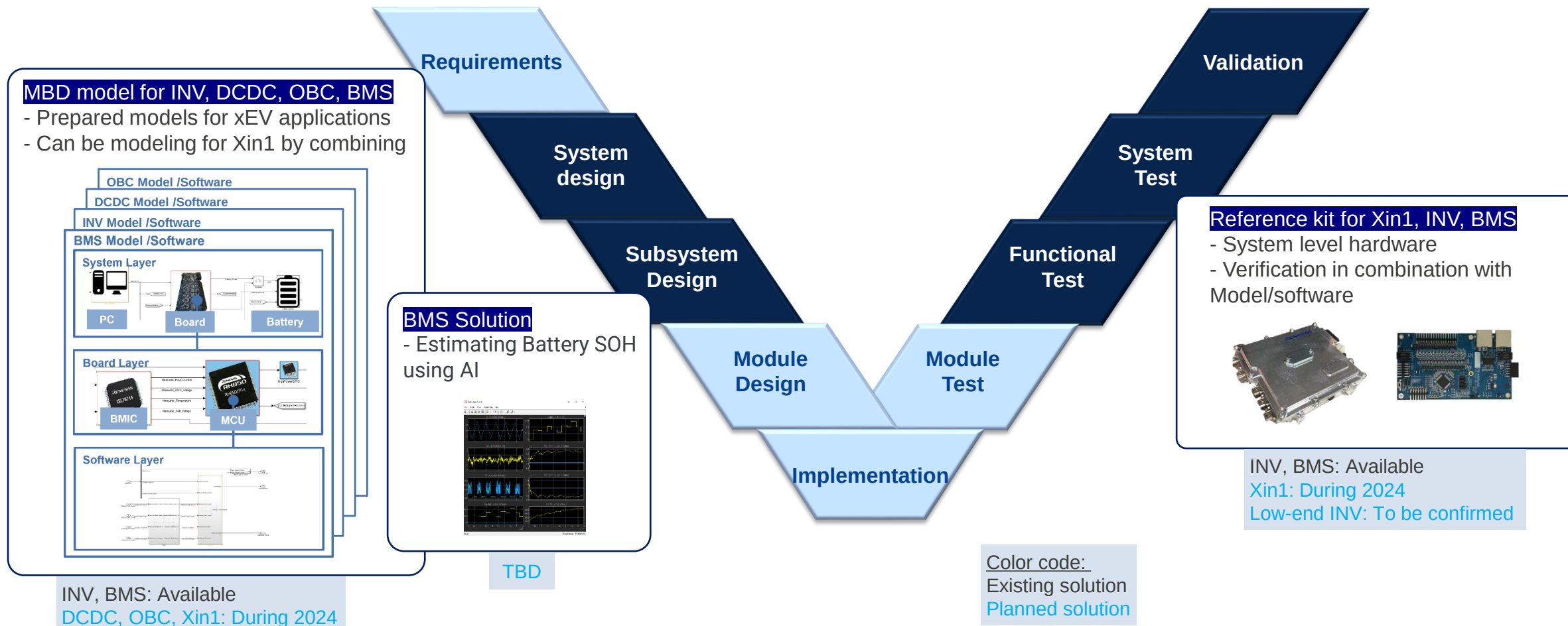
*Only U2B6 supports the EMU.



XEV REFERENCE DESIGN / SOFTWARE / MODELS

RENESAS XEV REFERENCE DESIGN / SOFTWARE / MODELS

Enhancement of reference design & eco-system for easy development

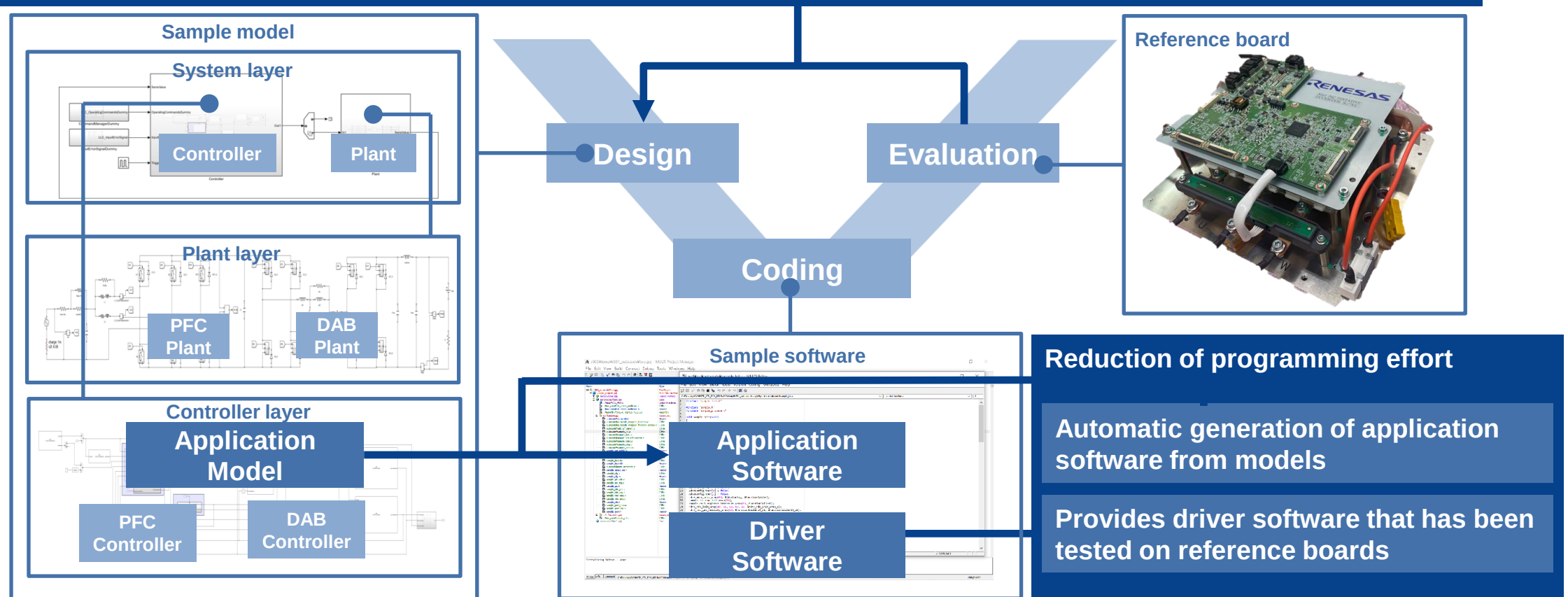


XEV ON BOARD CHARGER APPLICATION MODEL AND SOFTWARE

The purpose is to reduce development effort by providing sample models & software.
Collateral consists of Sample model, Sample software and Application note.

Reduction of defects discovered during the validation process reduces rework

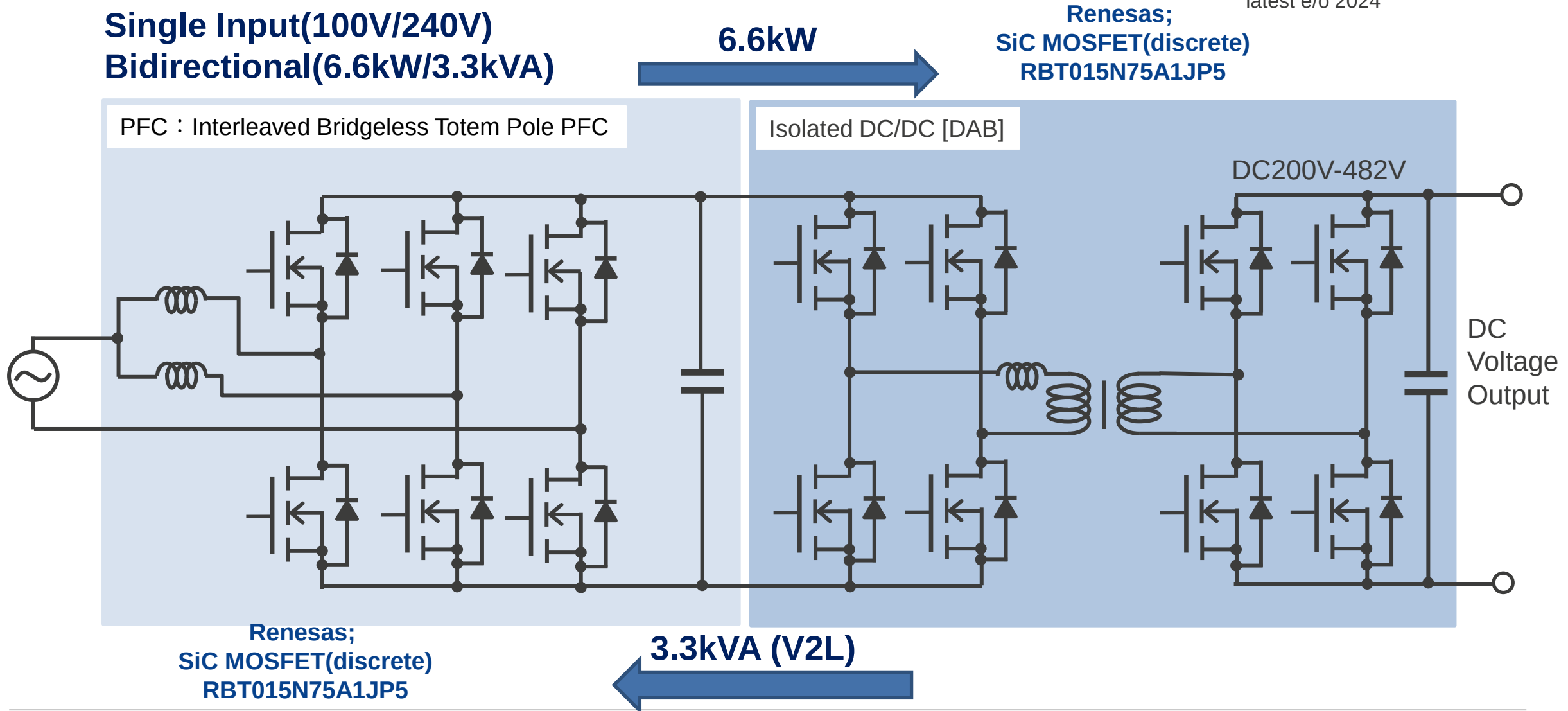
Simulation at the design stage enables early detection of system-level defects during development



XEV ON BOARD CHARGER APPLICATION MODEL AND SOFTWARE

ON BOARD CHARGER TOPOLOGY

- Status: under development with U2B6, target to be available by latest e/o 2024



OVER THE AIR UPDATE (OTA)

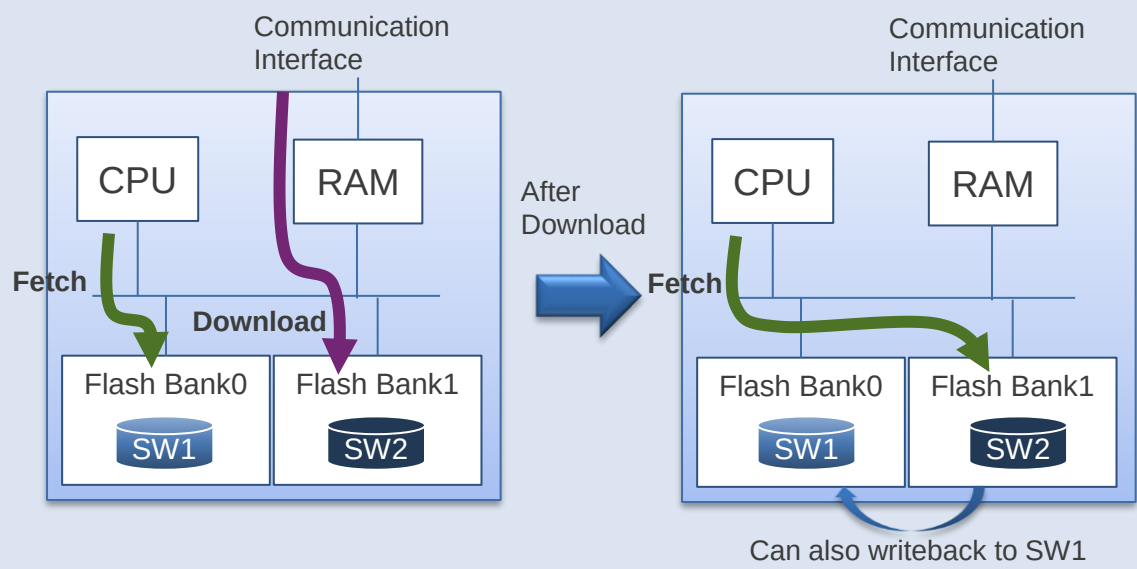
RH850/U2X SERIES

OTA

- Provides a No-Wait OTA System to meet the growing OTA needs
- Double-bank Flash system enables rewriting during program execution

■ Double Bank Flash System

Run from Bank 0 and Download to Bank 1 Run from Bank 1 and keep SW1 in Bank 0



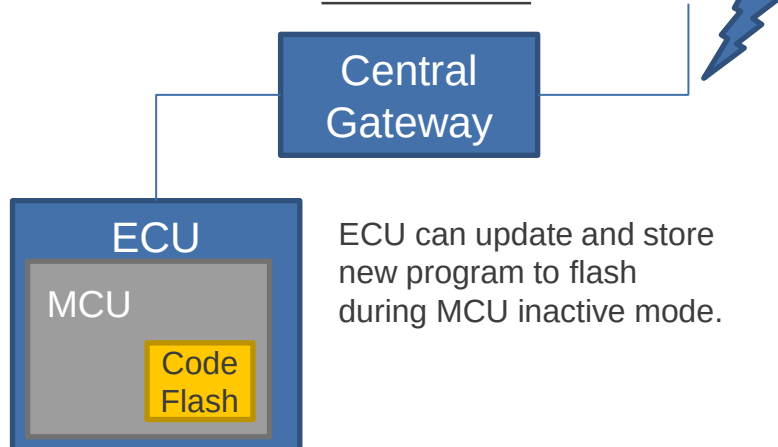
■ Independent Flash access

- Since Flash can be used as an independent two-sided surface, parallel access is possible.

		Flash Bank 0		Flash Bank 1	
		R	W/E	R	W/E
Flash Bank 0	R	✓	✓	✓	✓
	W/E	✓		✓	✓
Flash Bank 1	R	✓	✓	✓	✓
	W/E	✓	✓	✓	

OVER-THE-AIR (OTA) UPDATE CONCEPTS

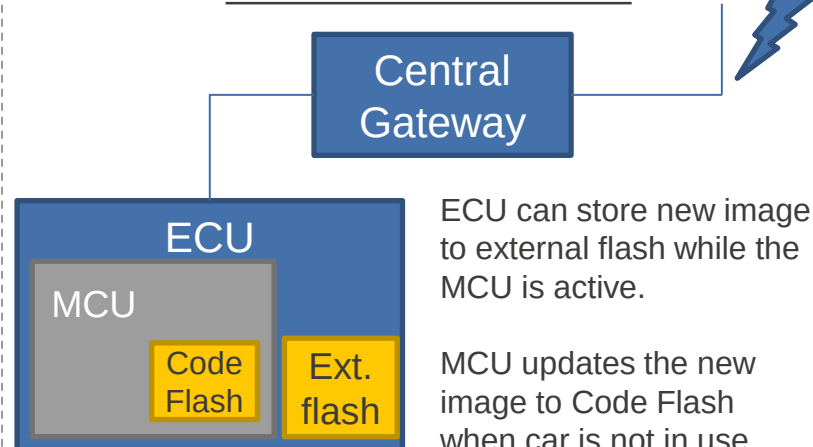
Wait OTA



Existing system

- Pro: No change for hardware
- Con: No back up image in ECU if update fails

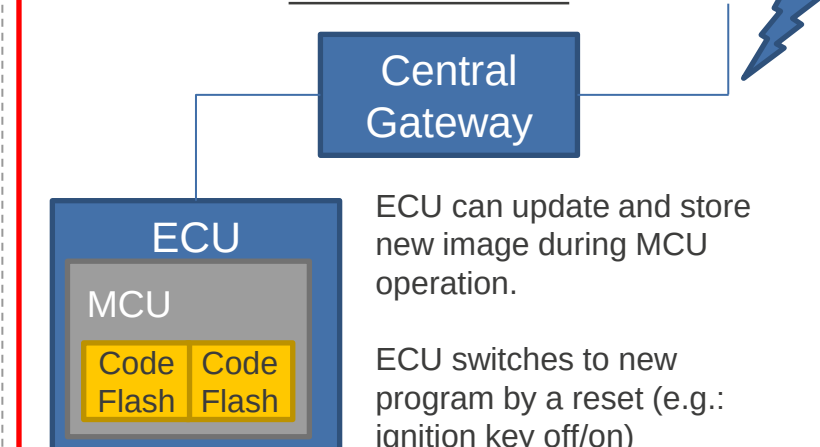
Semi-no wait OTA



External Flash

- Pro: ECU can store new program to external flash during MCU active mode
- Con: No back up program in ECU if update fails

No wait OTA



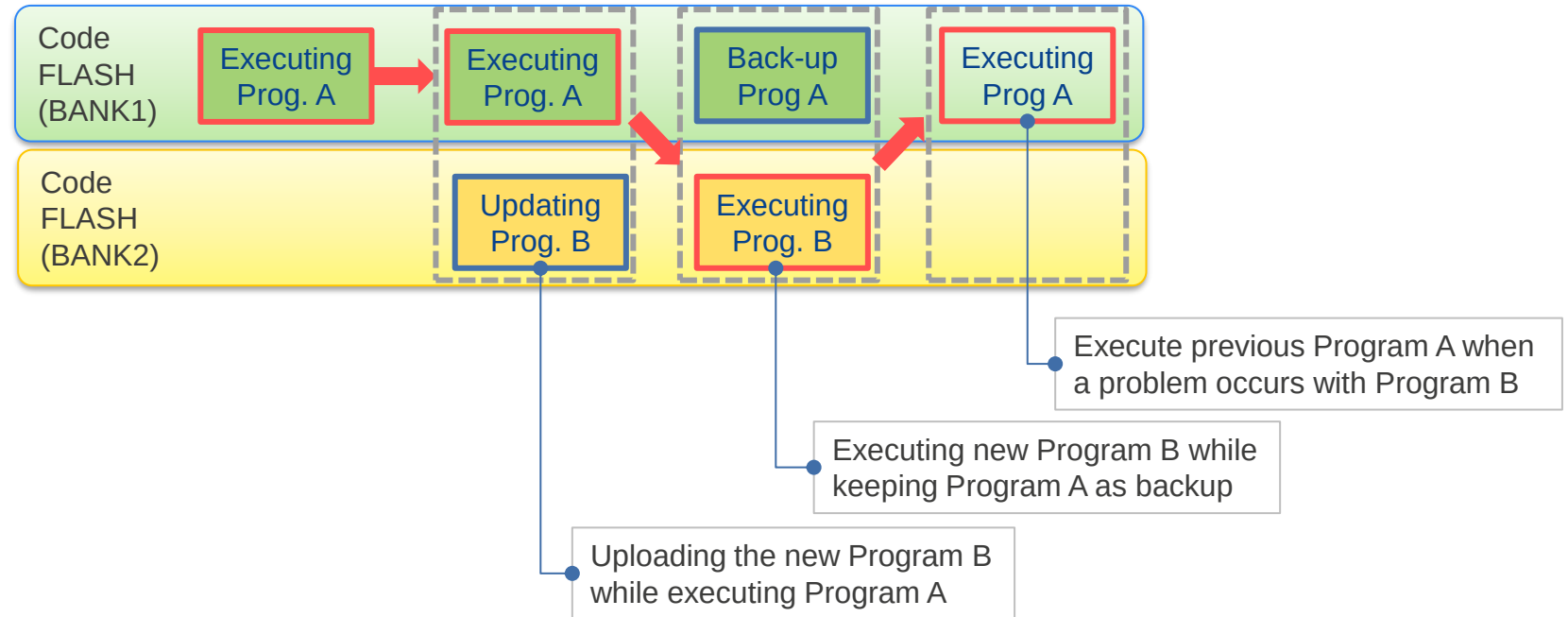
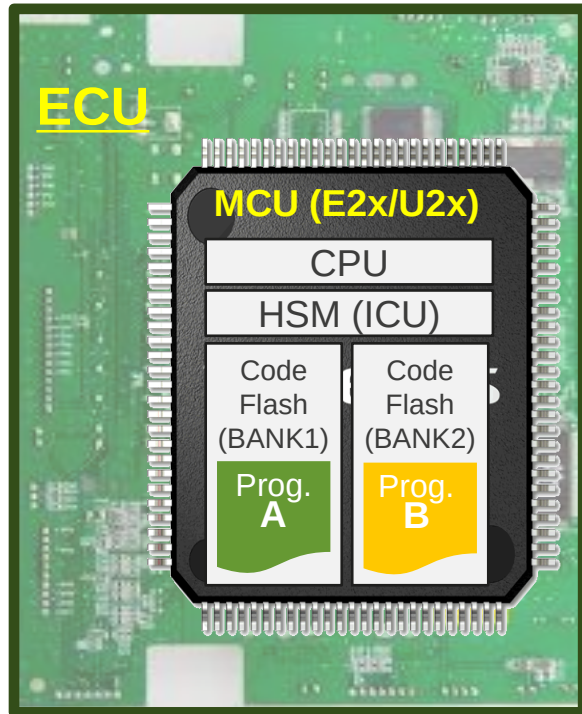
Dual bank embedded Flash

- Pro: ECU can store new image to Code Flash (another bank) during MCU active mode
- Pro: ECU can operate from original code if update fails

Available with RH850/E2x,U2x

OTA TECHNOLOGY (RH850/U2x, E2x)

→ FAST, SECURE AND ROBUST CHANGE TO NEW SOFTWARE



- The complete flash bank is available for OTA
- The flash bank can be used in a flexible way, i.e. OTA can be done with partial or complete bank memory

RH850 - OTA CAPABILITY SUMMARY

COMPARISON OF FEATURES

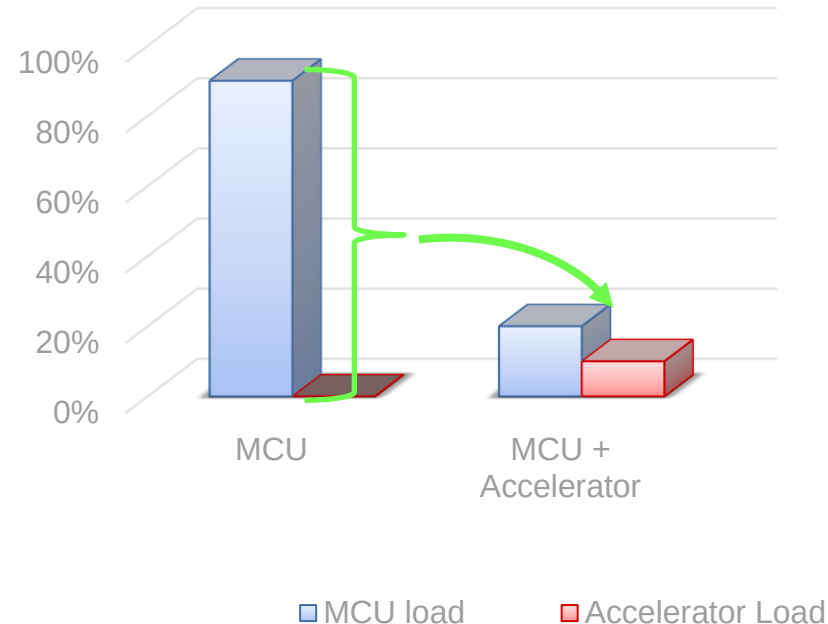


OTA Feature	F1K	P1H-C	E2M	E2UH/H	U2A/U2B
Code Flash (CF) Bank Swap	-	-	-	Yes, including secure CF	Yes, including secure CF
HW Configuration area	Single	Single	Single	Double with robust update	Double with robust update
Double secure CF region	-	-	Yes	Yes	Yes
CF BGO (R while P/E)	Yes (between different banks)	Yes (between different banks)	Yes (between different banks)	Yes (between different banks)	Yes (between different banks)
Application Operation	F1K	P1H-C	E2M	E2UH/H	U2A/U2B
Read from second CF bank	No for P/E of bank B	No for P/E of bank B	Possible while OTA updating	Possible while OTA updating	Possible while OTA updating
Second DF P/E while CF P/E	No	No	No	Yes, second DF is independent	Yes, second DF is independent
DF Read while CF P/E	No	Yes	No	Yes	Yes
DF P/E while CF P/E	No	No	No	Enhanced suspend function	Yes for DF of other Flash system; enhanced suspend function for same Flash system

A.I. HARDWARE ACCELERATOR

A.I. COMPUTATIONAL POWER

- Artificial Intelligence is very computationally intensive.
- This computation required is usually very streamlined.
- MCU cores are designed to be generic and not optimized for pure algorithm execution.
- It is very beneficial to have a HW accelerator that is optimized for dataflow-like computation.



WHY NEURAL NETWORKS OVER TRADITIONAL APPROACH?

- Many physical system are too complex to model with an equation-based approach*
- Neural Networks (NNs) offers an alternative approach, where the correlation between system inputs and outputs is derived by a set of experiments (training) rather than by an analysis of the system

$$\begin{aligned}b_f(x) &= \frac{b_{PM}(x)}{k_c} + b_{sl}(x) \\b_{PM}(x) &= \sum_{\mu} B_g b_{\mu} k_{s\mu} \cos(\mu \frac{\pi}{\tau} x) \\b_{sl}(x) &= -2\gamma \frac{g}{t_1} \sum_{k=1}^{\infty} k_{0k}^2 k_{sk} \cos(k \frac{2\pi}{t_1} x) \\&\quad \times \sum_{\mu} B_g b_{\mu} k_{s\mu} \cos(\mu \frac{\pi}{\tau} x) \\b_{sl}(x) &= -2\gamma \frac{g}{t_1} \sum_{k=1}^{\infty} k_{0k}^2 k_{sk} \cos(k \frac{2\pi}{t_1} x) \\&\quad \times \sum_{\mu} B_g b_{\mu} k_{s\mu} \cos(\mu \frac{\pi}{\tau} x) \\\gamma &= \frac{4}{\pi} \left[\frac{b_{14}}{2g} \arctan\left(\frac{b_{14}}{2g}\right) - \ln \sqrt{1 + \left(\frac{b_{14}}{2g}\right)^2} \right]\end{aligned}$$



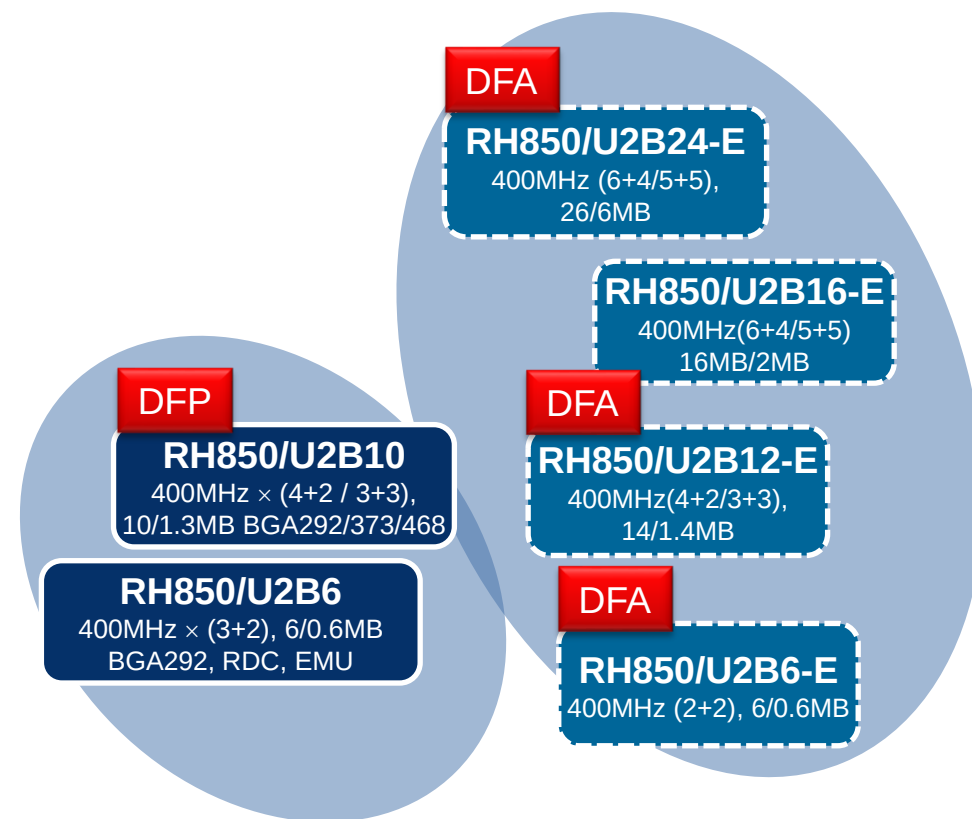
*https://www.researchgate.net/publication/4022645_Analytical_approach_to_cogging_torque_calculation_in_PM_brushless_motors

AI/ML USE CASES

APPLICATION	USE CASE
BMS	• Electro Impedance Spectroscopy (EIS), • Real time SOC/SOH monitoring (Accurate range detection), • Safe fast charging (AI based lithium plating detection with optimised high fidelity cell model)
MPC (Model Predictive maintenance)	• Tire thread wear detection • Brake pad wear detection • E-motor bearing wear detection
Virtual Sensors	• Sensorless or low cost sensor based e-motor, • Current sensor modelling OBC • E-motor position sensor
E-motor control	• Iron losses evaluation • E-motor characterization: building flux maps from limited information

GENERAL INTRODUCTION

- Real-time actuator control applications of the “next gen vehicle motion” categories are challenged by various new requirements that lead to increased algorithm complexity and generate the need for much higher performance levels (e.g. complex control algorithms for traction inverter or EU7 related exhaust control).
- To realize the much higher performance requirements including A.I. and M.L. functions based on conventional cost-balanced MCU solutions, specialized HW-accelerators will be required as co-processors.



DATA FLOW PROCESSOR

NEW MARKET TREND: HW ACCELERATION FOR COMPLEX CONTROL

Concept

- DFP-IP “DR1000C” developed by Denso
- HW-accelerator for fast execution of complex mathematical algorithms
 - Data-level and Task-level parallelism (MIMD)
 - Supporting control theory, signal processing, data-based models, physical models
 - Support of Machine Learning (ML) + Artificial Intelligence (AI)

Advantages

- High performance
 - Much faster than conventional CPU (Harvard) architecture
 - Up to 12.8GFLOPS and 1300DMIPS
- Flexible concept, freely programmable to meet future use cases
- Specifically designed for automotive
 - Filling performance gap between CPU core and highly parallel architecture (e.g. GPU)
 - Safety support to meet ASIL-D
- Established tool support (SDK, MATLAB), 3rd party support

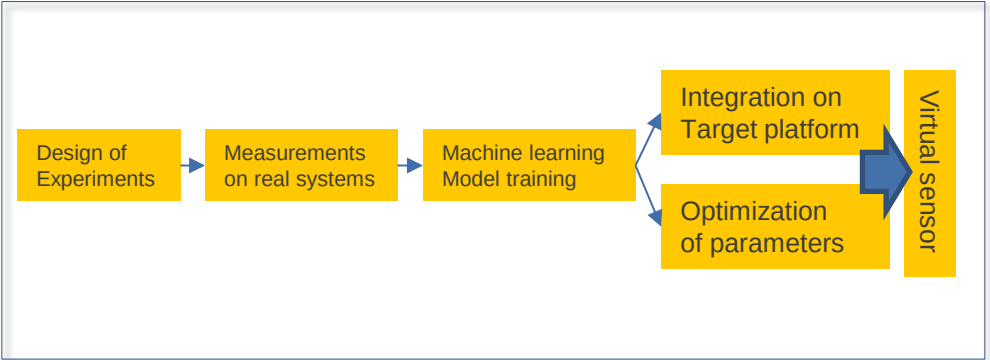


DFP Performance	Details	Total
GFLOPS	VPU: 400MHz x 2 (Mul-add instructions) x 16 (512b SMD)	12.8GFLOPS
DMIPS	SPU: 0.86DMIPS/MHz (per core) x 400MHz x 4cores	1376DMIPS

DFP meets a wide range of future application requirements

HW ACCELERATORS USE CASES

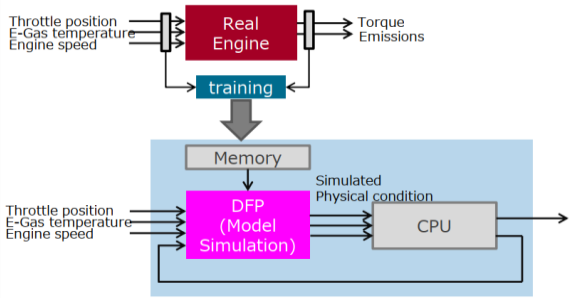
Virtual Sensor



Cost saving by virtual instead of real sensor

Radial Basis Function

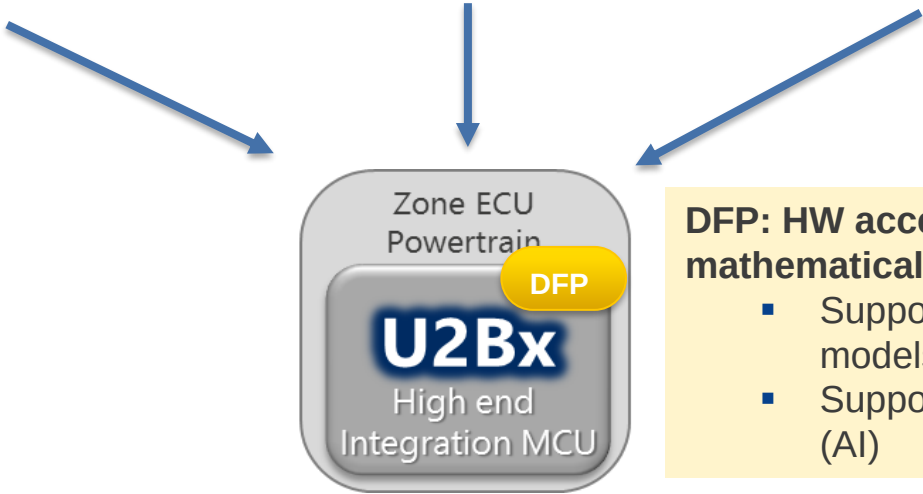
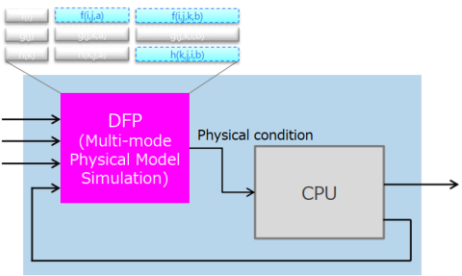
Real-time model running based on calibrated data by Radial Basis Function.



Accurate modeling to achieve EU7 emission targets

Combustion Control

Combustion control based on In-cylinder physical condition model. (e.g. Air inflow)



DFP: HW accelerator for fast execution of complex mathematical algorithms

- Supporting control theory, signal processing, data-based models, physical models
- Support of Machine Learning (ML) + Artificial Intelligence (AI)

DATA FLOW PROCESSOR (DFP)

- USE CASES -

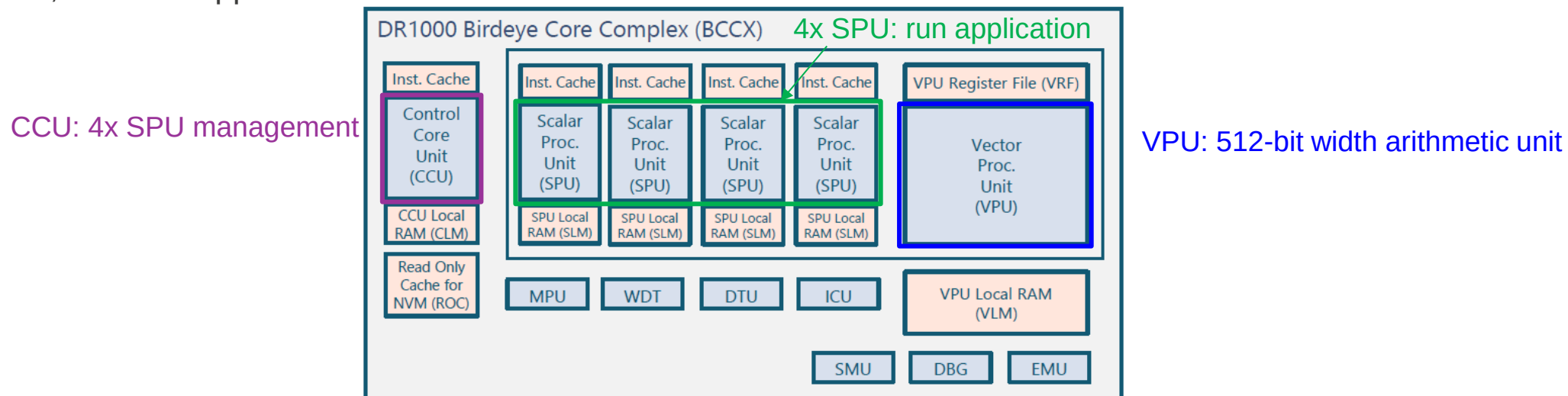
Segment	Application	Algorithm example
Autonomous Drive	Safety mechanism for AI decision	Rule based path planning
ADAS	Vision recognition	CNN
	Radar signal processing	CFAR, FFT
Powertrain (ICE)	Combustion control	Radial basis function
	After treatment control of catalytic converter	Physical model simulation
	Fuel injection control	Injector modelling
xEV / EPS	Brushless-motor: sensor-less position control	Model predictive control FNN
	BMS: State of Charge	Kalman Filter
Gateway	Intrusion detection system	Maximum entropy method

DATA FLOW PROCESSOR DETAILS

DATA FLOW PROCESSOR (DFP)

Concept

- This accelerator has capability high-performance processing using wide-width Vector Unit, focusing mainly on matrix operations, and this IP is provided by Denso (ex-NSITEXE) as Data Flow Processor (DFP: DR1000C).
- DFP has 4x CPU (SPU: Scalar Processor Unit) and 1x VPU (Vector Processor Unit), and VPU can be used highly efficiently to share by multiple cores, improving processing performance.
- Each SPU can execute different tasks independently under CCU management, which allows for complex processing.
- Software service
 - Matrix operation, Neural Network, Kalman Filter and FFT
- DFP has 16ch interrupt from Host CPU and Peripheral IP, the different software can be assigned to each interrupt.
- ASIL-B, ASIL-D support

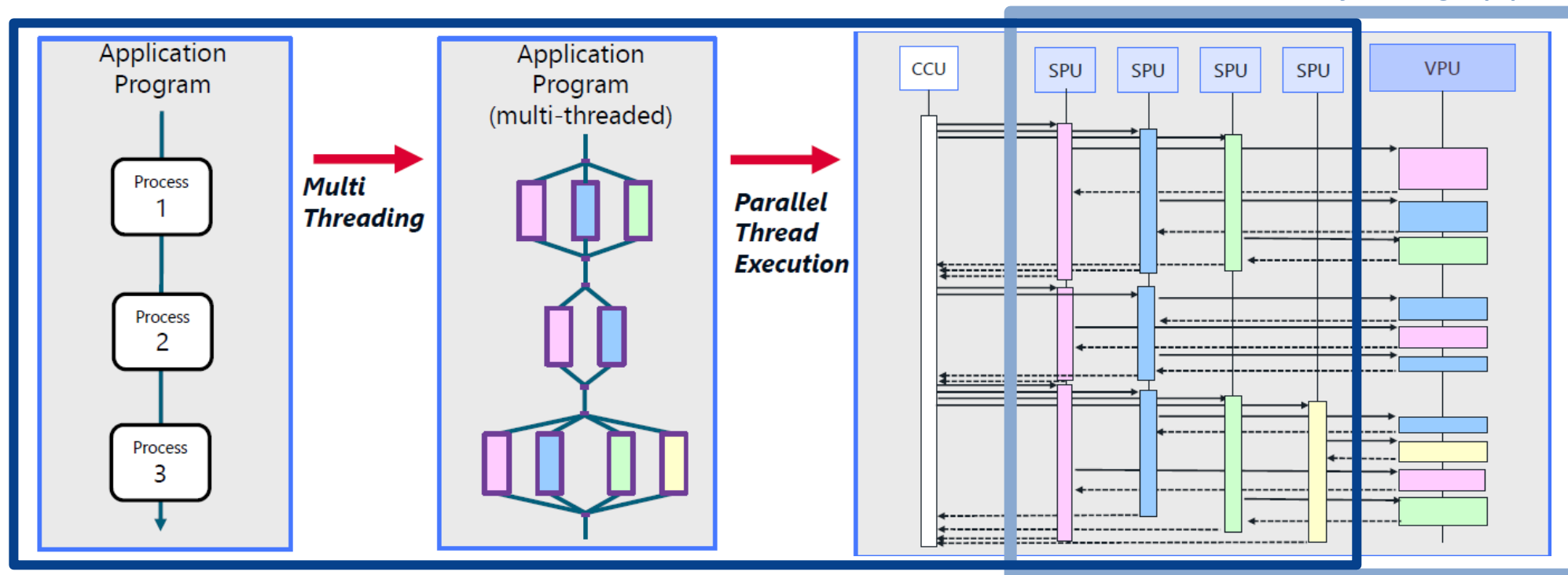


DATA FLOW PROCESSOR (DFP)

EXECUTION OF INDEPENDENT TASKS AND SHARING OF VPU BY EACH SPU

VPU is shared by each SPU and executes vector instructions from each SPU in FIFO order.

➡ VPU can execute continuously at High-pipeline utilization.



Each SPU can execute different tasks independently under CCU management.

➡ This allows for complex processing.

➤ DFP can use VPU efficiently and execute complex processing.

DFP SOFTWARE DEVELOPMENT ENVIRONMENT

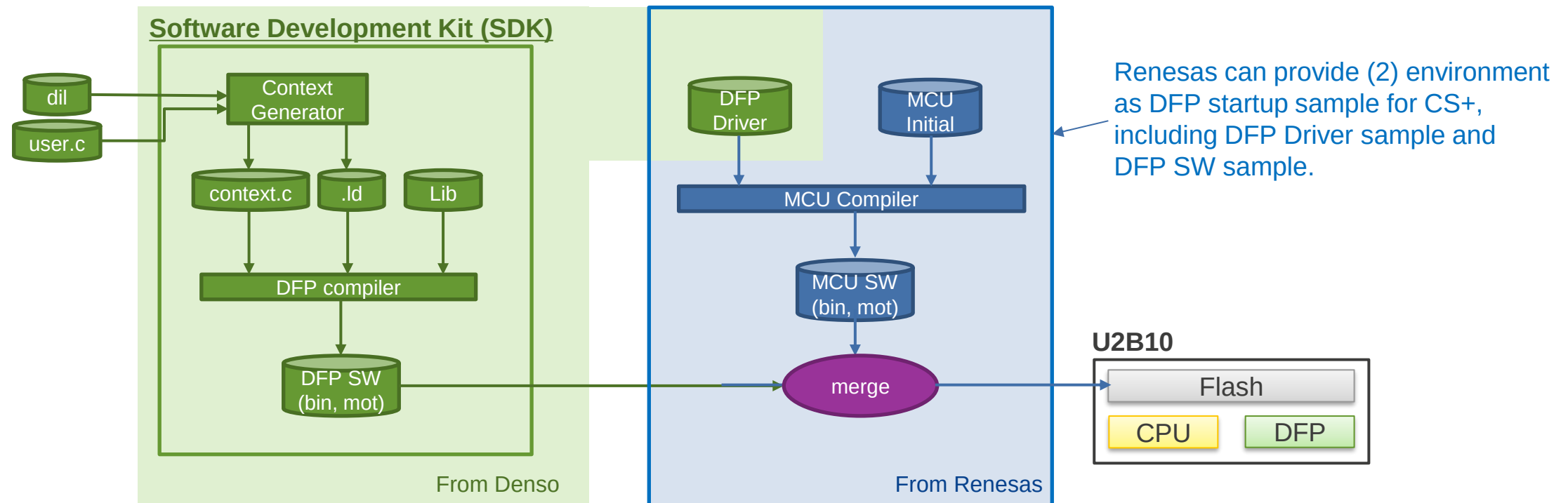
MCU OVERALL ENVIRONMENT

The following block diagram shows the overall DFP software development environment. There are 2 type of environment.

(1) DFP software by DFP-SDK (Software Development Kit) provided by Denso

(2) Host Driver by RH850 Compiler (e.g. CS+, GHS, etc)

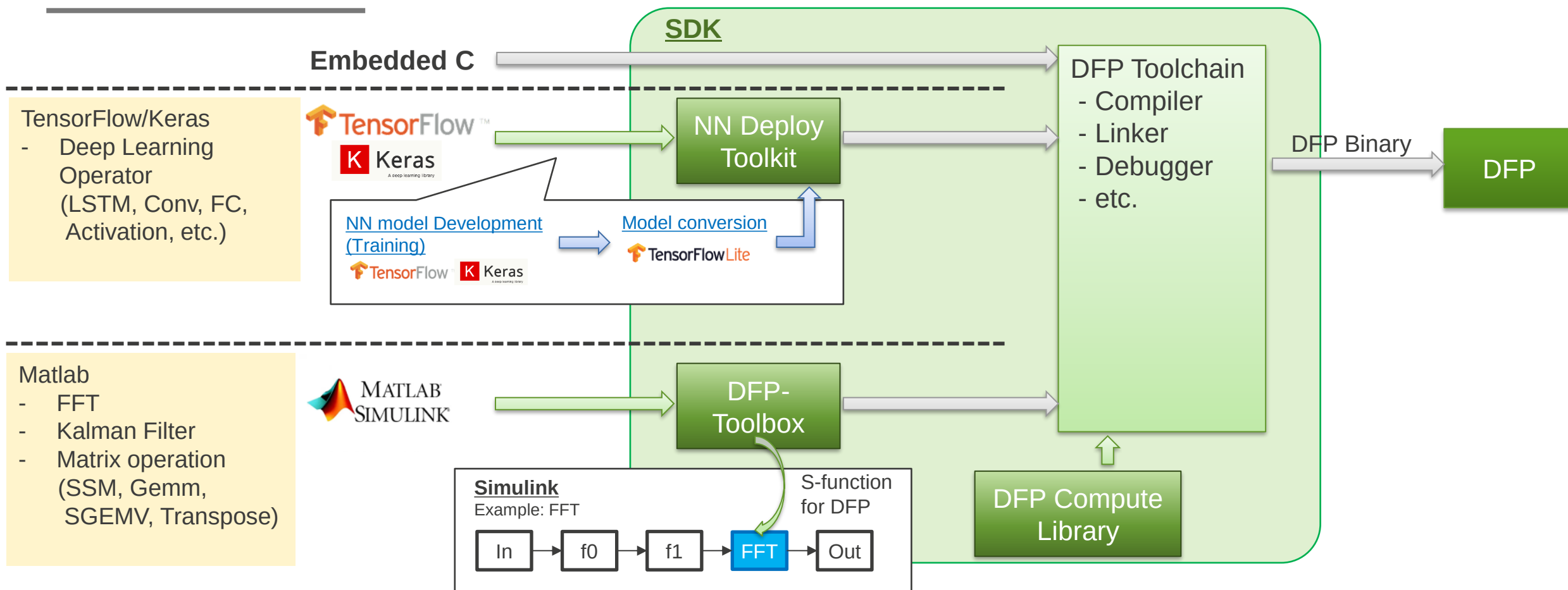
➡ Finally, it is necessary to merge each files ((1)+(2)), because DFP and RH850 core are different.



➤ DFP Software development flow is shown in the next slide

DFP SOFTWARE DEVELOPMENT ENVIRONMENT

DFP SOFTWARE DEVELOPMENT BY SDK



- **DFP SDK supports TF/Keras/Matlab model-based development**
- **For details, please refer to document “DR1000C(DFP) SDK and tools”**

DFP SOFTWARE DEVELOPMENT ENVIRONMENT

DEBUG ENVIRONMENT

Debug Environment	Debug tool			RH850			DFP			Remarks
	GUI (1)	GUI (2)	Emulator	Core debug	Resource Monitor	Trace	Core debug	Resource Monitor	Trace	
Standard Debugging	CS+ Note1	-	E2 Emulator Note1	✓	✓	-	-	- Note2	-	DFP is handled as a peripheral macro of a RH850.
DFP stand alone	SDK	-	E2 Emulator	-	-	-	✓	✓	-	Debugging the DFP requires a dedicated DFP debugger (SDK).
DFP combined with MCU	CS+	SDK	E2 Emulator	✓	✓	-	✓	✓	-	Debugging the DFP requires a dedicated DFP debugger (SDK).

Note1: This can be supported by debug-vendor (e.g. GHS, Lauterbach etc.), not only CS+.

Note2: DFP internal resources (DFPRAM, DFP registers) accessible from the RH850 core can be monitored.

DFP DELIVERABLES

Deliverables				Description/Contents	Note	Status	From	Providing
Manual	NSITEXE DR1000C User Manual			Hardware Manual		Available	Denso	Download from the following web page (Download) URL: https://www.nsitexe.com/en/download/
	NSITEXE DR1000C Software User’s Manual			Software Manual		Available		Please contact Denso for purchasing information and contract (Contact) URL: https://www.nsitexe.com/en/contact/
Development tool	SDK	Standard Package	DR1000C SDK Standard Package	Toolchain (Compiler, Debugger, Linker) ISS Simulator Runtime Thread Scheduler Sample Application		Available		
		Extension Package	DR1000C SDK Safety Package	Software Test Library Software Safety Manual		Available for Evaluation		
			NN Deploy Toolkit for DR1000C	Deployment tool for TensorFlow/Keras model	Includes DFP Compute Library	Available for Evaluation		
			DFP Toolbox for DR1000C	Deployment tool for MATLAB/Simulink	Includes DFP Compute Library	Available for Evaluation		
	CS+			Renesas integrated development environment (IDE)		Available	Renesas	Download from Renesas web page (Download) URL: https://www.renesas.com/us/en/software-tool/cs
	DFP startup sample			CS+ sample environment Include sample DFP Driver and sample DFP SW	DFP SW is created by Denso	Available	Renesas	Please contact Renesas
Training material	DR1000C(DFP) Concept and Features			Introduction of DFP concept		Available	Renesas (Created by Denso)	Please contact Renesas
	DR1000C(DFP) SDK and tools			Introduction of SDK/Tool Concept		Available		

DATA FLOW ARCHITECTURE

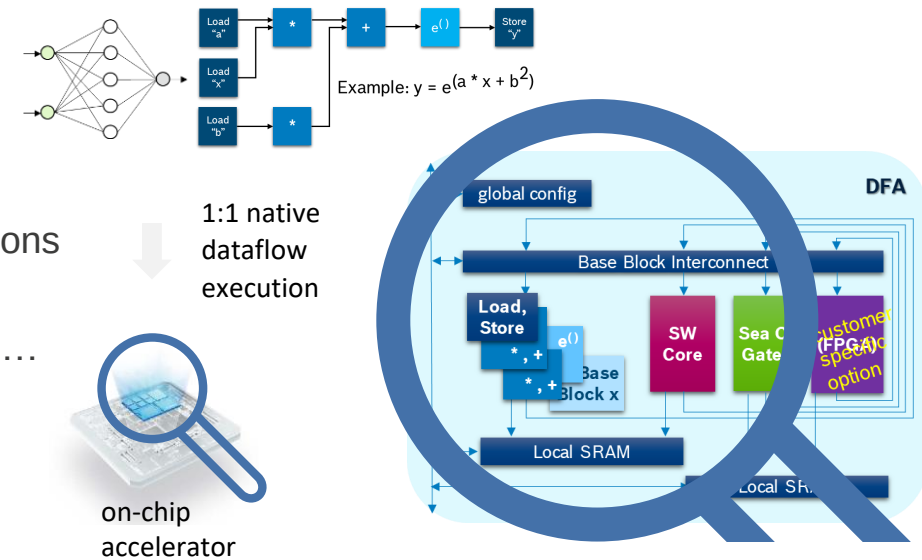
DFA – DATA FLOW ARCHITECTURE

ASIL-D SIMD, AI/ML ACCELERATION

Supported on U2Bx-E line-up

Description

- **Hardware accelerator for advanced mathematic algorithms.**
 - Artificial intelligence (AI), machine learning (ML)
 - Data-based modeling, signal processing, control theory, physical equations
- Native **dataflow execution** in **hardware**
- **Base blocks** for standard operations **LOAD, STORE, MAC, EXP, CORDIC, ...**
- SW Core and Sea of Gates (HW) for additional algorithms as option (not on U2Bx-E)
- **Safety support ASIL-D**



Benefits

- Highest performance
 - Fast calculation e.g., 15 μ s for FFT 512 points, float 32bit
 - Better “performance per size & power consumption” vs. SIMD
- Enabler for new computing intensive functions
 - Real “**AI algorithms** inside” with **floating point accuracy**
 - Flexibility for future algorithms by SW Core and Sea of Gates:

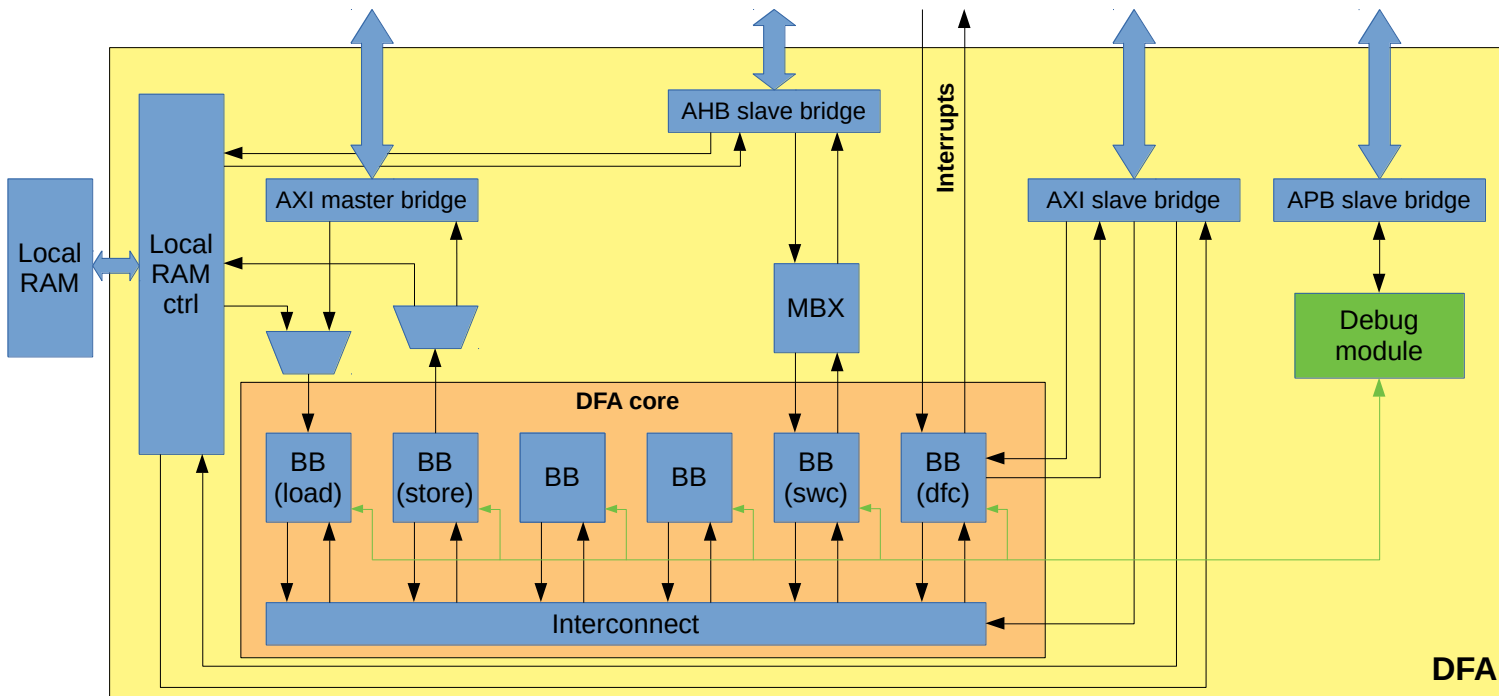
Supported Algorithms

- ✓ **Machine Learning:** Gaussian Process / Bayesian Regression,...
- ✓ **Neural Networks:** MLP, CNN, RBF, RNN..
- ✓ **Signal Processing:** FFT, FIR, IIR...
- ✓ **SIMD Acceleration:** matrices and vectors operations
- ✓ **Control Theory:** Solver for linear equations, Matrix inversion, Cholesky decomposition

DFA – DATA FLOW ARCHITECTURE

PRINCIPLE ARCHITECTURE

- Distributed architecture where several **BaseBlocks** operate in parallel (up to 14 complex operations per cycle).
- High-Bandwidth Local RAM to store input/output data.
- Intermediate data is moved directly between **BaseBlocks**, increasing processed data bandwidth far beyond RAM (or a traditional register file) bandwidth capability.
- Expandable computational capabilities using a **SoftWare Core** (not in U2Bx-E).

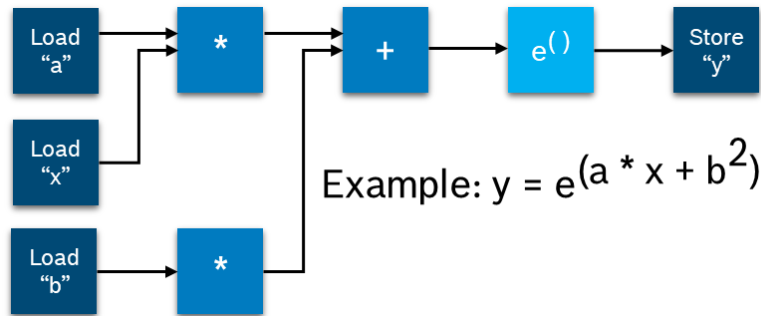


- μ C integration
 - Master/Slave peripheral
 - μ C Core can access global configuration, base blocks, SRAM
- Base Blocks (BB)
 - De-central intelligence
 - BBs send data directly to BBs
 - Data triggers execution
 - Interconnect routes data (between bus and crossbar)
- Dedicated Local SRAM
 - Banking for high throughput

DFA – DATA FLOW ARCHITECTURE

DATAFLOW CONCEPT

- Computing intensive algorithms are dataflow like → not well suited for standard $\mu C + \mu P$
- Sequence of operations and exchanged data



- Predefined **Base Blocks (BBs)** for base operations, adaptable functionality
 - 8x Load, 4x Store, 4x MAC, 4x AddSub, 2x EXP, 2x CORDIC, 1x DFC
- **Interconnect** connects BBs, forms dataflow
- **Fully pipelined** structure leads to an optimal throughput

DFA – DATA FLOW ARCHITECTURE

DIFFERENT ALGORITHMS EXECUTION WITH/ DFA

Algorithm	More Info on Algorithm	CPU only time (us)	CPU+DFA* time (us)	Time reduction
MLP	4-layer Neural Network, each layer being fully connected with sigmoid activation function. Total Parameters: 9953	377	62	83.55%
CNN	One layer of the CNN, Total Parameters: 864	16700	6800	59.28%
RNN	The selected Recurrent Neural Network implements a series forecast. It uses 6 times steps, with an input width of 25 samples, thus having a final output width of 20.	361	44	87.81%
RBF	Gaussian function was chosen as radial basis function. Total sample points: 1296	186	65	65.05%
FIR	The implemented Finite Impulse Response filter uses an input set of 400 samples with 75 taps, leading thus to a result width of 326 samples.	821	265	67.72%
FFT	The Cooley-Tukey FFT is the most common implementation of a Discrete Fourier Transform (DFT). This DFT has been used during the implementation. For this reference implementation, the input data is considered to be a set of 1024 complex single-precision floating point samples.	597	279	53.27%
VNorm	The Vector Normalization algorithm implements an array of simple scalar operations of the type $A * B + C$ with arrays containing 200 data items.	7	5	28.57%
MatrixMult	The implemented matrix multiplication algorithm performs multiplications between matrices of size 10 x 10.	55	20	63.64%

* All performance was measured while running the CPU at 400MHz, and the DFA HW accelerator at 200MHz. The software was compiled by using Green Hills Multi with compiler version 2021.1.5 and optimization flag `-Ogeneral`.

DFA – DATA FLOW ARCHITECTURE

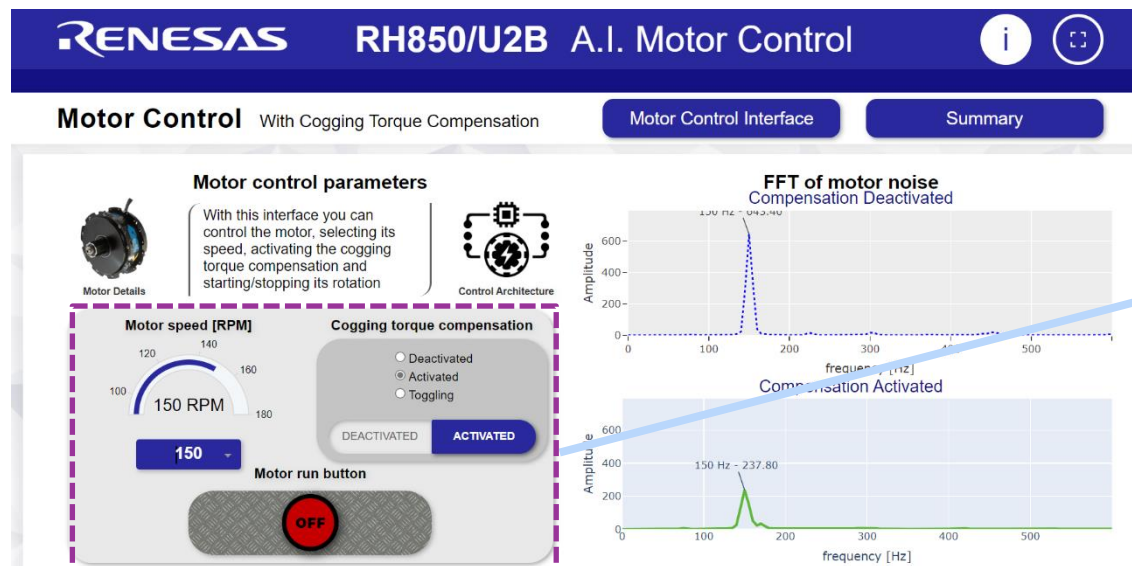
USE CASE EXAMPLES

Environment	Application	Algorithm
xEV vehicle	Insulation Monitoring Impedance measurement is used in xEV system to evaluate the insulation degree in high voltage system. For safety reason, EU and NA regulations that the high-voltage and low-voltage section of a system have to be isolated by a minimum impedance value.	FFT Fast Fourier Transform
xEV vehicle & thermal engine	Sensor Emulation Build of a mathematical model of a sensor needed to control a system rather than having a physical one.	RBF Radial Basis Function
Cyber security	Intrusion Detection System (IDS) Modern vehicles, including autonomous vehicles and connected vehicles, are increasingly connected to the external world, which enables various functionalities and services. However, the improving connectivity also increases the attack surfaces of the Internet of Vehicles (IoV), causing its vulnerabilities to cyber-threats. Neural networks can implement a learning Intrusion Detection System (IDS) can be used for IoV systems, specifically using using convolutional neural networks (CNNs)	CNN Convolutional Neural Network

DFA POC - SHOWCASE

- The PoC showcases E-Motor Vibration Control (via compensating Cogging Torque) by executing the NN based Torque compensation algorithm on DFA IP (U2B10-FCC chip used).
- Function: Via the GUI, the Torque compensation can be activated and deactivated. The GUI displays the frequency spectrum of the vibration in each case.
- Also, E-Motor noise reduction can be heard clearly as well as felt by touching plexi-case of the PoC

Cogging torque is the interaction between the permanent magnets in the rotor, and the slots of the stator. Because the rotor magnets are attracted to the stator teeth, the torque required to move the rotor changes with the relative position of the rotor to the stator. (Wikipedia)



INTEGRATION PLATFORM / ZONE-ECU

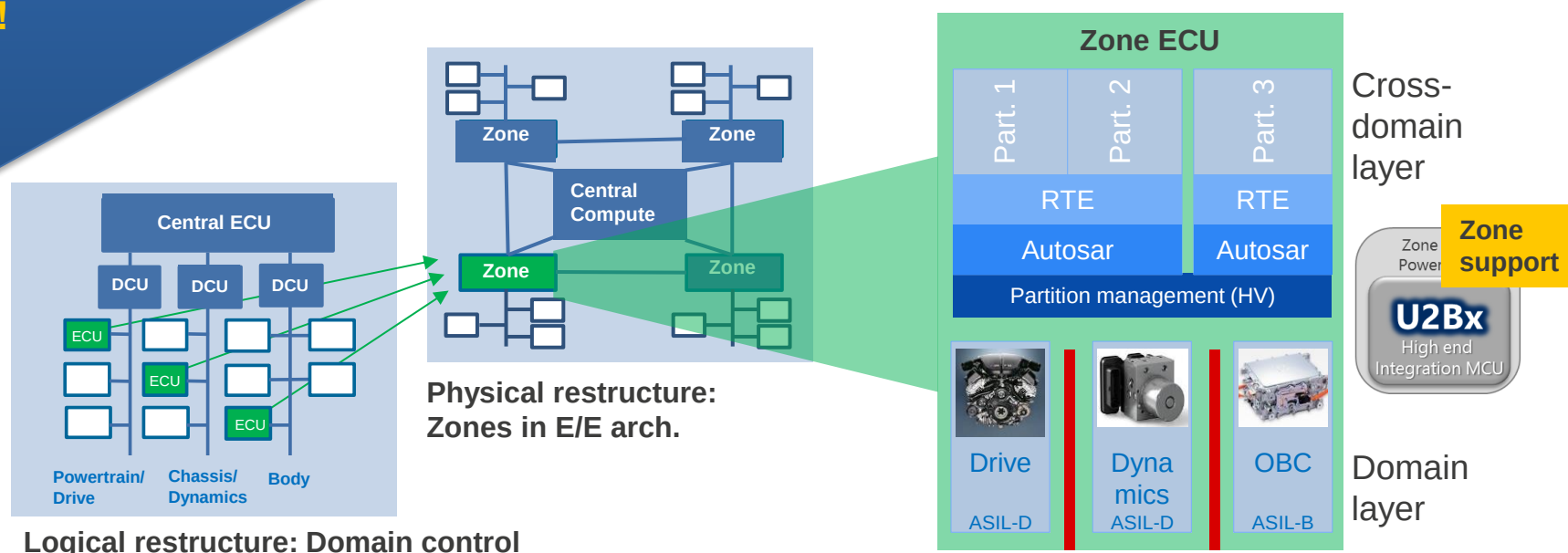
NEW MARKET TREND: INTEGRATION PLATFORM

MEGA-TIER ONE'S PLANNED INTRODUCTION FROM 2025/26 ONWARDS

Motivation

- Fewer ECUs in the car → Lower network complexity
- Reduction of wiring harness → Lower weight
- Reduced latency → Better control performance
- and of course, ... cost down!

Integration of applications as well as the new Cross-Domain approach for upcoming E/E architectures are causing a disruptive market change on a world-wide OEM basis, already today.



Freedom from Interference realized by QoS, MPU, Guards, Safety, Security

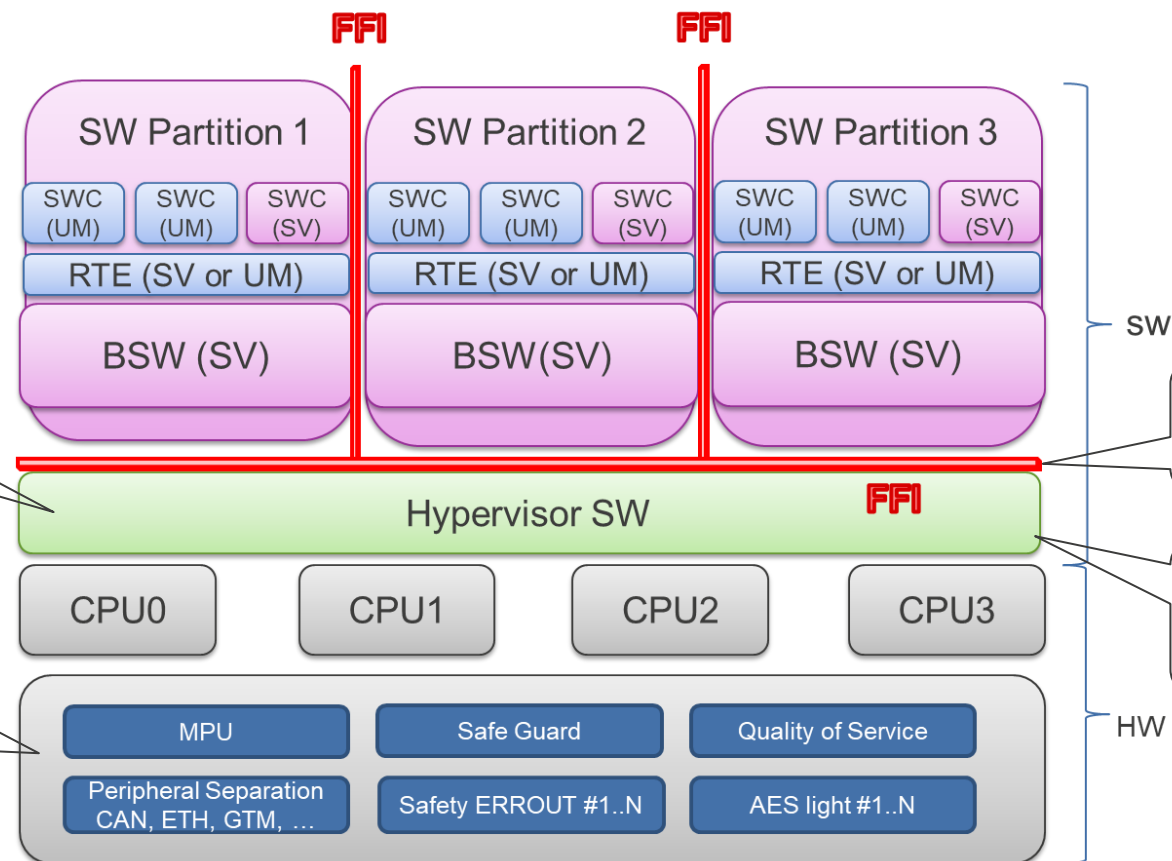
INTEGRATION PLATFORM - SW/HW CONCEPT ON MCU



Built-in Hardware Support for Hypervisor

- Privilege mgmt.
- Flexible Interrupt models
- Fast context switch
- Performance measurement

Built-in Hardware Support for System and Peripheral related Freedom-From-Interference



Freedom from Interference (FFI) between Applications secured by Hypervisor

HV-Software partners



Ensuring Space separation, Time separation and Real-timeness, to meet the requirements of ECU Integration of the Zone Era

RH850 VIRTUALIZATION

RH850 can realize a practical virtualization in automotive control by space/time separation systems where real-time-ness is prioritized.

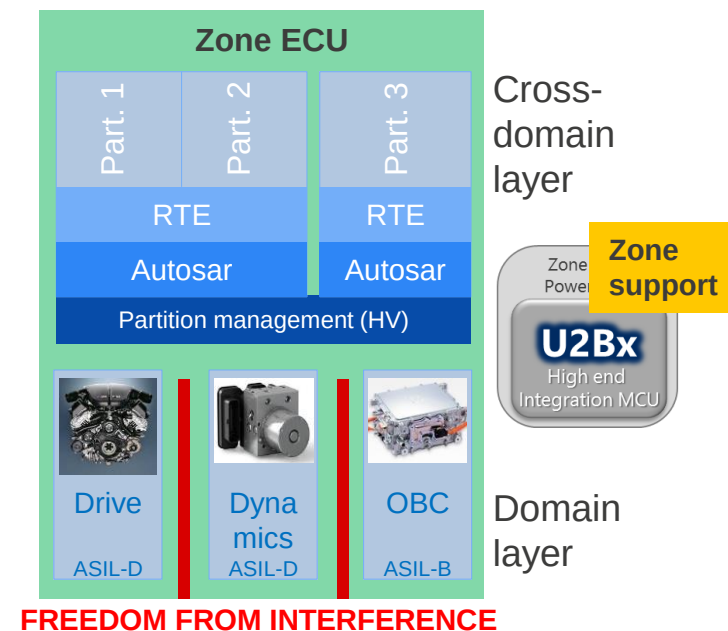
Items	RH850 virtualization-related function
<u>Space separation</u> Partitioning mechanism sufficient for multiple app separation	Management of entire systems by system protection ID
	Ring protection by CPU operating mode and privileges
	Separation of memory protection by operating mode
	Fine-grained (each ch) guard mechanisms on the slave side to separate peripheral functions
<u>Real-time-ness</u> Real-time-ness prioritized interrupt model High-speed VM switching	Interrupt model to Foreground VM without Hypervisor transition
	Interrupt transition to Background VM
	VM context high-speed save/restore instructions
<u>Time Separation</u> Execution time management mechanism to realize safety systems	VM execution time management by performance measurement function
	Measurement of how many times interrupts are accepted by the performance measurement function
	Low-latency timer function for VM monitoring/scheduling

U2B ZONE INTEGRATION SUPPORT BY HARDWARE

FEATURE OVERVIEW

U2x supports all necessary key HW-features to realize the integration of multiple SW-applications (partitions) while maintaining an utmost level of **FREEDOM FROM INTERFERENCE (FFI)**.

1. Hypervisor HW-assist function to enable a HV-OS in a high performance manner (fast context switching, HV interrupt concept) → **Value: Lower HV SW overhead, better real time capability**
2. Quality of Service (QoS): Latency monitor and regulation function for all bus masters to ensure minimum bandwidth → **Value: Full FFI achievement by HW, lower homologation cost**
3. Communication modules well distributed over peripheral bus groups to ensure best possible conflict-free access from SW-partitions → **Value: Full FFI achievement by HW, lower homologation cost, no need for SW-based protection features**
4. Safety: Individual Error output signal generation per partition to ensure individual treatment according to FuSa level → **Value: Full FFI achievement, lower homologation cost**
5. Security: Multiple instances of AES128 lockstep modules for conflict-free and deterministic secure/safe communication → **Value: Possible to meet fast boot time requirements even for the 26MB device. Meeting OEM requirements for secure communication speed.**
6. CAN XL: → **Value: Future proof interface to meet OEM requirements for larger payloads with deterministic messaging**



INTEGRATION PLATFORM – POC BY RENESAS



- **Webpage link:** [Zone-ECU Virtualization Solution Platform | Renesas](#)
- **Introduction video:** [RH850/U2x Virtualization Development Platform | Renesas Electronics Corporation](#)

Support Devices

RH850/U2Bx	32bit MCU
RH850/U2A16	32bit MCU

Recommended Boards

U2B MCU Piggyback Board
U2A MCU Starterkit

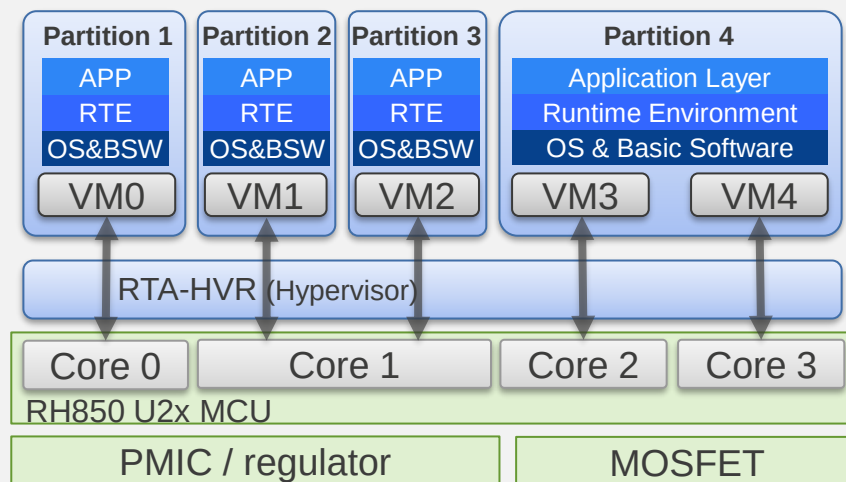
Recommended Devices

*RAA271084**2 ASIL D PMIC for RH850/U2x
*RAA271082**2 ASIL B PMIC

*2 Not contained in the reference board

System Concept

The Hypervisor allows a single microcontroller to run several Virtual Machines (VMs) independently and in parallel



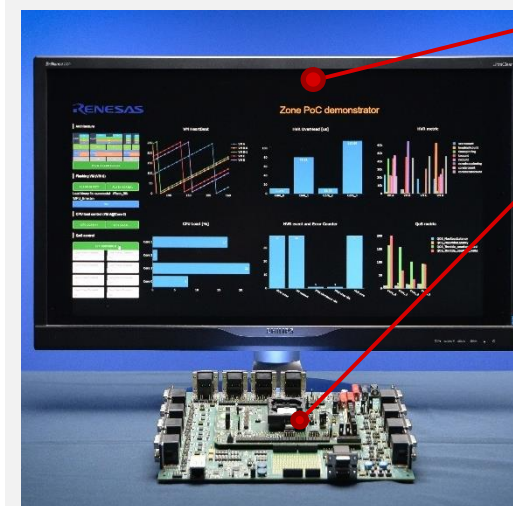
Software*1

*1 Eval. licenses required

- RTA-HVR (Hypervisor) Prototype
- RTA-CAR (Classic AUTOSAR SW Stack)
- CycurHSM (Security stack for ICU-Mx)
- Renesas U2x MCAL



Virtualization Solution Platform



GUI

- Based on Python 2.7 32 bit
- Tkiner GUI interface

Evaluation Board options

1) RH850/U2A16 Starterkit



Release planned in Q3 2022

2) RH850/U2Bx Piggyback board



Available

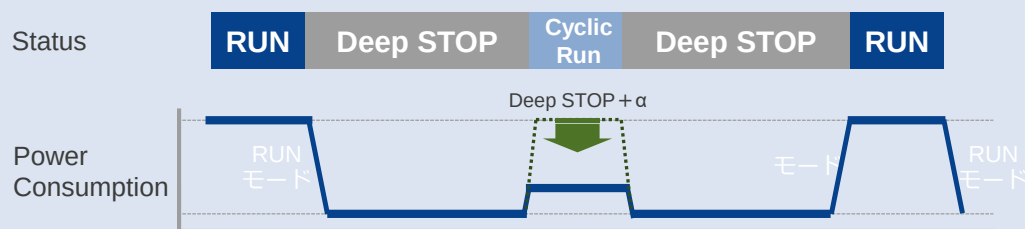
RH850/U2BX-E SERIES

LOW POWER TECHNOLOGY

- Highly efficient power consumption contributes to "greener" systems for both ECUs and vehicles
- Deep STOP mode allows wake-up **without using the CPU**. Achieves lower power consumption operation
- Proven 28nm process ensures low power consumption even in RUN mode

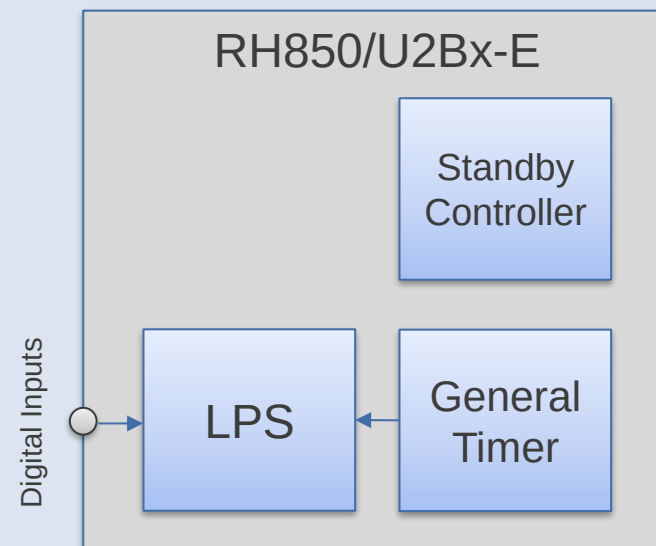
■ Standby power reduction by Deep STOP mode

- Deep STOP mode does not use the CPU for low-power standby operation. It is a mode that allows only peripheral functions to operate.
- The sequencer that operates during standby enables even lower power consumption operation.



■ Low Power Sampler (LPS)

- To monitor the external input without consuming CPU resources
- LPS can check the digital input ports
- User can set the interval time for checking

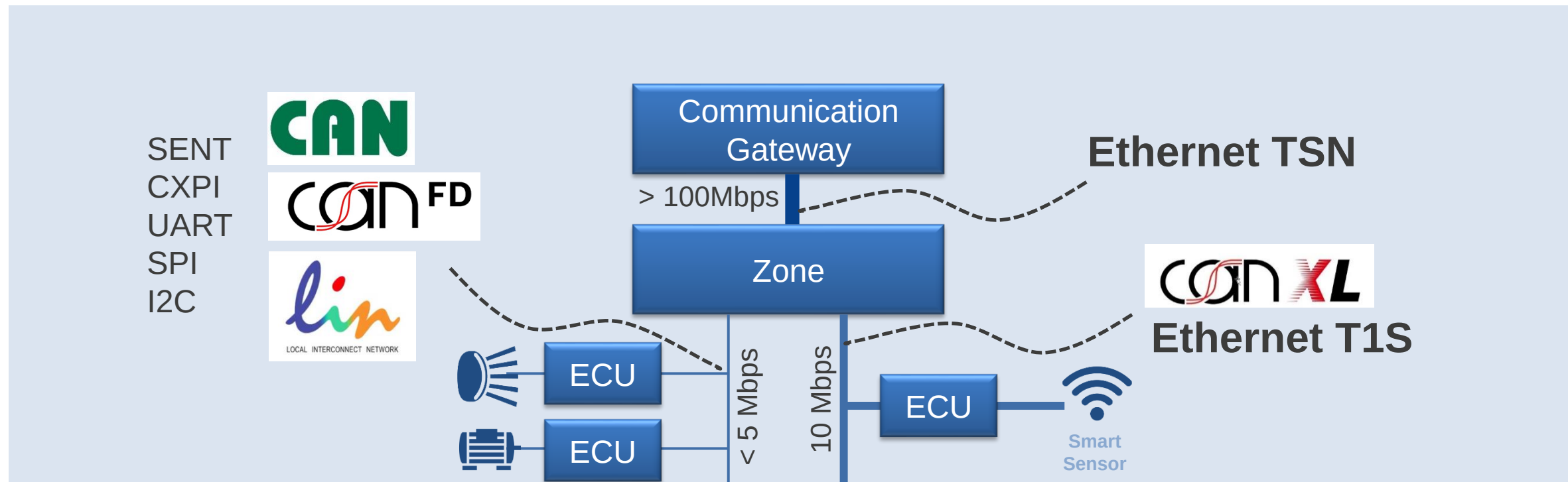


RH850/U2Bx/U2Bx-E

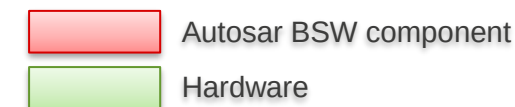
RICH COMMUNICATION INTERFACES

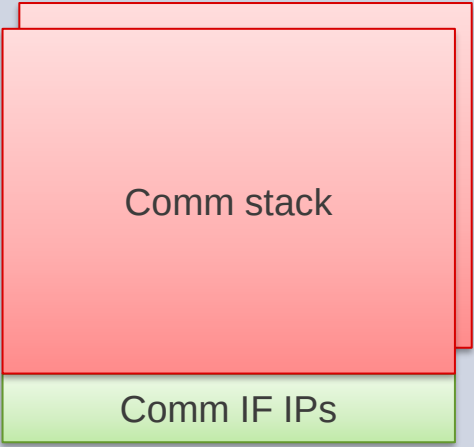
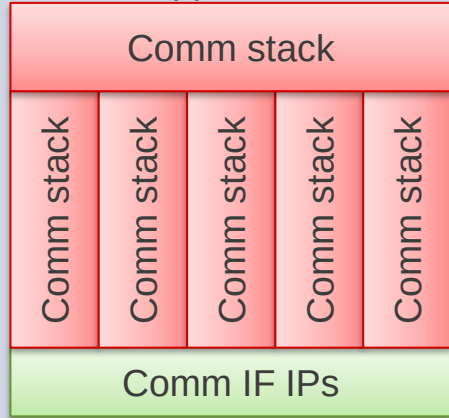
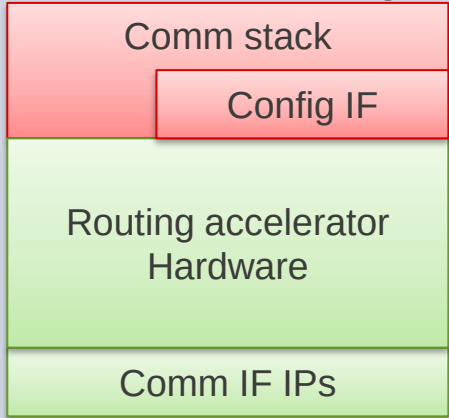
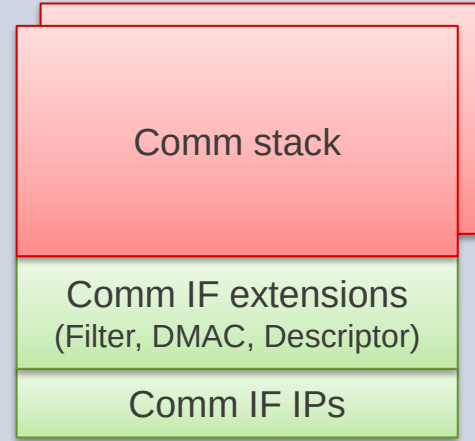
- Supporting communication interface needs as E/E architectures change
- Supports the latest interfaces: Ethernet TSN (100Mbps), Ethernet T1S, CAN-XL
- Supports legacy interfaces: CAN / CAN-FD, LIN, UART, CXPI, SPI, I2C

■ COMMUNICATION INTERFACES



WAYS ADDRESSING ROUTING DEMAND



Software based approach	Local CPU based approach	Dedicated Hardware based acceleration	Standard hardware based acceleration
+ Standard workflow + Hardware independent + Flexible and known method - High CPU load/ low latency - Limited usage of existing acceleration function 	+ ???? - No standard flow - A lot of IPC - Device dependent - Inflexible resource assignment - Firmware approach 	+ Zero CPU load for routing done in HW + Low latency - No standard flow - Device dependent - Inflexible resource assignment 	+ Integrated Autosar workflow + No specific hardware/ device required + Offloading of trivial tasks to CPU - Device dependent 
Standard Autosar approach	Supplier A approach	Hardware solution approach	Renesas' preferred approach for complex gateways

FUNCTIONAL SAFETY AND SECURITY SUPPORT PROGRAM FOR AUTOMOTIVE

FUNCTIONAL SAFETY AND SECURITY

SUPPORT PROGRAM FOR AUTOMOTIVE



Hardware

Safety/Security mechanisms
MCU, SoC, A&P



Software

CPU Core self-test
Safety/Security Software



Work products

Safety/Security analysis tool,
Report, etc.



Consulting

Workshop
Development support



SAFETY

RENESAS' FUNCTIONAL SAFETY ACTIVITIES

→ LONG EXPERIENCE IN FUNCTIONAL SAFETY FOR AUTOMOTIVE

Compliance of ISO26262 in a company organized manner

- Establishment of safety life cycle
- Involvement of safety activities in internal standardized development flow

Establishment of internal assessment organization

- Independent (I3 level) internal organization
- Confirmation measures at appropriate development stages
- Active members of ISO26262/ ISO19451 / IEC61508 expert in internal assessment organization

MCUs that comply to ISO26262 and evolution to SoC, A&P

- Worldwide No. 1 experience in semiconductor devices for automotive application
- Safety technologies reviewed and accepted by worldwide customers
- Efficient and sufficient safety mechanisms with company-owned technology

Strong influence in relevant standards

- ISO26262 WG08* / ISOPAS 19451 (UK / Germany / Japan)
- SAE (Germany), JSAE (Japan), Jaspar (Japan)

WP: Work Products

A&P: Analog & Power

* WG08 is the former WG16

SAFETY SUPPORT PROGRAM MENU (RL78 / RH850 / R-CAR)

Services	Price	
	RL78 / RH850	R-Car
Standard FuSa Deliverables: - Development Interface Report/Agreement (DIR/DIA) - Safety Requirement Specification (SRS) - Safety Application Note (SAN) - Safety Case Summary (SC) - Functional Safety Assessment Report Summary Version (FSA) - FMEDA (CAR Tool) *1 - Question & Answer on deliverables (excluding FMEDA, best effort)	Free This is delivered along with ASIL hardware product based on request and NDA.	
Safety Technical Support: - Per one year for one window person at customer (during business hours) - Per RL78 / RH850 / R-CAR Product Series *2 - Ticket based support (e.g., JIRA or support email) - Detailed discussion on safety requirements of the Renesas SEooC products - Hands-on support on CAR Tool - Calls/meetings (limited to 1h per week) This does not include custom topics (e.g., FMEDA contents modification based on customer's use case, advising on compliance to ISO26262 at system level). *3	 / (year·person)	 / (year·person)

*1: Regarding CAR Tool version (static/dynamic) and capability, please refer to page [FMEDA](#). Released “as-is” without warranty/liability.

*2: Regarding the product series definition, please refer to page [Series](#).

*3 :For further System FuSa consultancy and customization service of FMEDA, please check our Collaboration Partner: [Consultancy](#)

RENESAS SOFTWARE SAFETY SUPPORT

SW Licenses	(DLA / ELA ^{*1}) Development License Agreement Evaluation License Agreement	(MPL / SLA ^{*1}) Mass Production License Software License Agreement
SW Functional Safety Work Products	No ^{*2}	Yes
SW Functional Safety Support	No Support	Support ^{*3} on FuSa Work products content

^{*1}. DLA and MPL are license agreements for RH850 products. ELA and SLA are license agreements for R-CAR products.
Evaluation and development are permitted under DLA; for qualification and production purposes a valid MPL is required.
Evaluation is permitted under ELA; for development, qualification and production purposes a valid SLA is required.

^{*2}. Sample FuSa Work Products can be provided under DLA with additional Memorandum of Understanding (MoU) for RH850.

^{*3}. Covered by support and maintenance agreement (SMA)

FUNCTIONAL SAFETY

- SCOPE OF SUPPORT PROGRAM -

System safety concept
& Design information



Functional safety support program

◆ Assumed safety requirements

- ◆ Safety application note
- ◆ Verified SW driver

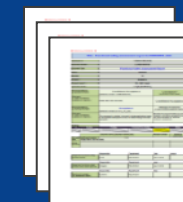


◆ Renesas device safety analysis data base

**FMEDA tool
(CAR Tool)**

◆ Consulting

◆ HW work product

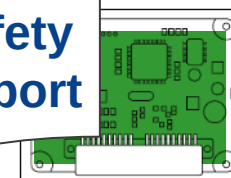


◆ SW work product

System SW

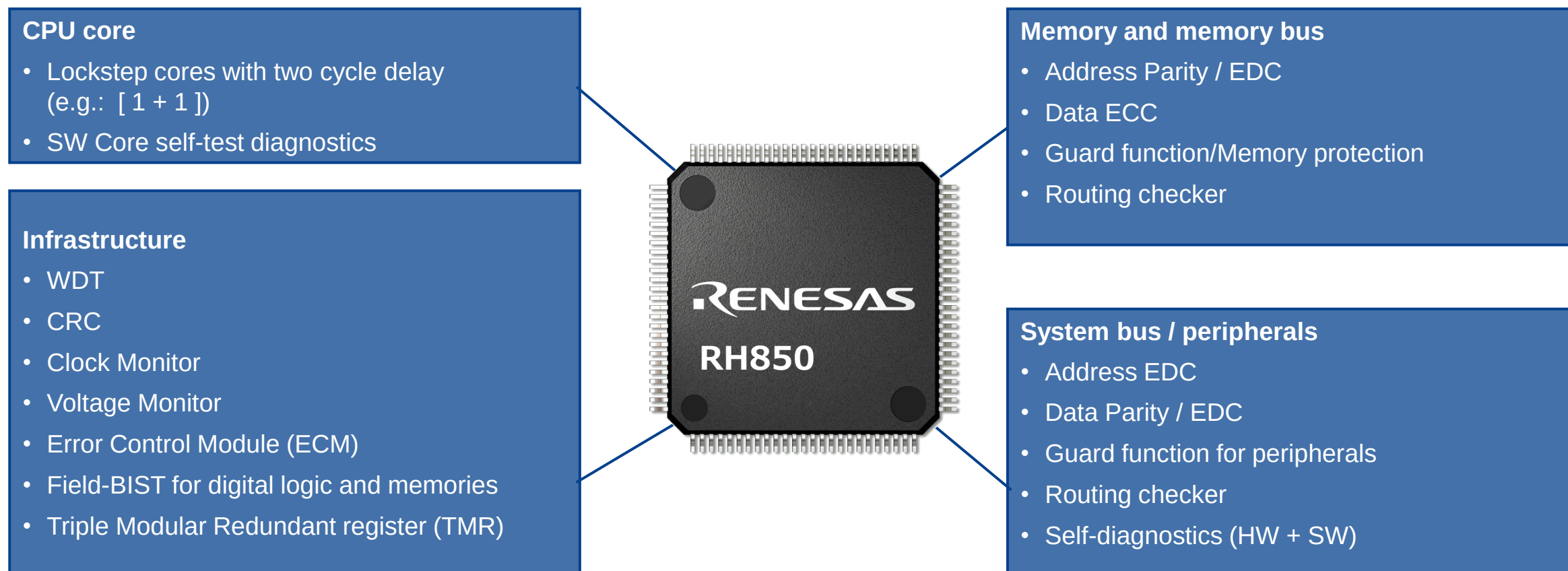


System safety
analysis report



FUNCTIONAL SAFETY

SAFETY MECHANISMS AGAINST ELEMENTS IN MCU



SECURITY

COMPETENCY IN SECURITY

Security Know-How

- Long expertise in smart card and embedded security
- Review of SHE specification
- Member of the Security WG within the European C2C-CC
- Dedicated security expert team in Renesas
- Established working relationship to major OEMs
- Established infrastructure to handle sensitive security designs

Customer Projects

- Custom product for locking applications at European OEM for three device generations
- Support for OEM's additional features for SHE in ICU-S IP
- Cooperation for ICU-M / HSM with leading European Tier 1
- Design-in of security enabled products for chassis, body, cluster, and gateway applications



ASSP for
key/lock

key/lock ASSP



RH850/F1x
RH850/P1x
RH850/D1x

ICU-S
SHE-compliant IP

2013



RH850/F1x
RH850/E2X

ICU-M
Embedded cryptographic MCU (HSM)

2015



RH850/U2A
RH850/U2B

ICU-MH
Embedded cryptographic MCU (HSM)
ECC, SHA-2, SHA-3*

2018

*U2B only

MARKET REQUIREMENTS

Guidance of security functions required for an Automotive HSM (Hardware Security Module)

	AUTOSAR (HIS) SHE	EVITA Light	EVITA Medium	EVITA Full
Assumed Use Case	Automotive ECU	Sensor/Actuator ECU	In-Vehicle ECU	V2X/GW ECU
Internal CPU	No (State Machine)	No (State Machine)	Yes	Yes
Code Flash	No	No	Yes	Yes
RAM	Yes	Optional	Yes	Yes
NVM	Yes	Optional	Yes	Yes
HW Crypto Algorithms	AES-128	AES-128	AES-128	AES-128 ECC-256 SHA-2*
Random Number Generator	AES-PRNG w/ TRNG seed	AES-PRNG w/ external seed	AES-PRNG w/ TRNG seed	AES-PRNG w/ TRNG seed

EVITA : <https://www.evita-project.org/>

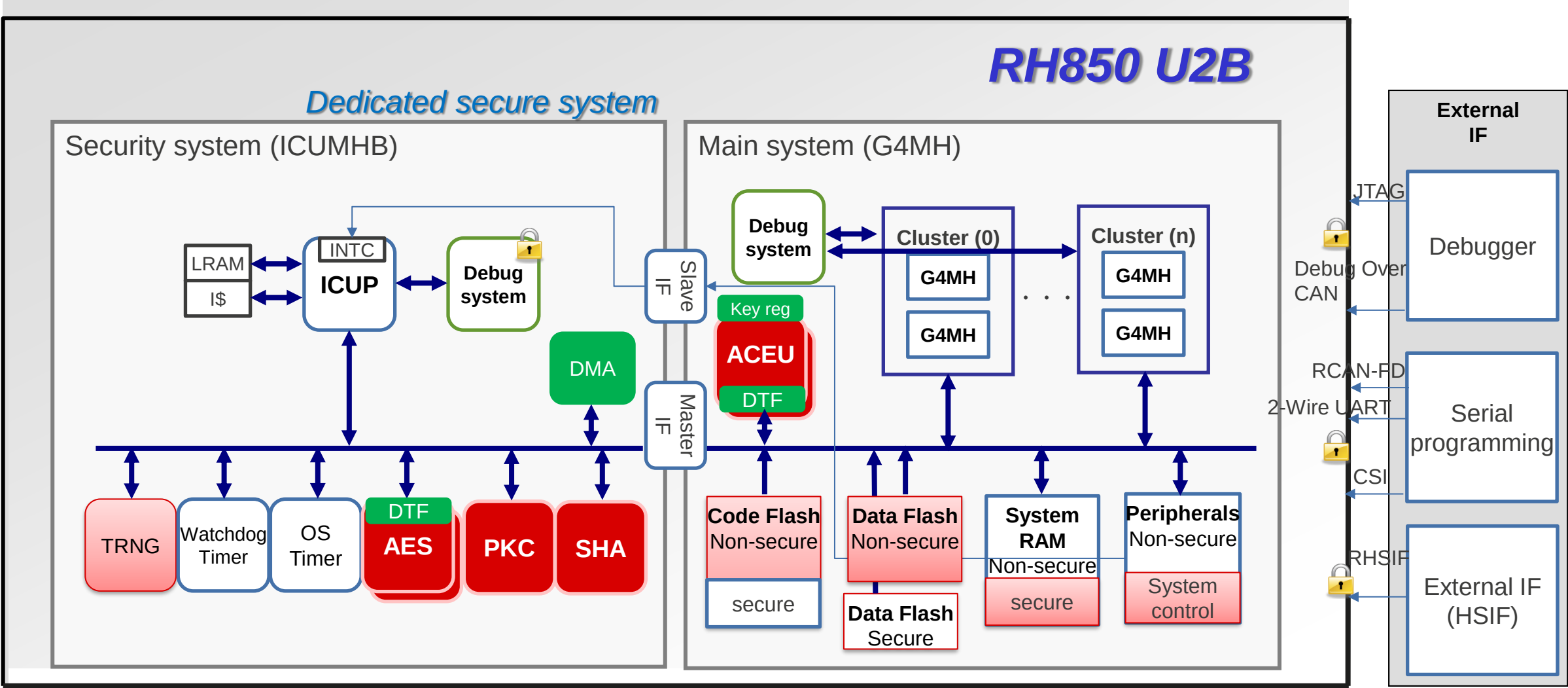
HIS SHE : Hersteller Initiative Software Secure Hardware Extension (**NOTE** : HIS SHE functional specification version 1.1 transferred ownership to AUTOSAR and was re-released in 2019 as AUTOSAR Specification of Secure Hardware Extensions version R19-11)

PRNG : Pseudo Random Number Generator (Also known as a DRBG : Deterministic Random Bit Generator)

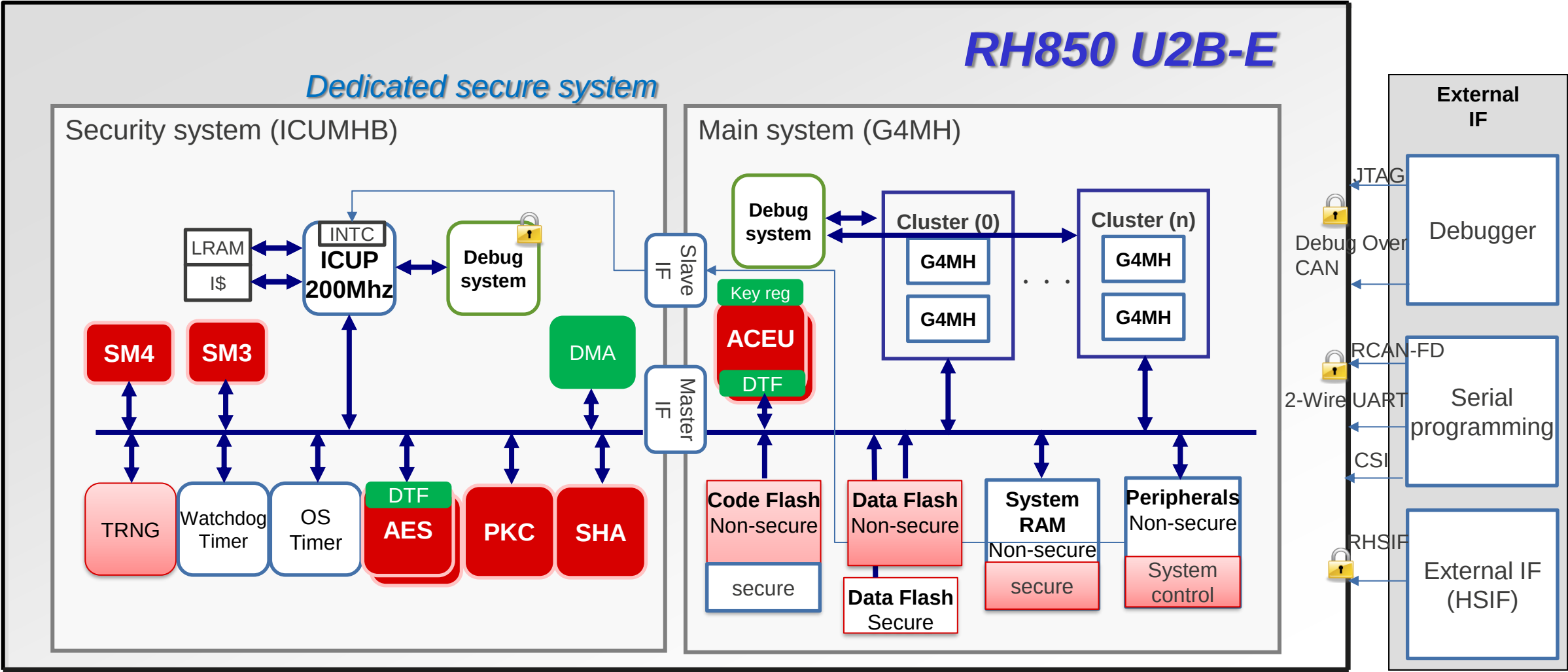
TRNG : True Random Number Generator (Also known as an NRBG : Non-deterministic Random Bit Generator)

* EVITA full, WHIRLPOOL, ICU-MH, SHA-2

ICUMHB IN THE MCU: A SIMPLIFIED VIEW



ICUMHB IN THE MCU: A SIMPLIFIED VIEW



ICU-M / MH COMPARISON TABLE (1/3)

IP	ICUMD	ICUMHA	ICUMHB	ICUMHB
Products	F1Kx	U2A	U2B/U2B-E	U2C
Process	RV40F	RV28F	RV28F	RV28F
CPU (ICUP)	G3K	G3K	G3K	G3K
LRAM	64KB	64KB	U2B/-FCC: 64KB U2B-E: 128KB	128KB
Instruction Cache	4KB	4KB	4KB	4KB
MPU	8 area	8 area	8 area	8 area
Frequency (max)	60MHz	100MHz	200MHz	160MHz
Code Flash (secure)	Shared	Shared	Shared	Shared
Data Flash (secure)	F1KM-S4: Shared	Dedicated U2A16/8/6: 64KB	Dedicated U2B24-FCC/U2B24-E: 160KB U2B16-E: 128KB U2B10/6: 64KB U2B12-E/6-E: 64KB	Dedicated U2C8/4/2: 64KB

ICU-M / MH COMPARISON TABLE (2/3)

Symmetric Crypto

IP	ICUMD	ICUMHA	ICUMHB	ICUMHB
Products	F1Kx	U2A	U2B/U2B-E	U2C
AES	AES-128	AES-128, 192, 256	AES-128, 192, 256	AES-128, 192, 256
AES supportive mode	ECB, CBC, CTR, CFB-128, OFB, GCM, XTS, CCM, CMAC, MP	ECB, CBC, CTR, CFB-128, OFB, GCM, XTS, CCM, CMAC, MP	ECB, CBC, CTR, CFB-128, OFB, GCM, XTS, CCM, CMAC, MP	ECB, CBC, CTR, CFB-128, OFB, GCM, XTS, CCM, CMAC, MP
Context	3	3	3	3
ROM key	3	3 (max 256-bit)	3 (max 256-bit)	3 (max 256-bit)
Dedicated access master	—	✓	✓	✓
SM4	—	—	U2B/-FCC: - U2B-E: SM4 128-bit	SM4 128-bit
SM4 supportive mode	—	—	U2B/-FCC: - U2B-E: ECB, CBC, CTR, CFB-128, OFB, GCM, CCM, CMAC	ECB, CBC, CTR, CFB-128, OFB, GCM, CCM, CMAC

ICU-M / MH COMPARISON TABLE (3/3)

OTHER CRYPTO ENGINES

(1): Increased performance
(2): Can be supported by ICUM FW

IP	ICU-MD	ICU-MHA	ICU-MHB	ICU-MHB
Products	F1Kx	U2A	U2B/U2B-E	U2C
TRNG	Supported	Supported Updated (*1)	AIS31 PTG.2 (target) Updated (*1)	AIS31 PTG.2 (target) Updated (*1)
SHA	— (*2)	SHA-1, SHA-224, SHA-256,	SHA-1, SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, SHA-512/256, SHA3-256, SHAKE256 SHA3-512, U2B-E: SM3	SHA-1, SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, SHA-512/256, SHA3-256, SHAKE256 SHA3-512, SM3
SipHash	— (*2)	— (*2)	— (*2)	— (*2)
PKCC	— (*2)	RSA2K	RSA2K/3K/4K	RSA2K/3K/4K
	— (*2)	ECC, Up to 576-bit	ECC, Up to 576-bit	ECC, Up to 576-bit

PQC ALGORITHM SUPPORT

IP	ICUMHA	ICUMHB	ICUMHB
Products	U2A	U2B/U2B-E	U2C
PQC Algorithm ➤ SPHINCS+-SHA2-128s ➤ CRYSTALS-Dilithium ➤ CRYSTALS-Kyber-768	✓ Kyber and Dilithium can be supported by only SW (HW is not supported). ✓ SPHINCS+-SHA2-128s can be supported by SW and HW (SHA2-256 HW). ✓ After SPHINCS+-SHA2-192s can be supported by only SW (SHA2-512 HW is not supported).	✓ Kyber, Dilithium, and SPHINCS+ can be supported by SW and HW (SHA2-256 HW, SHA2-512 HW, SHA3 HW).	✓ Kyber, Dilithium, and SPHINCS+ can be supported by SW and HW (SHA2-256 HW, SHA2-512 HW, SHA3 HW).

PQC ALGORITHM SUPPORT

ML-DSA (ex. Dilithium), ML-KEM (ex. Kyber), SLH-DSA (ex. Sphincs+)

PQC Algorithm	[ICUMD and before] RH850/X1x, E2x	[ICUMHA, ICUMXA] RH850/U2A, R-Car V3H	[ICUMHB, ICUMXB, RSIP-M] RH850/U2B, U2C, U2B-E, R-Car S4, V4x, R-Car X5x, U5x
ML-DSA (ex. Dilithium)	By only SW. (SHA2/SHA3 HW is not supported).	By only SW. (SHA3 HW is not supported).	By SW and HW. (SHA2-256 HW, SHA2-512 HW, SHA3 HW).
ML-KEM (ex. Kyber)			
SLH-DSA (ex. Sphincs+)		SPHINCS+-SHA2-128s can be supported by SW and HW. (SHA2-256 HW). After SPHINCS+-SHA2-192s can be supported by only SW. (SHA2-512 HW is not supported).	

This investigation does not include memory resource consumption

HASH FUNCTION REQUIRED FOR PQC ALGORITHM

- Hash function required for PQC algorithm
 - ML-KEM (ex. Kyber): **SHA3** is required regardless of key length
 - ML-DSA (ex. Dilithium): **SHA3** is required regardless of key length
 - SLH-DSA (ex. SPHINCS+): There are **SHA2** and **SHA3** versions (all combinations of key lengths are present)
 - There are a total of 6 things that use SHA2 in the latest Draft version.
 - SPHINCS+-SHA2-128s and SPHINCS+-SHA2-128f require **SHA-256**.
 - SPHINCS+-SHA2-192s and SPHINCS+-SHA2-192f require **SHA-512**.
 - SPHINCS+-SHA2-256s and SPHINCS+-SHA2-256f require **SHA-512**.

AES CRYPTO ENGINE UNIT (ACEU)

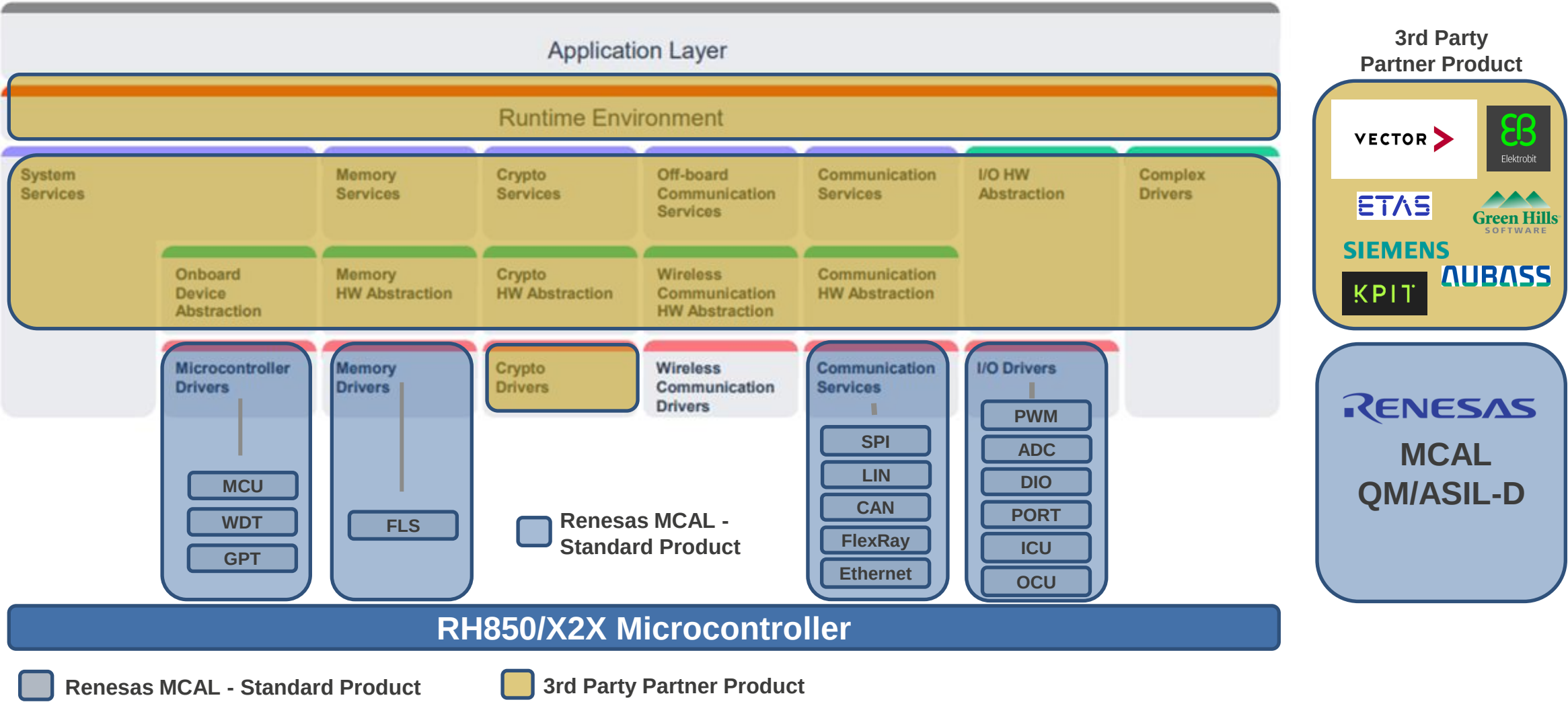
- Key size update (128, 192, 256-bit)
- Keys for ACEU are supplied from its own RAM area (KEYRAM).
 - Only ICUMHB can load keys from Secure Data Flash to KEYRAM (startup phase).
 - Non-secure domain master modules cannot read keys from KEYRAM.
- Same features as ICUM internal AES engine
 - 3x AES contexts, same AES modes supported (ECB, CBC, CTR, CFB-128, OFB, GCM, XTS, CCM, CMAC, MP)
 - Dedicated access master (Data Transfer Function: DTF)
 - Lock step
- ACEU can be configured as Secure Domain or application domain
 - Data transfer is then controlled by corresponding master
- Several ACEU channels available depending on device
 - U2B24-FCC: 2x ACEU
 - U2B24-E: 4x ACEU U2B16-E: 3x ACEU U2B12-E: 3x ACEU U2B6-E: 2x ACEU

AUTOSAR MCAL SW

```
function use_array(a) {  
  for (var c = 0; c < a.length; c++) {  
    use_array(a[c], a);  
  }  
  return a;  
}  
  
function new_user(a) {  
  for (var b = "", c = 0; c < a.length; c++) {  
    b += a[c] + " ";  
  }  
  return b;  
}
```

DOMAttrModified textInput input change key
INTOUE: " + a.unique);

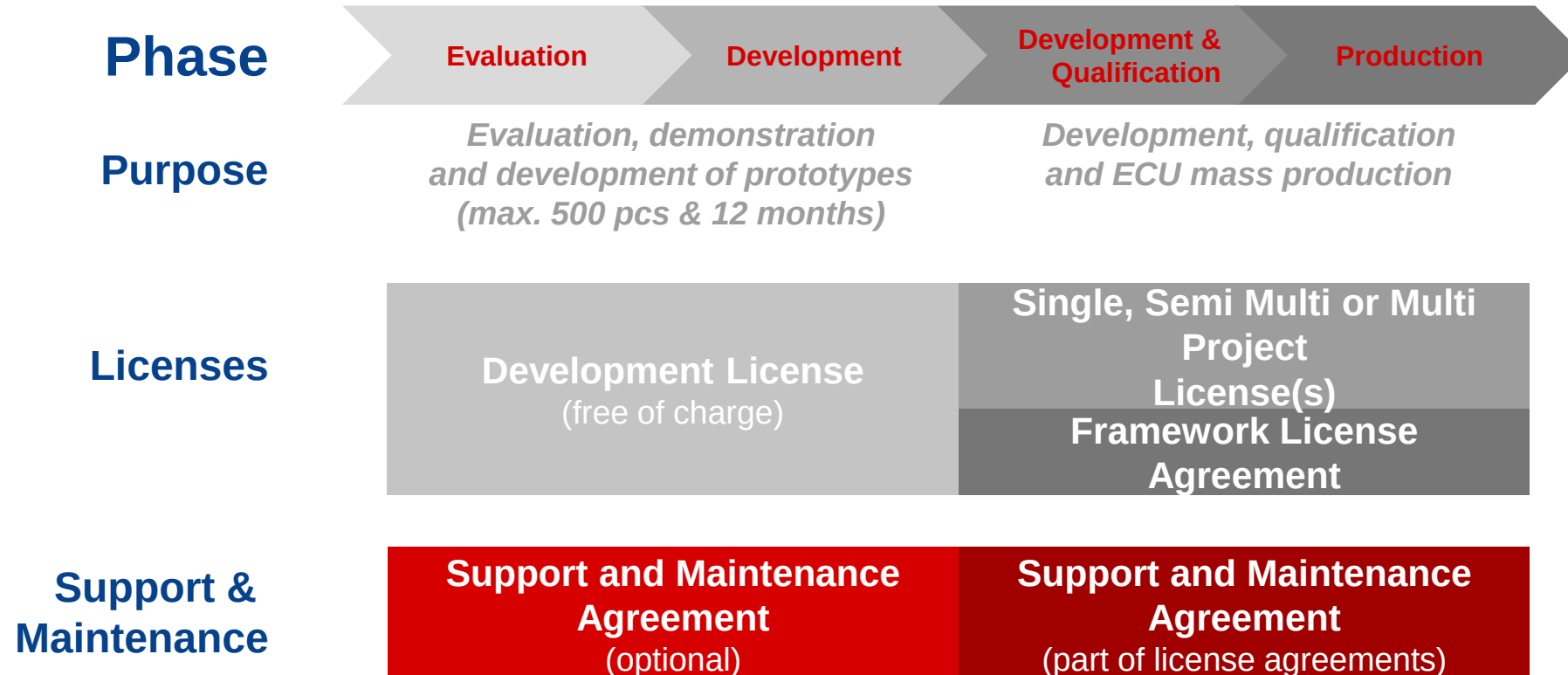
AUTOSAR CLASSIC - RENESAS MCU MCAL – WHAT’S INSIDE?



RENESAS – MCU MCAL PORTFOLIO

AUTOSAR Release Version	AUTOMOTIVE Segments / Applications				
	Body	Powertrain	HEV/EV	Chassis & Safety	Instrument Cluster
R22-11	RH850/U2Bx, U2Bx-E, U2C (ISO 26262 / FuSa compliant up to ASIL-D)				
R21-11	RH850/U2Bx (Intermediate development step)				
4.3.1	RH850/U2A, U2B24-FCC (ISO 26262 / FuSa compliant up to ASIL-D)				
4.0.3 / 4.2.2	RH850/X1x Products + RH850/E2x (ISO26262 / FuSa compliant up to ASIL-D)				

LICENSING & SUPPORT SCHEME



X2X MCAL LICENSE VERSIONS AND CONDITIONS

Development License (DLA)

- for evaluation, demonstration and development (incl. prototypes), must not be used for qualification and mass production
- all available software for all sub-series included (latest versions)
- no warranty, no liability, limited to 1 year and 500 units (prototypes)
- license free of charge
- support and maintenance optional

Free of Charge (FOC) „[Web Download Version](#)“

- for all OEMs, car models and applications
- valid for all devices within a series (eg. F1x) listed in the license agreement
- QM quality for mass production usage
- for world-wide commercial usage, unlimited in production volume & time
- no license fee
- no liability / warranty
- SW maintenance (may be reduced compared to commercial license)
- online available for instant download
- including source code
- support through Renesas forum

Single Project License (SPL)

- for only one single project
 - one ECU with one specific device (e.g. U2A16 176QFP) for one tier1 project / not for tier1 platform
- for world-wide commercial usage, unlimited in production volume & time
- one-time license fee
- support and maintenance mandatory for 1 year, optional for following years

Semi-Multi Project License (SMPL)

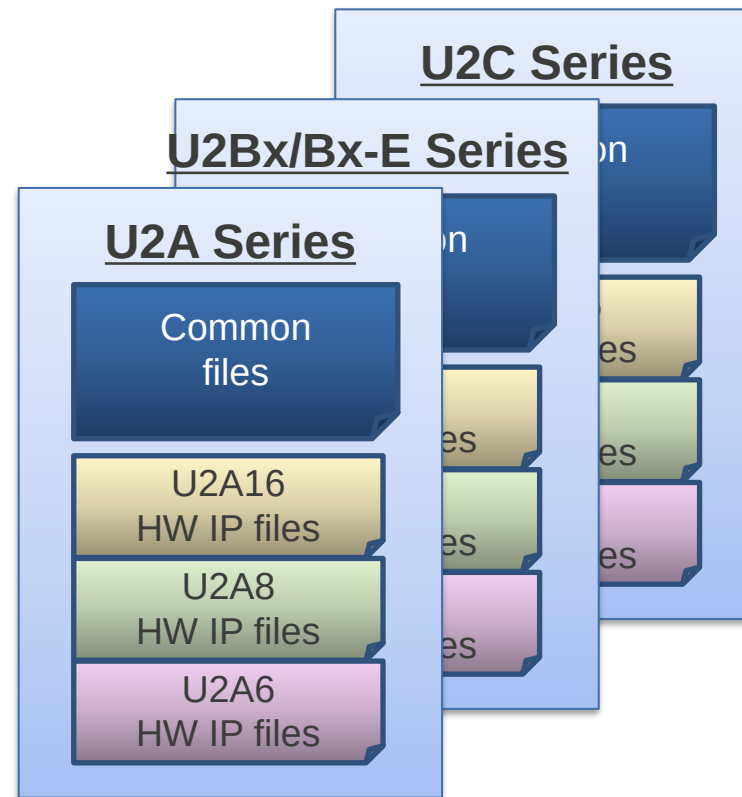
- for all OEMs, car models and applications
- for one MCU product line (e.g. U2A16)
- for world-wide commercial usage, unlimited in production volume & time
- one-time license fee
- support and maintenance mandatory for 1 year, optional for following years

Multi Project License (MPL)

- for all OEMs, car models and applications
- valid for all devices within a series (eg. U2Ax) listed in the license agreement
- for world-wide commercial usage, unlimited in production volume & time
- one-time license fee
- support and maintenance mandatory for 1 year, optional for following years

X2X - MCAL Package content

One Single MCAL Package supports all variants within Series!



MCAL DELIVERABLES

MCAL package content (For QM / ASIL)

Modules Overview

Getting Started Manual

Fixed Issues List (FIL)

Known-Issues-List (KIL)

AUTOSAR MCAL User's manual

Generation Tool User's manual

MCAL Source code

Generation Tool

Sample Application

Release Notes

Safety Package content (For ASIL MCAL)

Development Interface Report (DIR) or
Development Interface Agreement (DIA)

Functional Safety Assessment Report (FSA)

SW Safety Requirement Specification (SRS)

AUTOSAR MCAL Tool Summary Classification

MCAL Safety Case Application Note (SW-SAN)

MCAL Safety Case Summary

Project Plan (refined)

MCAL Software configuration

Release Notes

For ASIL customers, Safety WPs are
provided on top of the MCAL package

MCAL QUALITY & ASSESSMENT

- Renesas has long history in the automotive business and is a partner of TIER1/OEMs since years.
- Renesas established a SW development process that complies to ISO9001/IATF16949, CMMI L3 and A-SPIICE L2
- Renesas recently had an A-SPIICE assessment from other customer with overall L2 result

Classification	Definition		Customer's purpose example
	Specifications and design status	Qualification status	
α	Specifications status: Design specification undetermined Design status: Already part of the function / non-functional implementation Any of the above, or those that fall under both	Unfinished Software Design Qualification* ¹	• Specifications investigation • Partial functional evaluation
β	Specifications status: Finalized design specifications	Unfinished Software Qualification* ²	• Function / performance evaluation • Partial Quality evaluation*⁴
QM-MP	Design status: <u>All of the function / non-function is already implemented</u>	Software Qualification* ² has been completed	• Function / performance evaluation
ASIL-MP	Those that fall under <u>both of the above</u>	Software Qualification* ² and Functional Safety Assessment* ³ had been completed	• Quality evaluation • Apply to mass production

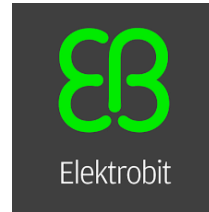
*1: Software development: Implementation standard for Software Design Qualification (RCT-JB5001-021)

*2: Operation standard for Software Qualification (RCT-JF1001)

*3: Implementation Standard for ISO 26262 Compliance of Software Development: Confirmation Measures (RCT-JB0019-005)

*4: Represents the part that has been assessed by the Software Qualification of the Quality Assurance Department.

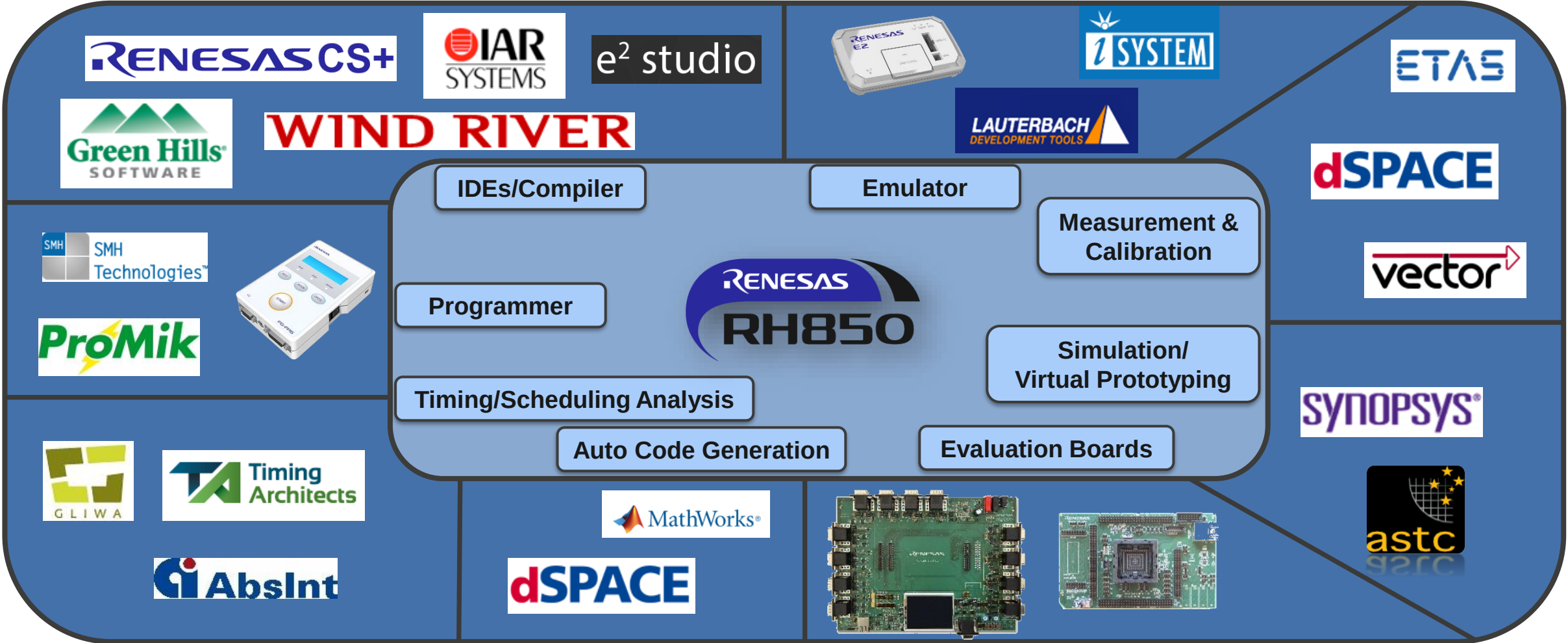
3RD PARTY AUTOSAR PARTNER



TOOLS

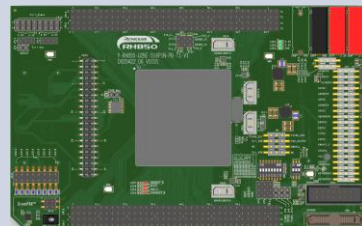
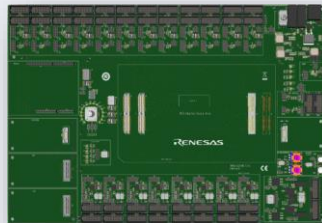


RH850 TOOLS ECO SYSTEM OVERVIEW



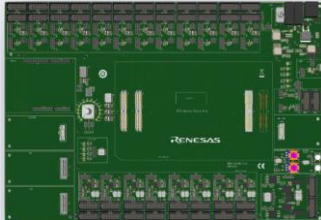
EVALUATION PLATFORM – RH850/U2B-E SOLUTION OVERVIEW

■ Modular design, available for all package variants

Supported Devices Board Type	RH850/U2Bx-E Series				
	BGA 292-pin		BGA 516-pin		
	RH850/U2B24-E EVA		RH850/U2B24-E EVA		
RH850/U2B-E Piggyback board with device socket (supports stand-alone operation)	 Y-RH850-U2BE-292PIN-PB-T1-V1		 Y-RH850-U2BE-516PIN-PB-T1-V1		
Main board (adds additional functionality, e.g. physical I/F for Ethernet, FlexRay, CAN and LIN)	 Y-RH850-X1X-MB-T1-V1		 Y-RH850-X1X-MB-T2-V1	 Y-RH850-X2X-MB-T1-V1	 Y-COMMON-MB-T1-V1

EVALUATION PLATFORM – RH850/U2B SOLUTION OVERVIEW

■ Modular design, available for all package variants

Supported Devices Board Type	RH850/U2Bx Series			
	BGA 292-pin	BGA 373-pin	BGA 468-pin	
	RH850/U2B10(-FCC) RH850/U2B6(-FCC)	RH850/U2B10(-FCC)	RH850/U2B24-FCC	
RH850/U2B Piggyback board with device socket (supports stand-alone operation)	 Y-RH850-U2B-292PIN-PB-T1-V1	 Y-RH850-U2B-373PIN-PB-T1-V1	 Y-RH850-U2B-468PIN-PB-T1-V1	
Main board (adds additional functionality, e.g. physical I/F for Ethernet, FlexRay, CAN and LIN)	 Y-RH850-X1X-MB-T1-V1	 Y-RH850-X1X-MB-T2-V1	 Y-RH850-X2X-MB-T1-V1	 Y-COMMON-MB-T1-V1

EASY MIGRATION FROM 40NM X1X TO 28NM U2X MCU FAMILY



MCU Architecture

Security

Safety

Peripherals

	 Series	 Series	 U2A/B	 U2Cx	
MCU Architecture	RH850/ G3x	RH850/ G3x	RH850/ G4x	RH850/ G4x	Up to 5x higher DMIPS performance with 28nm MCU
Security	EVITA Light-Mid	EVITA Light-Mid	EVITA Full	EVITA Full	Increased security to highest automotive standard “EVITA-Full”
Safety	ASIL-B	ASIL-D	ASIL-D	ASIL-B & ASIL-D	Full, flexible ISO26262 support in same 28nm MCU family
Peripherals	CAN/CANFD, FR LIN/UART, ADC TAUx, SENT, eMMC	CAN/CANFD, FR LIN/UART, ADC TAUx,TSG3, PSIx, SENT, GTM*	CAN/CANFD, FR LIN/UART, ADC TAUx, TSG3, GTM, PSIx, SENT, CXPI, eMMC, Hypervisor	CAN/CANFD, FR LIN/UART, ADC TAUx, TSG3, GTM, PSIx, SENT, CXPI, eMMC	Common peripherals with extended features as differences + additional features*: CAN-XL, Ethernet T1S, ISO21434, China Crypt, I3C, I2S,...

* Depending on device

[Renesas.com](https://www.renesas.com)